TOPIC 7 · VISCOUS FLOWS

Solved Problems: Official Viscous Flow Sheets VF2 to VF20

Worked solutions to the nineteen viscous-flow sheets of the course (VF2 to VF20), most of them past exam problems. Each statement is transcribed from the sheet and its figure redrawn; each solution follows the handwritten key of the sheet with every step written out: the simplified Navier-Stokes equations term by term, every integration with its constants fixed by the boundary conditions, the fluxes and stresses, the order-of-magnitude criteria for a slender or quasi-steady flow, and the checks. Where the key has a slip, the correction is shown as an erratum. References to Notes.pdf are to Chapter 7 (Viscous Flows).

PROBLEM VF2

Oscillating Wall under a Liquid Layer

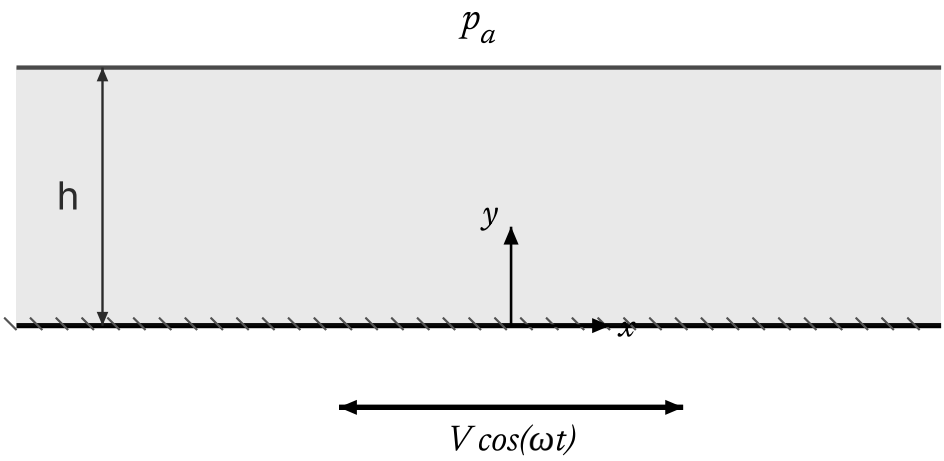
Original statement · VF2_w_solution.pdf

Unsteady unidirectional flow | Complex solution | Quasi-steady and Stokes limits

OFFICIAL PROBLEM STATEMENT

Consider the periodic motion induced in a liquid layer of thickness h when the bottom wall moves with velocity $V\cos(\omega t)$. Write the equation with boundary conditions that determines $v_x(y,t)$. Consider separately the cases $\omega h^2/\nu \gg 1$ and $\omega h^2/\nu \ll 1$. Obtain the general solution and show how it reduces to the limiting solutions studied earlier in the appriate limits.

The sheet prints "appriate"; it is kept. The top of the layer is a free surface at the pressure p_a, as the figure of the sheet shows.



Statement figure (redrawn from VF2_w_solution.pdf, p. 1): a liquid layer of thickness h with a free surface at p_a on a wall that oscillates with velocity V cos(ωt); y is normal to the wall and x along it

Source: VF2_w_solution.pdf (page 1: statement and first solution; pages 2 to 5: handwritten derivation). Notes.pdf, Chapter 7: unidirectional flow, Eq. 7.6; Stokes flow near a wall, Eqs. 7.26-7.31; oscillating wall of a channel, Eqs. 7.32-7.35.

Worked solution

Hypotheses, Data and Equation

- The liquid has constant ρ and μ. The flow is parallel to the wall: $\bar{v} = v_x(y,t)\,\bar{e}_x$, because continuity gives $\partial v_x/\partial x = 0$ (Notes.pdf, Eq. 7.1).
- **Pressure.** The y component of the momentum equation gives $0 = -\partial P/\partial y$ with $P = p + \rho g y$, so P is the same at every height. At the free surface $p = p_a$ and $y = h$, hence $P = p_a + \rho g h$, a constant: $\partial P/\partial x = 0$ and $P_l = 0$ (the free surface stays flat).
- **Free surface.** The air above exerts a negligible shear stress (its viscosity is neglected), so the shear stress of the liquid vanishes at $y = h$: $\partial v_x/\partial y = 0$.
- **Periodic solution.** The wall velocity is periodic and the sought solution is the permanent one, so no initial condition is needed (Notes.pdf, p. 78). With a forcing $V \sin \omega t$ one could impose $v_x = 0$ at $t = 0$, which is compatible.

Equation (7.6) with $P_l = 0$ and $\nu = \mu/\rho$, and its boundary conditions:

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}; \qquad y = 0 :\ v_x = V \cos \omega t, \qquad y = h :\ \frac{\partial v_x}{\partial y} = 0$$

Symbol	Meaning	SI unit
h	thickness of the layer	m
V, ω	amplitude and frequency of the wall velocity	m/s, s^{-1}
ν	kinematic viscosity	m^2/s
$\omega h^2/\nu$	ratio of the viscous time h^2/ν to ω^{-1}	1

General Solution with a Complex Function

Since $V \cos \omega t = \mathrm{Re}\left(V e^{\imath \omega t}\right)$, the linear problem is solved for a complex v^* with $v_x = \mathrm{Re}(v^*)$:

$$\frac{\partial v^*}{\partial t} = \nu \frac{\partial^2 v^*}{\partial y^2}; \qquad y = 0 :\ v^* = V e^{\imath \omega t}, \qquad y = h :\ \frac{\partial v^*}{\partial y} = 0$$

Look for $v^* = V e^{\imath \omega t} f(y)$. Substituting, $\imath \omega f = \nu f''$, that is

$$f'' - \frac{\imath \omega}{\nu} f = 0; \qquad f(0) = 1, \qquad f'(h) = 0$$

(in the key's non-dimensional form, $z = y/h$ and $2\Omega^2 = \omega h^2/\nu$, the problem has **one parameter**: $f'' - 2\imath\Omega^2 f = 0$ with $f(0) = 1$, $f'(1) = 0$).

Characteristic equation. $\lambda^2 = \imath \omega/\nu$. With $\imath = e^{\imath\pi/4} \cdot e^{\imath\pi/4}$, $\sqrt{\imath} = (1 + \imath)/\sqrt{2}$, so the two roots are $\pm\lambda$ with

$$\lambda = (1 + \imath)\sqrt{\frac{\omega}{2\nu}}, \qquad \lambda h = (1 + \imath)\,\Omega \quad \text{with} \quad \Omega = \sqrt{\frac{\omega h^2}{2\nu}}$$

(check: $\lambda^2 = (1 + \imath)^2\, \omega/(2\nu) = 2\imath\, \omega/(2\nu) = \imath\omega/\nu$).

Constants. $f = C_1 e^{\lambda y} + C_2 e^{-\lambda y}$. The wall condition gives $C_1 + C_2 = 1$. The condition at the free surface, $f'(h) = \lambda\left(C_1 e^{\lambda h} - C_2 e^{-\lambda h}\right) = 0$, gives $C_1 e^{\lambda h} = C_2 e^{-\lambda h}$. Substituting $C_2 = 1 - C_1$:

$$C_1\left(e^{\lambda h} + e^{-\lambda h}\right) = e^{-\lambda h} \implies C_1 = \frac{e^{-\lambda h}}{e^{\lambda h} + e^{-\lambda h}}, \qquad C_2 = 1 - C_1 = \frac{e^{\lambda h}}{e^{\lambda h} + e^{-\lambda h}}$$

Putting them in f:

$$f = \frac{e^{-\lambda h} e^{\lambda y} + e^{\lambda h} e^{-\lambda y}}{e^{\lambda h} + e^{-\lambda h}} = \frac{e^{\lambda (h-y)} + e^{-\lambda (h-y)}}{e^{\lambda h} + e^{-\lambda h}} = \frac{\cosh\left[\lambda (h-y)\right]}{\cosh(\lambda h)}$$

and the complex solution is

$$v^* = V e^{\imath \omega t}\, \frac{\cosh\left[(1+\imath)\Omega(1 - y/h)\right]}{\cosh\left[(1+\imath)\Omega\right]}, \qquad v_x = \mathrm{Re}(v^*)$$

Discrepancy with the official solution

The second boundary-condition equation of the key, boxed on page 1, reads $C_1 e^{(1+\imath)\sqrt{\omega/2\nu}\, h} - C_2 e^{-(1+\imath)\sqrt{\omega/2\nu}\, h} = 1$. The condition $f'(h) = 0$ gives 0 on the right-hand side, as derived above, and the coefficients C_1, C_2 that the key writes just below are the ones that follow from 0, not from 1. It is a slip in the key; the final solution is not affected.

The Real Solution

With $s = \Omega(1 - y/h)$ and $\cosh(s + \imath s) = \cosh s \cos s + \imath \sinh s \sin s$:

$$f = \frac{A + \imath B}{C + \imath D}, \qquad A = \cosh s \cos s,\ B = \sinh s \sin s,\ C = \cosh \Omega \cos \Omega,\ D = \sinh \Omega \sin \Omega$$

Multiplying numerator and denominator by $C - \imath D$ gives $f = [(AC + BD) + \imath(BC - AD)]\,/(C^2 + D^2)$. The real part of $e^{\imath\omega t} f$ is $[(AC + BD)\cos\omega t - (BC - AD)\sin\omega t]\,/(C^2 + D^2)$ (since $\mathrm{Re}[(\cos\omega t + \imath\sin\omega t)(a + \imath b)] = a\cos\omega t - b\sin\omega t$). Also $C^2 + D^2 = (\cosh 2\Omega + \cos 2\Omega)\,/2$, which is the denominator $e^{2\Omega} + e^{-2\Omega} + 2\cos 2\Omega$ of the key up to the factor 4. Then

$$\frac{v_x}{V} = \frac{(AC + BD)\cos\omega t - (BC - AD)\sin\omega t}{C^2 + D^2}$$

At $y = 0$, $s = \Omega$, so $A = C$ and $B = D$ and $v_x/V = \cos\omega t$, as the wall condition requires. The key obtains the same function as a sum of four terms of the form $e^{\pm\Omega z}\cos(\Omega z \pm \omega t + \dots)$, multiplying by the conjugate of the denominator.

Limit $\omega h^2/\nu \ll 1$: Quasi-steady Solution

Here $|\lambda h| = \sqrt{2}\,\Omega \ll 1$, so the viscous time h^2/ν is much shorter than ω^{-1} and the unsteady term is negligible. The equation reduces to $\partial^2 v_x/\partial y^2 = 0$ with $v_x = V\cos\omega t$ at $y = 0$ and $\partial v_x/\partial y = 0$ at $y = h$. Integrating twice, $v_x = A(t)y + B(t)$; the condition at $y = h$ gives $A = 0$ and the wall gives $B = V\cos\omega t$:

$$v_x = V\cos\omega t$$

The liquid layer moves like a solid: with no shear stress at the free surface, nothing opposes the motion. The exact solution gives the same: $\cosh(x) \simeq 1 + x^2/2$ for small x, so $f \simeq 1 + \frac{\lambda^2}{2}\left[(h - y)^2 - h^2\right] = 1 - \frac{\imath\omega}{2\nu}\,y\,(2h - y) \to 1$. The first correction (computed here, an **extension**) is $v_x \simeq V\left[\cos\omega t + \frac{\omega}{2\nu}\,y\,(2h - y)\sin\omega t\right]$, of relative size $\omega h^2/\nu \ll 1$ (it uses $\mathrm{Re}\left[\imath e^{\imath\omega t}\right] = -\sin\omega t$).

Limit $\omega h^2/\nu \gg 1$: Stokes Layer

Here $\Omega \gg 1$: the thickness h is much larger than the penetration length $\delta = \sqrt{\nu/\omega}$. The sheet first writes $\partial v_x/\partial t = 0$ with two question marks: the right argument is not that the layer is at rest, but that the motion is confined to a thin layer. Balancing $\omega V \sim \nu V/\delta^2$ gives $\delta \sim \sqrt{\nu/\omega} \ll h$, and the problem becomes the one of Stokes (Notes.pdf, Eq. 7.26):

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}; \qquad y = 0: \; v_x = V \cos \omega t, \qquad y \gg \sqrt{\nu/\omega}: \; v_x \to 0$$

whose solution (7.31) is

$$v_x = V \, e^{-y/\sqrt{2\nu/\omega}} \, \cos\left(\omega t - \frac{y}{\sqrt{2\nu/\omega}} \right)$$

It follows from the exact solution: $\cosh[\lambda(h-y)]/\cosh(\lambda h) = \left[e^{-\lambda y} + e^{-\lambda(2h-y)}\right] / \left[1 + e^{-2\lambda h}\right]$, and $\left|e^{-2\lambda h}\right| = e^{-2\Omega} \to 0$, $\left|e^{-\lambda(2h-y)}\right| \le e^{-\Omega}$ for $y \le h$, so $f \simeq e^{-\lambda y}$ with $\lambda = (1+\imath)/\sqrt{2\nu/\omega}$, and the real part of $V e^{\imath\omega t} e^{-\lambda y}$ is the expression above. The thickness h **disappears** from the result.

The key solves the same limit in the non-dimensional variable $\eta = y/\delta$ with $\delta^2\omega/\nu = 1$: $f'' - \imath f = 0$, $f(0) = 1$, $f(\infty)$ bounded, so the root with growing exponential has a zero coefficient ($A = 0, B = 1$) and $f = e^{-(1+\imath)\eta/\sqrt{2}}$.

Answers and Checks

- Problem: $\partial v_x/\partial t = \nu \, \partial^2 v_x/\partial y^2, v_x(0,t) = V \cos \omega t, \partial v_x/\partial y(h,t) = 0$.
- General solution: $v_x = \mathrm{Re}\left\{ V e^{\imath\omega t} \cosh\left[(1+\imath)\Omega(1-y/h)\right] / \cosh\left[(1+\imath)\Omega\right] \right\}$ with $\Omega^2 = \omega h^2/(2\nu)$.
- $\omega h^2/\nu \ll 1$: $v_x = V \cos \omega t$ (rigid motion). $\omega h^2/\nu \gg 1$: Stokes solution, independent of h.
- Boundary conditions of the general solution: at $y = 0$, $f = 1$; at $y = h$, $\sinh[\lambda(h-y)] = 0$, so $f' = 0$.
- Dimensions: $\omega h^2/\nu$ is $T^{-1}L^2/(L^2T^{-1}) = 1$; $\sqrt{2\nu/\omega}$ is a length.

- Check against the official key: VF2_w_solution.pdf gives the same equation, complex solution, real part, rigid limit and Stokes limit, with the slip in the second boundary-condition equation noted above; the minus sign that the key writes before $e^{-\eta/\sqrt{2}}e^{\imath(\omega t-\eta/\sqrt{2})}$ on page 4 (a reading of the handwriting) also disappears in the real part it boxes.

PROBLEM VF3

Annular Flow with a Moving Outer Cylinder

Original statement · VF3.pdf

Annulus | Poiseuille + Couette | Zero flux

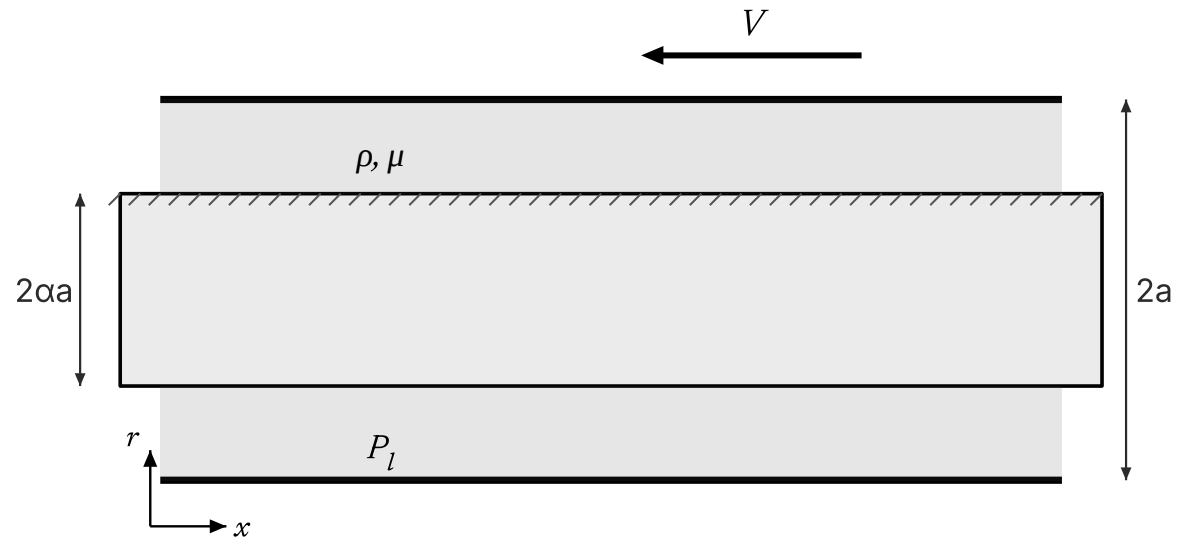
OFFICIAL PROBLEM STATEMENT

A fluid of density ρ and viscosity μ is confined between two infinitely long coaxial cylinders of radii a and αa, with $\alpha < 1$, aligned in the x direction. The fluid is forced to move steadily due to the presence of a known reduced pressure gradient $P_l = -\partial P/\partial x$ and also due to the backward motion of the outer cylinder, with velocity V.

1. Determine the fluid velocity $v_x(r)$.

2. Obtain the power needed to move the outer cylinder.

3. Calculate the value of V for which the volume flux in the pipe is identically zero.



Statement figure (redrawn from VF3.pdf): liquid of density ρ and viscosity μ between a rod of radius αa and a cylinder of radius a, along x, driven by the pressure gradient P_l; the outer cylinder moves backward with velocity V

Source: VF3.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 7: unidirectional flow, Eq. 7.7; steady flow in a circular pipe, Eqs. 7.13-7.16.

Worked solution

Hypotheses, Data and Unknowns

- Steady, fully developed flow of a liquid with constant μ: $\bar{v} = v_x(r)\bar{e}_x$ (Notes.pdf, Eq. 7.1, with axial symmetry); P_l is a known constant (Notes.pdf, p. 74).
- The inner body, a rod of radius αa, is at rest; the outer cylinder of radius a moves backward, that is, with velocity $-V\bar{e}_x$.
- "Backward" means against the pressure-driven flow: $P_l > 0$ drives the liquid towards $+x$.
- The power is taken per unit length of cylinder, since the cylinders are infinitely long.

Symbol	Meaning	SI unit
a, αa	outer and inner radius ($\alpha < 1$)	m
P_l	reduced pressure gradient, $-\partial P/\partial x$	Pa/m
V	speed of the outer cylinder (towards $-x$)	m/s
μ	viscosity	kg/(m s)

Velocity Profile

Equation (7.7) in steady state, with the conditions at the two walls:

$$0 = P_l + \frac{\mu}{r}\frac{d}{dr}\left(r\frac{dv_x}{dr}\right); \qquad r = \alpha a:\ v_x = 0, \qquad r = a:\ v_x = -V$$

Multiplying by r/μ and integrating once: $r\,dv_x/dr = -P_l r^2/(2\mu) + C_1$. Dividing by r and integrating again:

$$v_x = -\frac{P_l}{4\mu}r^2 + C_1 \ln r + C_2$$

The term $C_1 \ln r$ cannot be dropped: the axis is not part of the domain, so there is no regularity condition. It is convenient to write the solution with the inner wall as origin of the logarithm, $v_x = -\frac{P_l}{4\mu}\left(r^2 - \alpha^2 a^2\right) + C \ln \frac{r}{\alpha a}$, which satisfies $v_x(\alpha a) = 0$ for any C. The outer wall fixes C:

$$-V = -\frac{P_l}{4\mu} a^2 \left(1 - \alpha^2\right) + C \ln \frac{1}{\alpha} \implies C = \frac{-V + \dfrac{P_l}{4\mu} a^2 \left(1 - \alpha^2\right)}{\ln\left(\alpha^{-1}\right)}$$

$$v_x = -\frac{P_l}{4\mu}\left[r^2 - (\alpha a)^2\right] + \left[-V + \frac{P_l}{\mu}\frac{\left(1 - \alpha^2\right) a^2}{4}\right] \frac{\ln\left(r/\alpha a\right)}{\ln\left(\alpha^{-1}\right)}$$

Check: at $r = \alpha a$ the logarithm is zero and the bracket $r^2 - (\alpha a)^2$ vanishes; at $r = a$ the first term is $-P_l a^2(1 - \alpha^2)/(4\mu)$ and the second is $-V + P_l a^2(1 - \alpha^2)/(4\mu)$, whose sum is $-V$.

Power Needed to Move the Outer Cylinder

The force per unit area of the liquid on the outer wall is $\tau_p = \bar{\bar{\tau}}' \cdot \bar{n}$ with $\bar{n}$ pointing from the wall into the liquid, $\bar{n} = -\bar{e}_r$, so $\tau_p = -\mu \left(\partial v_x / \partial r\right)_{r=a}$ (the convention of (7.16)). Differentiating,

$$\frac{dv_x}{dr} = -\frac{P_l}{2\mu} r + \frac{C}{r} \implies \tau_p = \frac{P_l a}{2} - \frac{\mu C}{a} = \frac{P_l a}{2} - \frac{\mu}{a}\frac{-V + \dfrac{P_l}{4\mu} a^2 \left(1 - \alpha^2\right)}{\ln\left(\alpha^{-1}\right)}$$

$$\tau_p = \frac{P_l a}{2}\left[1 - \frac{1 - \alpha^2}{2 \ln\left(\alpha^{-1}\right)}\right] + \frac{\mu V}{a \ln\left(\alpha^{-1}\right)}$$

The liquid pushes the cylinder with the force $2\pi a\,\tau_p$ per unit length, in the direction $+x$. The external agent that moves the cylinder at velocity $-V\bar{e}_x$ must balance it, with the force $-2\pi a\,\tau_p\,\bar{e}_x$, so the power it supplies per unit length is force times velocity:

$$W = (-2\pi a\tau_p)\,(-V) = \tau_p\,V\,2\pi a$$

(power per unit length of cylinder, in W/m).

Zero Volume Flux

The flux through the annulus is $Q = \int_{\alpha a}^{a} 2\pi r\, v_x\, dr$. With $G = P_l/(4\mu)$, $b = \alpha a$, $\mathcal{L} = \ln(\alpha^{-1})$ and the two integrals (Barrow's rule):

$$\int_b^a r\left(r^2 - b^2\right) dr = \left[\frac{r^4}{4} - \frac{b^2 r^2}{2}\right]_b^a = \frac{\left(a^2 - b^2\right)^2}{4}$$

$$\int_b^a r \ln\frac{r}{b}\, dr = \left[\frac{r^2}{2}\ln\frac{r}{b} - \frac{r^2}{4}\right]_b^a = \frac{a^2}{2}\mathcal{L} - \frac{a^2 - b^2}{4}$$

(the second uses $\int r\ln r\, dr = \frac{r^2}{2}\ln r - \frac{r^2}{4}$, checked by differentiating). With $D_a = a^2 - b^2 = a^2(1-\alpha^2)$ and $C = (-V + GD_a)/\mathcal{L}$:

$$\frac{Q}{2\pi} = -G\frac{D_a^2}{4} + C\left(\frac{a^2}{2}\mathcal{L} - \frac{D_a}{4}\right) = 0$$

The key stops at the equivalent integral equation, boxed, $\frac{V}{\ln(\alpha^{-1})}\int_{\alpha a}^{a} r\ln\frac{r}{\alpha a}\, dr = \frac{P_l}{4\mu}\int_{\alpha a}^{a}\left[\frac{(1-\alpha^2)a^2\ln(r/\alpha a)}{\ln(\alpha^{-1})} - r^2 + (\alpha a)^2\right] r\, dr$, and does not evaluate it. Evaluating it (computed here) and solving for V:

$$(-V+GD_a)\left(\frac{a^2}{2}-\frac{D_a}{4\mathcal{L}}\right)=G\frac{D_a^2}{4}\implies V=GD_a\,\frac{\dfrac{a^2}{2}-\dfrac{D_a}{4}-\dfrac{D_a}{4\mathcal{L}}}{\dfrac{a^2}{2}-\dfrac{D_a}{4\mathcal{L}}}$$

and, using $\frac{a^2}{2}-\frac{D_a}{4}=\frac{a^2(1+\alpha^2)}{4}$ and multiplying numerator and denominator by $4\mathcal{L}/a^2$:

$$V=\frac{P_l a^2}{4\mu}\left(1-\alpha^2\right)\frac{\left(1+\alpha^2\right)\ln\left(\alpha^{-1}\right)-\left(1-\alpha^2\right)}{2\ln\left(\alpha^{-1}\right)-\left(1-\alpha^2\right)}$$

Answers and Checks

- Velocity: $v_x=-\frac{P_l}{4\mu}\left[r^2-(\alpha a)^2\right]+\left[-V+\frac{P_l}{4\mu}(1-\alpha^2)a^2\right]\ln(r/\alpha a)/\ln(\alpha^{-1})$.
- Power per unit length: $W=2\pi a\,V\left\{\frac{P_l a}{2}\left[1-\frac{1-\alpha^2}{2\ln(\alpha^{-1})}\right]+\frac{\mu V}{a\ln(\alpha^{-1})}\right\}$.
- Zero flux: $V=\frac{P_l a^2}{4\mu}(1-\alpha^2)\frac{(1+\alpha^2)\ln(\alpha^{-1})-(1-\alpha^2)}{2\ln(\alpha^{-1})-(1-\alpha^2)}$.
- Limit $\alpha\to 0$ (no rod): $\ln(\alpha^{-1})\to\infty$ and $V\to\frac{P_l a^2}{4\mu}\cdot\frac{1}{2}=\frac{P_l a^2}{8\mu}$; directly, a pipe with a wall moving at $-V$ has $Q=\pi a^4 P_l/(8\mu)-V\pi a^2=0$, the same V.
- Limit $\alpha\to 1$ (thin gap $h=a(1-\alpha)$): with $\epsilon=1-\alpha$, $\ln(\alpha^{-1})\simeq\epsilon+\epsilon^2/2+\epsilon^3/3$, the numerator of the last fraction is $\simeq 2\epsilon^3/3$ and its denominator $\simeq 2\epsilon^2$, so $V\to\frac{P_l a^2}{4\mu}\,2\epsilon\,\frac{\epsilon}{3}=\frac{P_l h^2}{6\mu}$, which is the plane result: $Q=P_l h^3/(12\mu)-Vh/2=0$.
- Dimensions: $P_l a^2/\mu$ is $ML^{-2}T^{-2}\cdot L^2/(ML^{-1}T^{-1})=LT^{-1}$, a velocity; $\tau_p V$ per unit length is $(\mathrm{N/m^2})(\mathrm{m/s})(\mathrm{m})=\mathrm{W/m}$.
- Check against the official key: VF3.pdf gives the same velocity profile, the same τ_p and $W=\tau_p V\,2\pi a$, and, for the zero flux, the integral condition (not evaluated).

PROBLEM VF4

Pipe with a Partly Inserted Bar

Original statement · VF4.pdf

Slender flow criterion | Pipe and annulus in series | Matching the pressure

OFFICIAL PROBLEM STATEMENT

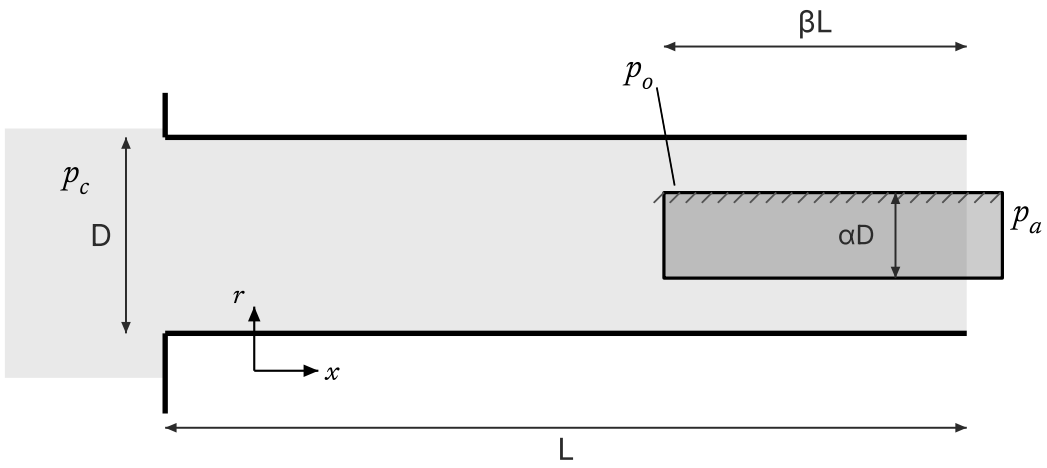
The liquid located in a container at pressure p_c discharges to the atmosphere, where the pressure is $p_a < p_c$, through a cylindrical pipe of diameter D and length $L \gg D$. To control the volume flux Q that circulates, a cylindrical bar of diameter αD is inserted coaxially from the exit, as shown in the figure, to partially block the liquid flow in the final stretch of the pipe.

1. Give the criterion for the motion in the tube to be dominated by viscosity.

2. Determine, in terms of the value of the pressure at the end of the tube p_o, the velocity field found for $0 < x/L < (1-\beta)$ and for $(1-\beta) < x/L < L$, where βL is the insertion length.

3. Obtain the volume flux Q and the value of p_o as a function of β.

On the sheet, "contained in a deposit" is struck out by hand and replaced by "located in a container", and the printed second range is written $(1-\beta) < x/L < L$, a typing slip for $(1-\beta) < x/L < 1$.



Statement figure (redrawn from VF4.pdf): a container at p_c discharging through a pipe of diameter D and length L; a coaxial bar of diameter αD enters from the exit along the last stretch βL; p_o is the pressure where the bar starts and p_a the exit pressure

Source: VF4.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 7: slender flows dominated by viscosity, Eqs. 7.40-7.46; pipe of variable section, Eqs. 7.47-7.51; entrance region, Eqs. 7.55-7.63.

Worked solution

Hypotheses, Data and Unknowns

- The liquid has constant ρ and μ; the flow is steady and axisymmetric.
- If the criterion of Part 1 holds, the flow is fully developed (Poiseuille) everywhere except in short regions of length of order D at the entrance and at the junction $x = (1-\beta)L$, and the pressure variations in them are small compared with the loss along the pipe (Notes.pdf, Eqs. 7.56 and 7.63). So $P(0) = p_c$ and the pressure at the end of the first stretch equals the one at the beginning of the second, p_o.
- Gravity is absorbed in the reduced pressure P; the gravity of a horizontal pipe does not matter, and the symbols p_c, p_o, p_a are used for the reduced pressures (as in the key).
- Let $a = D/2$, $\Delta p = p_c - p_a$ and $\mathcal{G} = \pi D^4/(128\mu)$, so that $\mathcal{G} = \pi a^4/(8\mu)$.

Symbol	Meaning	SI unit
D, αD	diameters of the pipe and of the bar	m
L, βL	length of the pipe and of the inserted bar	m
p_c, p_o, p_a	pressure in the container, at the bar tip, at the exit	Pa
Q	volume flux	m^3/s

Criterion for Viscous Dominance

The flow is dominated by viscosity when the convective acceleration is much smaller than the viscous force, Eq. (7.41), and the flow is steady here, so (7.40) is satisfied:

$$\frac{\rho v_L D}{\mu}\,\frac{D}{L} \ll 1$$

The velocity v_L is not known, so assume that viscosity dominates and estimate it from the balance of pressure and viscous forces, Eq. (7.44), $\Delta_L P \sim \mu v_L L/D^2$ with $\Delta_L P \sim p_c - p_a$:

$$v_L \sim \frac{D^2}{L}\,\frac{p_c - p_a}{\mu}$$

Substituting in the criterion:

$$\frac{\rho D}{\mu}\,\frac{D}{L}\,\frac{D^2}{L}\,\frac{p_c - p_a}{\mu} = \frac{\rho D^4\,(p_c - p_a)}{\mu^2 L^2} \ll 1$$

which only contains known data.

Velocity Fields

First stretch, $0 < x < (1-\beta)L$: circular pipe. As in (7.13)-(7.14), with a constant reduced pressure gradient $dP/dx = -P_{l1}$ and radius $a = D/2$:

$$v_x = \frac{P_{l1}}{4\mu}\left(a^2 - r^2\right) = \frac{P_{l1}}{16\mu}\left(D^2 - 4r^2\right)$$

The gradient is constant because the section is constant and Q is uniform, so $P_{l1} = (p_c - p_o)/\left[(1-\beta)L\right]$. In terms of the key's notation, $dP/dx = -(p_c - p_o)/\left[(1-\beta)L\right]$.

Second stretch, $(1-\beta)L < x < L$: annulus. The liquid is between the bar ($r = \alpha a$) and the pipe wall ($r = a$), both at rest, with $P_{l2} = (p_o - p_a)/(\beta L)$. Integrating (7.47) twice as in VF3, with $v_x(\alpha a) = v_x(a) = 0$:

$$v_x = \frac{P_{l2}}{4\mu}\left(a^2 - r^2\right) + C\ln\frac{r}{a}; \qquad 0 = \frac{P_{l2}}{4\mu}a^2\left(1-\alpha^2\right) + C\ln\alpha \implies C = -\frac{P_{l2}a^2\left(1-\alpha^2\right)}{4\mu\ln\alpha}$$

$$v_x = \frac{P_{l2}D^2}{16\mu}\left[1 - \left(\frac{2r}{D}\right)^2 - \frac{1-\alpha^2}{\ln\alpha}\ln\frac{2r}{D}\right], \qquad P_{l2} = \frac{p_o - p_a}{\beta L}$$

Check: at $r = D/2$ the bracket is $1 - 1 - 0 = 0$; at $r = \alpha D/2$ it is $1 - \alpha^2 - \frac{1-\alpha^2}{\ln \alpha}\ln\alpha = 0$.

Volume Flux

Pipe. From (7.15), $Q = \frac{\pi a^4}{8\mu} P_{l1}$, so

$$Q = \mathcal{G}\,\frac{p_c - p_o}{(1-\beta)L} \tag{1}$$

Annulus. With $s = r/a$, $Q = \int_{\alpha a}^{a} 2\pi r\, v_x\, dr = 2\pi a^2\,\frac{P_{l2}a^2}{4\mu}\, I$ with $I = \int_\alpha^1 s\left[1 - s^2 - k\ln s\right] ds$ and $k = (1-\alpha^2)/\ln\alpha$. Barrow's rule for each term:

$$\int_\alpha^1 s\,ds = \frac{1-\alpha^2}{2}, \qquad \int_\alpha^1 s^3 ds = \frac{1-\alpha^4}{4}, \qquad \int_\alpha^1 s\ln s\,ds = \left[\frac{s^2}{2}\ln s - \frac{s^2}{4}\right]_\alpha^1 = -\frac{1-\alpha^2}{4} - \frac{\alpha^2}{2}\ln\alpha$$

$$I = \frac{1-\alpha^2}{2} - \frac{1-\alpha^4}{4} + k\left[\frac{1-\alpha^2}{4} + \frac{\alpha^2}{2}\ln\alpha\right] = \frac{1-\alpha^2}{2} - \frac{1-\alpha^4}{4} + \frac{\left(1-\alpha^2\right)^2}{4\ln\alpha} + \frac{\alpha^2\left(1-\alpha^2\right)}{2}$$

and since $\frac{1-\alpha^2}{2} + \frac{\alpha^2(1-\alpha^2)}{2} = \frac{1-\alpha^4}{2}$, $I = \frac{1}{4}\left[\left(1-\alpha^4\right) - \frac{(1-\alpha^2)^2}{\ln(\alpha^{-1})}\right]$. Therefore

$$Q = \frac{\pi a^4 P_{l2}}{8\mu}\, f(\alpha) = \mathcal{G}\,\frac{p_o - p_a}{\beta L}\, f(\alpha), \qquad f(\alpha) = 1 - \alpha^4 - \frac{\left(1-\alpha^2\right)^2}{\ln\left(\alpha^{-1}\right)} \tag{2}$$

(f is the ratio of the flux of the annulus to that of the empty pipe for the same P_l: $f \to 1$ for $\alpha \to 0$ and $f \to 0$ for $\alpha \to 1$.) The key writes the same function as $f(\alpha) = 1 - \alpha^2(2 - \alpha^2) - \frac{(1-\alpha^2)\left[1-\alpha^2-2\alpha^2\ln(\alpha^{-1})\right]}{\ln(\alpha^{-1})}$; expanding, $1 - 2\alpha^2 + \alpha^4 + 2\alpha^2(1-\alpha^2) - (1-\alpha^2)^2/\ln(\alpha^{-1}) = 1 - \alpha^4 - (1-\alpha^2)^2/\ln(\alpha^{-1})$, the same.

Flux and Pressure at the Junction

The flux is the same in both stretches (Notes.pdf, Eq. 7.72 and the discussion of (7.49)). Equating (1) and (2):

$$\frac{p_c - p_o}{1-\beta} = f\,\frac{p_o - p_a}{\beta}$$

With $u = p_o - p_a$ and $p_c - p_o = \Delta p - u$: $\beta(\Delta p - u) = f(1-\beta)u$, so $u = \dfrac{\beta\,\Delta p}{\beta + f(1-\beta)}$. Then

$$Q = \mathcal{G}\,\frac{f\,u}{\beta L} = \frac{\pi D^4}{128\mu L}\,\frac{p_c - p_a}{1 + \beta\,(1-f)\,/f}, \qquad p_o = p_c - \frac{(1-\beta)\,(p_c - p_a)}{1 + \beta\,(1-f)\,/f}$$

(the second from $p_c - p_o = \Delta p - u = \dfrac{f(1-\beta)\Delta p}{\beta + f(1-\beta)}$, dividing numerator and denominator by f).

Discrepancy with the official solution

The two boxed results of the key have $1 + \dfrac{\beta f}{1-f}$ in the denominator, with $f = f(\alpha)$ of its own equation (2). That combination is not the one that follows from (1) and (2). Two limits prove it. For $\alpha \to 0$ (no bar) $f \to 1$, and the key's denominator $1 + \beta f/(1-f)$ tends to infinity, so $Q \to 0$, when the flow must be that of the empty pipe; the corrected denominator $1 + \beta(1-f)/f$ tends to 1 and gives $Q = \pi D^4 \Delta p/(128\mu L)$. For $\alpha \to 1$ (the bar fills the pipe) $f \to 0$ and the key gives the empty-pipe flux, the corrected formula $Q \to 0$. The calculation above is kept.

Answers and Checks

- Viscous dominance: $\rho D^4 (p_c - p_a)/(\mu^2 L^2) \ll 1$.
- Velocities: $v_x = \frac{(p_c - p_o)}{16\mu(1-\beta)L}\left(D^2 - 4r^2\right)$ for $0 < x < (1-\beta)L$ and $v_x = \frac{(p_o - p_a)D^2}{16\mu\beta L}\left[1 - (2r/D)^2 - \frac{1-\alpha^2}{\ln\alpha}\ln(2r/D)\right]$ for $(1-\beta)L < x < L$.
- Flux: $Q = \frac{\pi D^4}{128\mu L}\frac{p_c - p_a}{1+\beta(1-f)/f}$; pressure at the junction: $p_o = p_c - \frac{(1-\beta)(p_c - p_a)}{1+\beta(1-f)/f}$.

- Limits: $\beta \to 0$ (no bar) gives $Q = \pi D^4 \Delta p / (128 \mu L)$ and $p_o \to p_c - \Delta p = p_a$; $\beta \to 1$ (bar over the whole pipe) gives $Q = \mathcal{G} f \, \Delta p / L$ and $p_o \to p_c$, all the loss in the annulus; $f \to 1$ gives the empty pipe whatever β.
- Dimensions: $\pi D^4 \Delta p / (\mu L)$ is $L^4 \cdot M L^{-1} T^{-2} / (M L^{-1} T^{-1} \cdot L) = L^3 T^{-1}$, a volume flux; $\rho D^4 \Delta p / (\mu^2 L^2)$ is $M L^{-3} L^4 \, M L^{-1} T^{-2} / (M^2 L^{-2} T^{-2} L^2) = 1$.
- Check against the official key: VF4.pdf gives the same criterion, the same fields, the same flux in each stretch (1) and (2) and the same $f(\alpha)$; its last two boxed results differ from the corrected ones, as explained in the discrepancy above.

PROBLEM VF5

Discharge of Two Immiscible Liquids from a Tube

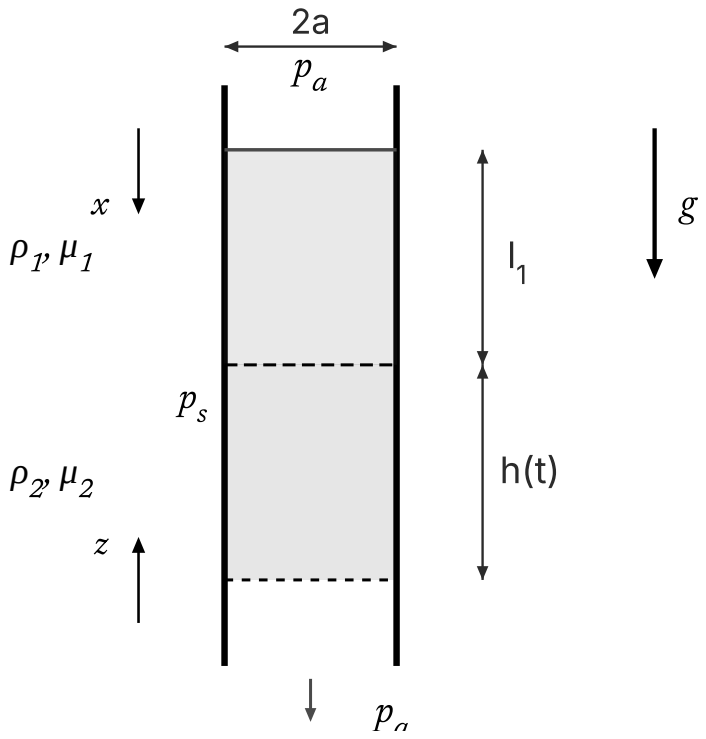
Original statement · VF5.pdf

Quasi-steady Poiseuille flow | Two immiscible liquids | Discharge time

OFFICIAL PROBLEM STATEMENT

A vertical tube of radius a open at the top contains initially two columns of heights l_1 and l_2 of two inmiscible liquids with different densities ρ_1 and ρ_2 and viscosities μ_1 and μ_2. At a given instant of time, the bottom end of the pipe opens up, releasing the fluid 2 to the atmosphere. During the discharge process, determine the volume flux as a function of time as well as the pressure distribution along the pipe. Calculate the time for complete discharge of the bottom liquid column.

The sheet prints "inmiscible". The figure labels the upper liquid (ρ_1, μ_1) of height l_1, the lower liquid (ρ_2, μ_2) of height $h(t)$, the pressure p_s at the interface, p_a above and below, and the axis x pointing down from the top.



Statement figure (redrawn from VF5.pdf): a vertical tube of radius a, open at both ends once the bottom is released, with liquid 1 (ρ_1, μ_1) of height l_1 above liquid 2 (ρ_2, μ_2) of height h(t); p_s is the pressure at the interface

Source: VF5.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: steady flow in a circular pipe, Eqs. 7.13-7.15; slender flows dominated by viscosity, Eqs. 7.40-7.46; pipe flow, Eqs. 7.47-7.51.

Worked solution

Hypotheses, Data and Unknowns

- **Viscous regime.** The flow is dominated by viscosity if the convective and local accelerations are negligible against the viscous force (Eqs. 7.40 and 7.41). With the velocity estimated by the balance of the weight of the column and the viscous force, $V_L \sim \rho g a^2/\mu$, the criterion is (computed in the first section of the solution)

$$\frac{\rho^2 g\, a^4}{\mu^2\, l} \ll 1$$

with $l \sim l_1 \sim l_2$ the length of the columns (the key writes the characteristic density and viscosity without subscript). Under it the flow is quasi-steady Poiseuille flow in each column. - The liquids are incompressible and immiscible, so the interface moves with them; the upper column keeps its length l_1 (it slides down as a plug as the lower column drains) and the lower column has the length $h(t)$, with $h(0) = l_2$. - Pressure at the open top and at the open bottom: p_a. The interface pressure $p_s(t)$ is unknown. - The x axis points down from the top and z points up from the bottom (figure); gravity acts along $+x$.

Symbol	Meaning	SI unit
a	radius of the tube	m
ρ_1, μ_1, l_1	density, viscosity, height of the upper liquid	kg/m^3, kg/(m s), m
$\rho_2, \mu_2, l_2, h(t)$	same for the lower liquid; $h(0) = l_2$	kg/m^3, kg/(m s), m
Q, p_s	volume flux and interface pressure	m^3/s, Pa

Criterion for Viscous Dominance

The local acceleration over the viscous term is $\dfrac{a^2/(\mu/\rho)}{t_0}$ with $t_0 \sim l/V_L$ the time of variation (the time to travel the column); the convective one is $\dfrac{\rho V_L a}{\mu}\dfrac{a}{l}$ (Eqs. 7.40-7.41). They are the same number, and with $V_L \sim \Delta P\, a^2/(\mu l)$, $\Delta P \sim \rho g l$ (the pressure difference that drives the flow is the weight of the column), $V_L \sim \rho g a^2/\mu$:

$$\frac{\rho V_L a}{\mu}\,\frac{a}{l} \sim \frac{\rho}{\mu}\,\frac{\rho g a^2}{\mu}\,\frac{a^2}{l} = \frac{\rho^2 g a^4}{\mu^2 l} \ll 1$$

Flux in Each Column

In a column of length ℓ of the pipe the flow goes down, along $+x$, and the reduced pressure $P = p - \rho g x$ (x down: the potential energy falls as x grows) has the uniform gradient along the flow $P_l = (P_{top} - P_{bottom})/\ell = (p_{top} - p_{bottom} + \rho g \ell)/\ell$, so that the weight of the column helps the flow. The profile is the Poiseuille one (Eq. 7.14), $v_x = \frac{P_l}{4\mu}(a^2 - r^2)$, with the flux (Eq. 7.15), $Q = \frac{\pi a^4}{8\mu} P_l$.

Upper liquid. From the free surface ($p = p_a$) to the interface ($p = p_s$), of length l_1:

$$Q = \frac{\pi a^4}{8\mu_1}\,\frac{p_a - p_s + \rho_1 g l_1}{l_1}$$

Lower liquid. From the interface ($p = p_s$) to the open end ($p = p_a$), of length h:

$$Q = \frac{\pi a^4}{8\mu_2}\,\frac{p_s - p_a + \rho_2 g h}{h}$$

Mass conservation of an incompressible liquid in a tube of constant section gives the same Q in both columns, so the pressure drops are

$$p_a - p_s + \rho_1 g l_1 = \frac{8\mu_1 l_1}{\pi a^4}\,Q, \qquad p_s - p_a + \rho_2 g h = \frac{8\mu_2 h}{\pi a^4}\,Q$$

Volume Flux and Evolution of the Interface

Adding the two equations cancels the unknown p_s:

$$(\rho_1 l_1 + \rho_2 h)\, g = \frac{8}{\pi a^4}\,(\mu_1 l_1 + \mu_2 h)\, Q \implies Q = \frac{\pi a^4 g}{8}\,\frac{\rho_1 l_1 + \rho_2 h}{\mu_1 l_1 + \mu_2 h}$$

This gives Q as a function of the position h of the lower interface. The lower column loses the volume $dV = -\pi a^2\, dh$ in dt, so $Q = -\pi a^2\, dh/dt$. Substituting:

$$(\rho_1 l_1 + \rho_2 h)\, g = -\frac{8}{a^2}\,(\mu_1 l_1 + \mu_2 h)\,\frac{dh}{dt}$$

Separating variables and integrating from $h = l_2$ at $t = 0$ (Barrow's rule on the left, with $\int_0^t dt' = t$):

$$\int_{l_2}^{h} \frac{\mu_1 l_1 + \mu_2 h'}{\rho_1 l_1 + \rho_2 h'}\, dh' = -\frac{g a^2}{8}\, t$$

The key leaves the left side unevaluated. Writing $\mu_1 l_1 + \mu_2 h' = \frac{\mu_2}{\rho_2}\,(\rho_1 l_1 + \rho_2 h') + l_1\left(\mu_1 - \frac{\mu_2 \rho_1}{\rho_2}\right)$ the integrand is $\frac{\mu_2}{\rho_2} + \frac{l_1(\mu_1\rho_2 - \mu_2\rho_1)}{\rho_2}\,\frac{1}{\rho_1 l_1 + \rho_2 h'}$ and, since $\int \frac{dh'}{\rho_1 l_1 + \rho_2 h'} = \frac{1}{\rho_2}\ln(\rho_1 l_1 + \rho_2 h')$ (computed here):

$$\frac{\mu_2}{\rho_2}\,(h - l_2) + \frac{l_1\,(\mu_1\rho_2 - \mu_2\rho_1)}{\rho_2^2}\,\ln\frac{\rho_1 l_1 + \rho_2 h}{\rho_1 l_1 + \rho_2 l_2} = -\frac{g a^2}{8}\, t$$

which gives $t(h)$ explicitly, and $h(t)$ implicitly; $Q(t)$ follows from $Q(h)$ above.

Pressure Distribution

The pressure at the interface follows from the upper column:

$$p_s - p_a = \rho_1 g l_1 - \frac{8\mu_1 l_1}{\pi a^4} Q = \rho_1 g l_1 - \mu_1 l_1 g\,\frac{\rho_1 l_1 + \rho_2 h}{\mu_1 l_1 + \mu_2 h} = \frac{g\,l_1 h\,(\mu_2\rho_1 - \mu_1\rho_2)}{\mu_1 l_1 + \mu_2 h}$$

Inside each column the actual pressure is linear in x: with $P_l = (p_{top} - p_{bottom} + \rho g \ell)/\ell$, $p = p_{top} - P_l x + \rho g x = p_{top} - (p_{top} - p_{bottom})x/\ell$, because the weight term $\rho g x$ cancels. With x measured down from the top,

$$p(x) = p_a + (p_s - p_a)\,\frac{x}{l_1} \quad (0 < x < l_1), \qquad p(x) = p_s + (p_a - p_s)\,\frac{x - l_1}{h} \quad (l_1 < x < l_1 + h)$$

(it is p_a at both ends; the interface is the only point where it differs, by $p_s - p_a$, positive if $\mu_2\rho_1 > \mu_1\rho_2$).

Complete Discharge Time

At $h = 0, t = t_d$. From the explicit $t(h)$ above:

$$t_d = \frac{8}{g a^2}\left[\frac{\mu_2 l_2}{\rho_2} + \frac{l_1\,(\mu_1\rho_2 - \mu_2\rho_1)}{\rho_2^2}\,\ln\left(1 + \frac{\rho_2 l_2}{\rho_1 l_1}\right)\right]$$

Answers and Checks

- Viscous dominance: $\rho^2 g a^4/(\mu^2 l) \ll 1$.
- Volume flux: $Q = \frac{\pi a^4 g}{8}\,\frac{\rho_1 l_1 + \rho_2 h}{\mu_1 l_1 + \mu_2 h}$, with $h(t)$ defined by $\int_{l_2}^{h} \frac{\mu_1 l_1 + \mu_2 h'}{\rho_1 l_1 + \rho_2 h'}\,dh' = -\frac{g a^2}{8} t$.
- Pressure: linear in each liquid, with the interface value $p_s - p_a = \frac{g l_1 h(\mu_2\rho_1 - \mu_1\rho_2)}{\mu_1 l_1 + \mu_2 h}$.
- Discharge time: the expression above.

- Limit of equal liquids ($\mu_1 = \mu_2 = \mu$, $\rho_1 = \rho_2 = \rho$): $p_s = p_a$, $Q = \pi a^4 \rho g/(8\mu)$ (constant, independent of h and l_1) and $t_d = 8\mu l_2/(\rho g a^2) = \pi a^2 l_2/Q$, the time to empty a volume $\pi a^2 l_2$ at that flux.
- Limit $l_1 \to 0$ (only liquid 2): $Q = \frac{\pi a^4 \rho_2 g}{8\mu_2}$ and $t_d = \frac{8\mu_2 l_2}{\rho_2 g a^2}$.
- Dimensions: $\pi a^4 \rho g/\mu$ is $L^4 \cdot ML^{-3} \cdot LT^{-2}/(ML^{-1}T^{-1}) = L^3T^{-1}$; $\rho^2 g a^4/(\mu^2 l)$ is $M^2L^{-6} \cdot LT^{-2} \cdot L^4/(M^2L^{-2}T^{-2} \cdot L) = 1$.
- Check against the official key: VF5.pdf gives the same criterion, profiles and fluxes, the equation for p_s, $(\rho_1 l_1 + \rho_2 h)g = \frac{8}{\pi a^4}(\mu_1 l_1 + \mu_2 h)Q$ and the integral relation boxed above; it stops there, so the explicit integral, the pressure field and the discharge time are computed here.

PROBLEM VF6

Oil Rising in a Vertical Tube

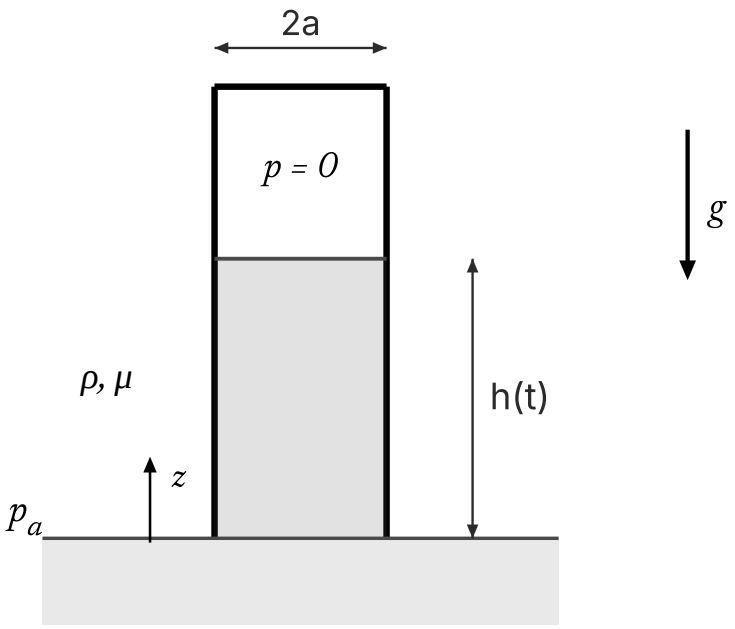
Original statement · VF6.pdf

Quasi-steady Poiseuille flow | Rising column | Force on the pipe

OFFICIAL PROBLEM STATEMENT

The bottom end of an empty vertical tube of radius a, closed at the top, is put in contact with a pool of oil. Because of the ambient overpressure, the liquid begins to flow into the tube, forming a column of increasing height $h(t)$ whose evolution in time is to be investigated assuming that the motion is dominated by viscosity. In particular:

- Obtain the value of the height h_∞ corresponding to the equilibrium position, reached asymptotically for large times.
- Give the condition that determines whether the motion is dominated by viscosity.
- Obtain the evolution of $h(t)$ as well as the pressure distribution along the pipe $p(x,t)$ for $0 < x < h$.
- Compute the force acting on the pipe as a function of time $\bar{F} = F_z(t)\bar{e}_z$.



Statement figure (redrawn from VF6.pdf): a vertical tube of radius a, closed at the top, with its lower end in a pool at p_a; the oil column of height h(t) has p = 0 above it; z is the vertical axis from the pool

Source: VF6.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: steady flow in a circular pipe, Eqs. 7.13-7.16; slender flows, Eqs. 7.40-7.41; pipe flow, Eqs. 7.47-7.51. The sheet answers the four

bullets in the margin order 1) to 4).

Worked solution

Hypotheses, Data and Unknowns

- The tube is evacuated above the liquid (the figure marks $p = 0$ there), so the free surface of the column is at pressure zero; the pool is at the atmospheric pressure p_a at the level of the lower end, which is the origin of the vertical axis z (the sheet writes $x \equiv z$ for the axial coordinate).
- The liquid is incompressible with constant ρ and μ; the column is slender, $h \gg a$ for most of the process.
- Reduced pressure: $P = p + \rho g z$.
- The atmosphere also acts on the outer face of the closed top end of the tube (this is what the sheet's force balance includes).

Symbol	Meaning	SI unit
a	radius of the tube	m
ρ, μ	density, viscosity of the oil	kg/m^3, kg/(m s)
p_a	ambient pressure	Pa
$h(t)$, h_∞	height of the column and its final value	m

Equilibrium Height

When $t \to \infty$ the liquid is at rest and the reduced pressure is uniform, $P = p + \rho g z = \text{constant}$. At the bottom ($z = 0$) it equals p_a and at the free surface ($z = h_\infty, p = 0$) it equals $\rho g h_\infty$:

$$p_a = \rho g h_\infty \implies h_\infty = \frac{p_a}{\rho g}$$

Criterion for Viscous Dominance

The viscous force dominates the convective acceleration (and the local acceleration, since $t_0 \sim L/V_L$) when, Eq. (7.41),

$$\frac{\rho V_L a}{\mu}\,\frac{a}{L} \ll 1$$

with the characteristic length $L \sim h_\infty$ and the velocity estimated by assuming that viscosity dominates, Eq. (7.44), $V_L \sim \frac{a^2}{L}\frac{p_a}{\mu}$ (the pressure difference that drives the flow is p_a):

$$\frac{\rho a^4 p_a}{\mu^2 h_\infty^2} = \frac{\rho a^4 p_a}{\mu^2}\,\frac{\rho^2 g^2}{p_a^2} = \frac{\rho^3 g^2 a^4}{\mu^2 p_a} \ll 1$$

(the key writes the characteristic transverse length as D; here it is the radius a, which only changes a numerical factor).

Evolution of the Column

Poiseuille flow in the column (Eqs. 7.13-7.15), with $P_l = -\partial P/\partial z$ uniform along the column at each instant:

$$v_z = \frac{P_l}{4\mu}\left(a^2 - r^2\right), \qquad Q = \frac{\pi a^4}{8\mu} P_l$$

The reduced pressure falls from $P(0) = p_a$ to $P(h) = \rho g h$ over the length h, so $P_l = (p_a - \rho g h)/h$. The mean velocity of the column is the velocity of its free surface, $Q = \pi a^2\, dh/dt$ (the column grows, so the flux into the tube is positive upward; the key writes $-\pi a^2 dh/dt$ and corrects the sign by hand). Hence

$$\frac{dh}{dt} = \frac{a^2}{8\mu}\,\frac{p_a - \rho g h}{h} = \frac{a^2 p_a}{8\mu}\,\frac{1 - h/h_\infty}{h}$$

Separating variables, with $h(0) = 0$ and $u = h/h_\infty$ ($h\,dh = h_\infty^2\,u\,du$):

$$\int_0^h \frac{h'\,dh'}{1 - h'/h_\infty} = h_\infty^2 \int_0^{h/h_\infty} \frac{u\,du}{1-u} = \frac{a^2 p_a}{8\mu}\,t, \qquad \int \frac{u\,du}{1-u} = \int \left(-1 + \frac{1}{1-u}\right) du = -u - \ln(1-u)$$

$$\frac{a^2 p_a}{8\mu h_\infty^2}\,t = \ln\frac{1}{1 - h/h_\infty} - \frac{h}{h_\infty}$$

Discrepancy with the official solution

The key writes this relation, and the equation of dh/dt before it, without the factor a^2: $\frac{P_a}{8\mu h_\infty^2} t = \ln\frac{1}{1-h/h_\infty} - \frac{h}{h_\infty}$. That cannot be right dimensionally: $p_a t/\mu$ is dimensionless, so the left side has the dimensions L^{-2} while the right side is dimensionless; and $dh/dt = (p_a/8\mu)(1 - h/h_\infty)/h$ has the dimensions $T^{-1}L^{-1}$ instead of LT^{-1}. The factor a^2 comes from $Q = \pi a^2 dh/dt$ and $Q = \pi a^4 P_l/(8\mu)$, whose ratio is a^2 (the key strikes out the πa^2 in that step). The same slip appears in the pressure-gradient lines of the key, written $8\mu Q/(\pi a^2)$, where the Poiseuille flux gives $8\mu Q/(\pi a^4)$. The results above keep the factor.

As $t \to \infty$, $\ln\frac{1}{1-h/h_\infty} \to \infty$ and $h \to h_\infty$. For short times $\ln\frac{1}{1-u} - u \simeq \frac{u^2}{2}$, so $h \simeq a\sqrt{p_a t/(4\mu)}$: the column grows as $\sqrt{t}$ (the laminar, slender-flow law), independent of gravity.

Pressure Distribution

The reduced pressure is linear, $P = p_a - P_l z$, with $P_l = (p_a - \rho g h)/h$. The actual pressure $p = P - \rho g z$:

$$p(z,t) = p_a - \frac{p_a - \rho g h}{h}\, z - \rho g z = p_a - \frac{p_a}{h}\, z = p_a \left(1 - \frac{z}{h(t)}\right), \qquad 0 < z < h$$

linear from p_a at the pool to zero at the top of the column; the weight terms cancel (the key writes the equivalent $P + \rho g z = p_a - \frac{8\mu Q}{\pi a^4} x$, boxed, with the same slip in the power of a).

Force on the Pipe

The liquid acts on the pipe through the shear stress of the wall. With the normal from the wall into the liquid, $\bar{n} = -\bar{e}_r$, $\tau_p = -\mu \left(\partial v_z/\partial r\right)_{r=a} = P_l a/2$ (Eq. 7.16), along $+\bar{e}_z$ (the rising liquid drags the wall upward). Over the lateral area $2\pi a h$:

$$F_{visc} = 2\pi a h \cdot \frac{a}{2}\, \frac{p_a - \rho g h}{h} = \pi a^2 \left(p_a - \rho g h\right)$$

The pressure inside the tube above the liquid is zero, and the atmosphere acts on the outer face of the closed top end with the force $-p_a \pi a^2 \bar{e}_z$ (the closed end is a disc of area πa^2). The total force:

$$\bar{F} = \pi a^2 \left(p_a - \rho g h\right) \bar{e}_z - p_a \pi a^2 \bar{e}_z = -\rho g h\, \pi a^2\, \bar{e}_z$$

i.e. $F_z(t) = -\rho g \pi a^2 h(t)$: the weight of the column (downward), which is a good check: the force on the pipe equals the weight of the liquid it holds, since the liquid is quasi-static in the vertical momentum balance of the whole column ($\pi a^2 p_a$ at the bottom, minus the weight, equals the shear transmitted to the wall).

Answers and Checks

- Equilibrium height: $h_\infty = p_a/(\rho g)$.
- Viscous dominance: $\rho^3 g^2 a^4/(\mu^2 p_a) \ll 1$.
- Height: $\frac{a^2 p_a}{8\mu h_\infty^2} t = \ln \frac{1}{1-h/h_\infty} - \frac{h}{h_\infty}$; pressure: $p = p_a(1 - z/h)$.
- Force on the pipe: $\bar{F} = -\rho g \pi a^2 h(t)\, \bar{e}_z$.
- Dimensions: $a^2 p_a t/(\mu h_\infty^2)$ is $L^2 \cdot ML^{-1}T^{-2} \cdot T/(ML^{-1}T^{-1}L^2) = 1$.
- Limits: at $t \to 0$, $h \to a\sqrt{p_a t/(4\mu)}$; at $t \to \infty$, $h \to h_\infty$ and $F_z \to -\pi a^2 p_a$ (the atmosphere on the top end, with the shear zero).
- Check against the official key: VF6.pdf gives $h_\infty = p_a/(\rho g)$, the criterion $\rho^3 g^2 D^4/(\mu^2 p_a) \ll 1$, the integral for $h(t)$ and $\bar{F} = -\rho g h \pi a^2 \bar{e}_z$; the key's $h(t)$ relation lacks the factor a^2, as noted above.

PROBLEM VF7

Discharge from a Conical Pipe

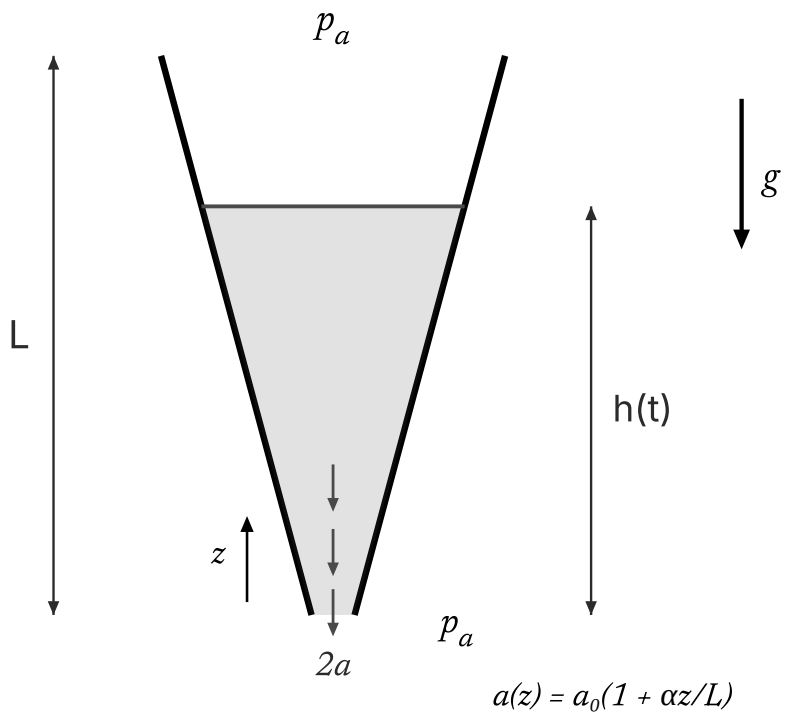
Original statement · VF7.pdf

Pipe of variable section | Quasi-steady discharge | Discharge time

OFFICIAL PROBLEM STATEMENT

The vertical conical pipe of length L shown in the figure has a radius that increases linearly from the lower end according to $a(z) = a_o(1 + \alpha z/L)$, with $a_o \ll 1$ and $\alpha \sim O(1)$. The pipe, open at the top, is initially full of a liquid of viscosity μ and density ρ. At a given time, the lower end opens up and the liquid discharges to the atmosphere under the action of gravity. Give the condition for the motion to be dominated by viscosity and, in that case, obtain the evolution of $h(t)$ as well as the time needed to complete the discharge.

The sheet writes $a_o \ll 1$; the condition is $a_o \ll L$ (a slender pipe), since a_o is a length.



Statement figure (redrawn from VF7.pdf): a vertical conical pipe of length L, radius a_0 at the lower end and increasing linearly upward, open at both ends, with the free surface at the height h(t) above the lower end

Source: VF7.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: slender flows, Eqs. 7.40-7.46; pipe of variable section, Eqs. 7.47-7.51.

Worked solution

Hypotheses, Data and Unknowns

- The pipe is slender and its radius changes slowly ($\alpha \sim O(1)$ and $a_o \ll L$), so the flow is a local Poiseuille flow (Eq. 7.48) and the flux Q does not depend on z (Eq. 7.49).
- z is measured upwards from the lower end. The free surface is at $z = h(t)$, with $h(0) = L$ (the pipe is full); the lower end is open to the atmosphere, $p = p_a$, and so is the free surface.
- Reduced pressure $P = p + \rho g z$.

Symbol	Meaning	SI unit
$a(z) = a_o(1 + \alpha z/L)$	local radius	m
L	length of the pipe	m
$h(t)$	height of the free surface above the lower end	m
Q	volume flux (downward)	m^3/s

Criterion for Viscous Dominance

As in VF5, the convective and local accelerations are negligible against the viscous force (Eqs. 7.40-7.41) when

$$\frac{\rho V_L a_o}{\mu}\,\frac{a_o}{L} \ll 1, \qquad V_L \sim \frac{\rho g a_o^2}{\mu} \implies \frac{\rho^2 g\, a_o^4}{\mu^2 L} \ll 1$$

(V_L is the velocity that the weight of the column gives with the viscous resistance, Eq. 7.44; the largest radius is of order a_o since $\alpha \sim O(1)$).

Flux and Pressure

The pressure gradient along the pipe follows from the local Poiseuille law (7.49), $Q = \frac{\pi a^4}{8\mu} P_l$ with the flow downwards, so that the reduced pressure grows upwards: $dP/dz = 8\mu Q/(\pi a^4)$. Between the open lower end ($z = 0, p = p_a, P = p_a$) and the free surface ($z = h, p = p_a, P = p_a + \rho g h$), integrating (Eq. 7.50):

$$\rho g h = \frac{8\mu Q}{\pi} \int_0^h \frac{dz}{a^4(z)} = \frac{8\mu Q}{\pi a_o^4} \int_0^h \frac{dz}{(1 + \alpha z/L)^4}$$

With $w = 1 + \alpha z/L, dz = (L/\alpha)\, dw$, and Barrow's rule:

$$\int_0^h \frac{dz}{(1 + \alpha z/L)^4} = \frac{L}{\alpha} \left[-\frac{1}{3w^3} \right]_1^{1+\alpha h/L} = \frac{L}{3\alpha} \left[1 - \frac{1}{(1 + \alpha h/L)^3} \right]$$

$$\rho g h = \frac{8\mu Q L}{3\pi \alpha a_o^4} \left[1 - \frac{1}{(1 + \alpha h/L)^3} \right]$$

(the factor α in the denominator is in the next line of the key: the boxed line of the sheet is partly struck out in the scan).

Evolution of the Free Surface

The volume that leaves in dt is the one lost by the column, $dV = -\pi a^2(h)\, dh$, so $Q = -\pi a^2(h)\, dh/dt$ with $a(h) = a_o(1 + \alpha h/L)$. Substituting, with $w = 1 + \alpha h/L$:

$$\rho g h = -\frac{8\mu L}{3\alpha a_o^2} \left[1 - \frac{1}{w^3} \right] w^2 \frac{dh}{dt} = -\frac{8\mu L}{3\alpha a_o^2} \, \frac{w^3 - 1}{w} \, \frac{dh}{dt}$$

(since $\left(1 - w^{-3}\right) w^2 = w^2 - w^{-1} = (w^3 - 1)/w$). Separating variables, with $h = L$ at $t = 0$:

$$\frac{3\alpha}{8}\,\frac{\rho g a_o^2}{\mu L}\,t = -\int_L^h \frac{w^3-1}{h'\,w}\,dh'$$

To evaluate the integral (computed here) set $s = \alpha h'/L$, so $dh'/h' = ds/s$ and $w = 1+s$; as $(1+s)^3 - 1 = s\left(s^2+3s+3\right)$ and $s^2+3s+3 = (s+2)(s+1)+1$:

$$\frac{(1+s)^3-1}{s(1+s)} = \frac{s^2+3s+3}{1+s} = s+2+\frac{1}{1+s}, \qquad \int\left(s+2+\frac{1}{1+s}\right)ds = S(s) = \frac{s^2}{2}+2s+\ln(1+s)$$

$$\frac{3\alpha\rho g a_o^2}{8\mu L}\,t = S(\alpha) - S\!\left(\frac{\alpha h}{L}\right), \qquad S(s) = \frac{s^2}{2}+2s+\ln(1+s)$$

which gives $t(h)$ explicitly, and $h(t)$ implicitly.

Time to Complete the Discharge

At $h = 0$, $S(0) = 0$:

$$t_d = \frac{8}{3\alpha}\,\frac{\mu L}{\rho g a_o^2}\left[\frac{\alpha^2}{2}+2\alpha+\ln(1+\alpha)\right]$$

The key stops at $t_d = \frac{8}{3\alpha}\frac{\mu L}{\rho g a_o^2}\int_0^L \frac{(1+\alpha h/L)^3-1}{h(1+\alpha h/L)}\,dh$; the value of the integral is $\alpha^2/2 + 2\alpha + \ln(1+\alpha)$, as computed.

Answers and Checks

- Viscous dominance: $\rho^2 g a_o^4/(\mu^2 L) \ll 1$.
- Evolution: $\frac{3\alpha\rho g a_o^2}{8\mu L}t = S(\alpha) - S(\alpha h/L)$, $S(s) = s^2/2 + 2s + \ln(1+s)$; the flux is $Q = -\pi a_o^2(1+\alpha h/L)^2\,dh/dt$.
- Discharge time: $t_d = \frac{8\mu L}{3\alpha\rho g a_o^2}\left[\frac{\alpha^2}{2}+2\alpha+\ln(1+\alpha)\right]$.

- Limit $\alpha \to 0$ (cylindrical pipe): $\frac{\alpha^2}{2} + 2\alpha + \ln(1+\alpha) = 3\alpha + \alpha^3/3 + \ldots$, so $t_d \to \frac{8\mu L}{\rho g a_o^2}$; directly, the column of constant length h has $P_l = \rho g$ and $Q = \pi a_o^4 \rho g/(8\mu)$, a constant mean velocity $a_o^2 \rho g/(8\mu)$, and L divided by it is the same.
- Dimensions: $\mu L/(\rho g a_o^2)$ is $ML^{-1}T^{-1} \cdot L/(ML^{-3} \cdot LT^{-2} \cdot L^2) = T$; α and S are dimensionless.
- Check against the official key: VF7.pdf gives the same criterion, pressure relation (with α in its denominator after the factor the scan strikes), differential equation and integral relations for $h(t)$ and t_d, without evaluating the integral.

PROBLEM VF8

Lifting a Cylinder from an Oil Film

Original statement · VF8.pdf

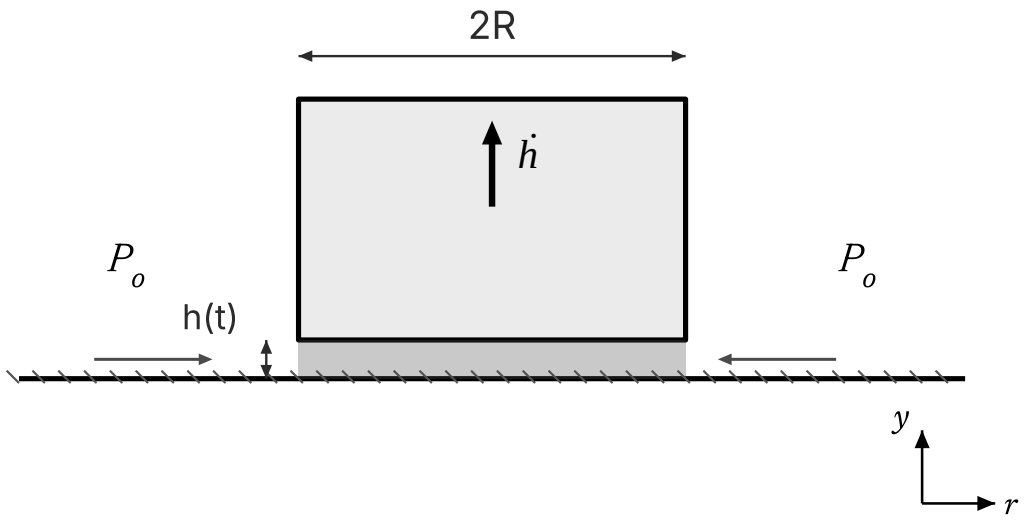
Squeeze film | Radial Poiseuille flow | Adhesion force

OFFICIAL PROBLEM STATEMENT

We want to lift the cylindrical body shown in the figure. A layer of oil exists between the body and the horizontal solid surface. The radial inflow caused by the lifting motion of the body is associated with the existence of underpressures in the liquid film, inducing an adhesion force to be computed here for the case of constant velocity $\dot{h} = dh/dt$.

- Show that the liquid motion in the liquid film between the body and the solid surface is dominated by viscosity for values of h much smaller than $\nu/\dot{h}$.
- Obtain the radial velocity profile v_r as a function of the radial pressure gradiente $P_l = -\partial P/\partial r$.
- Using continuity, derive an equation for the pressure distribution $P(r,t)$.
- Compute the pressure distribution as well as the adhesion force acting on the body.

The sheet prints "gradiente" and "pressuse".



Statement figure (redrawn from VF8.pdf): a cylindrical body of radius R lifted with velocity ḣ from a horizontal surface; the oil film of thickness h(t) between them is fed by a radial inflow from the outer pressure P_o

Source: VF8.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: slender films, Eqs. 7.64-7.71; the wedge effect, Eq. 7.72; forces, Eq. 7.76.

Worked solution

Hypotheses, Data and Unknowns

- Axisymmetric film: $\bar{v} = (v_r, v_y)$ with y the distance to the solid surface and r the radial coordinate from the axis; the film is slender, $h \ll R$.
- The body moves up with the constant velocity $\dot{h} = dh/dt$ and its lower face stays flat, so $v_y = \dot{h}$ there; the oil is in contact with the surface at $y = 0$ ($v_y = 0$). At the edge $r = R$ the pressure is the outer one, P_o (the key writes P_0).
- Gravity is absorbed in the reduced pressure P.

Symbol	Meaning	SI unit
R	radius of the body	m
$h(t)$, $\dot{h}$	thickness of the film and its rate	m, m/s
μ, ν	viscosity of the oil	kg/(m s), m^2/s

Criterion for Viscous Dominance

By continuity in the film, $\partial(r v_r)/\partial r + r\,\partial v_y/\partial y = 0$ gives $v_r/R \sim v_y/h \sim \dot{h}/h$, so the radial velocity scale is $V_L \sim R\dot{h}/h$ and the time scale of the process is $t_0 \sim h/\dot{h}$. Local acceleration against viscous force (Eq. 7.67) and convective acceleration against viscous force (Eq. 7.68):

$$\frac{O(\rho\,\partial v_r/\partial t)}{O(\mu \nabla^2 \bar{v})} = \frac{h^2/\nu}{t_0} \sim \frac{h\dot{h}}{\nu}, \qquad \frac{O(\rho \bar{v}\cdot\nabla\bar{v})}{O(\mu\nabla^2\bar{v})} = \frac{V_L h}{\nu}\,\frac{h}{R} \sim \frac{R\dot{h}}{h}\,\frac{h}{\nu}\,\frac{h}{R} = \frac{h\dot{h}}{\nu}$$

Both are small when $h\dot{h}/\nu \ll 1$, that is

$$h \ll \frac{\nu}{\dot{h}}$$

Radial Velocity Profile

With negligible inertia and uniform pressure across the film, the radial momentum equation is (Eq. 7.69)

$$0 = -\frac{\partial P}{\partial r} + \mu \frac{\partial^2 v_r}{\partial y^2}; \qquad y = 0: \; v_r = 0, \qquad y = h: \; v_r = 0$$

Integrating twice, with $P_l = -\partial P/\partial r$ independent of y: $\mu v_r = -P_l y^2/2 + C_1 y + C_2$, $C_2 = 0, C_1 = P_l h/2$:

$$v_r = \frac{P_l}{2\mu}\, y\,(h - y) = -\frac{h^2}{2\mu}\frac{\partial P}{\partial r}\,\frac{y}{h}\left(1 - \frac{y}{h}\right)$$

The flux per unit length of circumference is $\int_0^h v_r\, dy = P_l h^3/(12\mu)$ (as in Eq. 7.11).

Equation for the Pressure

Integrate the continuity equation $\frac{\partial (r v_r)}{\partial r} + \frac{\partial (r v_y)}{\partial y} = 0$ across the film:

$$\int_0^h \left[\frac{\partial (r v_r)}{\partial r} + \frac{\partial (r v_y)}{\partial y}\right] dy = \frac{\partial}{\partial r}\left(r \int_0^h v_r\, dy\right) + r\,[v_y(h) - v_y(0)] = 0$$

(the Leibniz term with dh/dr is zero because h does not depend on r). With $v_y(0) = 0$ and $v_y(h) = \dot{h}$ (kinematic condition on the moving body):

$$-\frac{\partial}{\partial r}\left(r\,\frac{h^3}{12\mu}\,\frac{\partial P}{\partial r}\right) + r\dot{h} = 0; \qquad r = 0: \; \frac{\partial P}{\partial r} = 0, \qquad r = R: \; P = P_o$$

Pressure Distribution and Adhesion Force

Integrating from the axis (Barrow, with $r\,\partial P/\partial r = 0$ at $r = 0$): $-r\frac{h^3}{12\mu}\frac{\partial P}{\partial r} + \dot{h}\frac{r^2}{2} = 0$, so $\frac{\partial P}{\partial r} = \frac{6\mu\dot{h}}{h^3}\,r$, and integrating from R to r with $P(R) = P_o$:

$$P - P_o = \frac{6\mu\dot{h}}{h^3}\,\frac{r^2 - R^2}{2} = \frac{3\mu}{h^3}\,\dot{h}\left(r^2 - R^2\right)$$

For $\dot{h} > 0$ (lifting), $P < P_o$ in the whole film: an **underpressure**, largest on the axis, $P(0) - P_o = -3\mu\dot{h}R^2/h^3$.

The oil acts on the base of the body with the normal pointing from the body into the liquid, $\bar{n} = -\bar{e}_y$, so the force is $-\int (P - P_o)\bar{n}\,d\sigma = \int (P - P_o)\,d\sigma\,\bar{e}_y$ (the uniform P_o gives no net force on the closed body):

$$\bar{F} = \int_0^R 2\pi r\,(P - P_o)\,dr\,\bar{e}_y = \frac{6\pi\mu\dot{h}}{h^3}\left[\frac{r^4}{4} - \frac{R^2 r^2}{2}\right]_0^R \bar{e}_y = -\frac{3\pi}{2}\,\mu\,\frac{R^4}{h^3}\,\dot{h}\,\bar{e}_y$$

The force is downward: the underpressure **holds the body back** (adhesion), and grows as h^{-3} when the film is thin and with the velocity of lifting.

Answers and Checks

- Viscous dominance: $h \ll \nu/\dot{h}$ (and $h \ll R$).
- Radial profile: $v_r = -\frac{h^2}{2\mu}\frac{\partial P}{\partial r}\frac{y}{h}\left(1 - \frac{y}{h}\right)$; equation: $-\frac{\partial}{\partial r}\left(r\frac{h^3}{12\mu}\frac{\partial P}{\partial r}\right) + r\dot{h} = 0$.
- Pressure: $P - P_o = \frac{3\mu\dot{h}}{h^3}(r^2 - R^2)$; force: $\bar{F} = -\frac{3\pi}{2}\mu\frac{R^4}{h^3}\dot{h}\,\bar{e}_y$.
- Dimensions: $\mu R^4\dot{h}/h^3$ is $ML^{-1}T^{-1}\cdot L^4\cdot LT^{-1}/L^3 = MLT^{-2}$, a force.
- Consistency: the total volume flux leaving through $r = R$ is $2\pi R\cdot\left(-P_l h^3/(12\mu)\right)$ at R, with $P_l = -6\mu\dot{h}R/h^3$ there: $2\pi R\cdot\frac{6\mu\dot{h}R}{h^3}\cdot\frac{h^3}{12\mu} = \pi R^2\dot{h}$, the volume swept by the body (negative sign: inflow), as continuity requires.
- Check against the official key: VF8.pdf gives the same scale analysis, profile, pressure equation, $P - P_o = 3\mu\dot{h}(r^2 - R^2)/h^3$ and $F = -\frac{3\pi}{2}\mu R^4\dot{h}/h^3$.

PROBLEM VF9

Conveyor Belt Pumping on an Incline

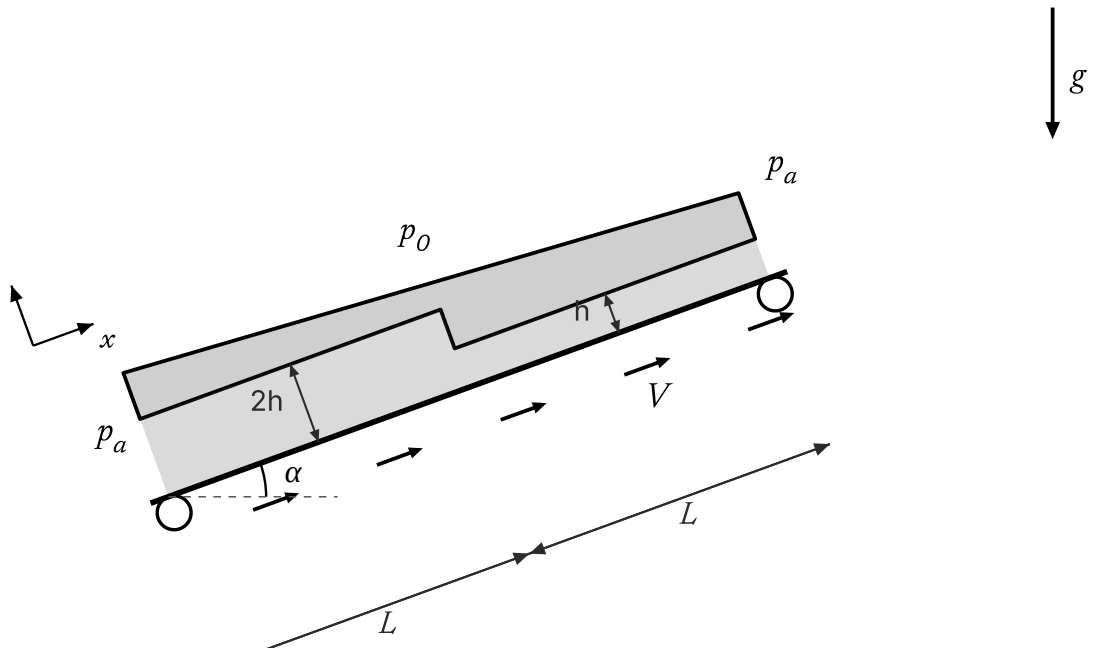
Original statement · VF9.pdf

Couette + Poiseuille | Two stretches in series | Power and energy balance

OFFICIAL PROBLEM STATEMENT

The conveyor belt of velocity V shown in the figure is used to pump a liquid of density ρ and viscosity μ along a straight inclined channel of angle α including two stretches of equal length L with different heights $2h$ and h, respectively.

- Give the conditions for the liquid-film motion to be dominated by viscosity.
- Compute the pressure distribution along the channel $p(x)$.
- Determine the volume flux Q as well as the minimum value of V for which $Q > 0$.
- Obtain the power needed to drive the conveyor belt.



Statement figure (redrawn from VF9.pdf): an inclined channel at the angle α between the conveyor belt, which moves up the slope with velocity V, and a fixed wall with a step (gap 2h over a length L, then h over a length L); both ends are at p_a and p_0 is the pressure at the step

Source: VF9.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.8-7.12; wedge effect, Eqs. 7.69-7.75.

Worked solution

Hypotheses, Data and Unknowns

- The belt is the lower wall of the channel (the line between the two rollers in the figure) and moves up the slope, along $+x$, with velocity V; the upper wall is fixed. x runs along the belt and y is normal to it, from the belt towards the fixed wall.
- Gap: $2h$ for $0 < x < L$ and h for $L < x < 2L$. Both ends of the channel are at the atmospheric pressure p_a (the inlet at the bottom reservoir, the outlet at the top).
- The flow is steady and slender, so in each stretch the pressure is uniform across the gap and the reduced pressure $P = p + \rho g x \sin\alpha$ (with $Z = x\sin\alpha$ at $y \to 0$) has a constant gradient.
- Let p_0 be the pressure at the step, $x = L$.

Symbol	Meaning	SI unit
V	velocity of the belt	m/s
h, $2h$	gaps of the two stretches	m
L, α	length of each stretch, angle of the channel	m, rad
p_a, p_0	pressure at the ends and at the step	Pa
Q	volume flux per unit width	m^2/s

Criterion for Viscous Dominance

For a slender film (Eqs. 7.67-7.68) with the velocity scale V, the largest gap $2h \sim h$ and the length L:

$$\frac{\rho V h}{\mu}\,\frac{h}{L} \ll 1$$

The flow is steady, so the unsteady condition (7.67) holds.

Velocity and Flux in Each Stretch

The reduced pressure at the ends: $P(0) = p_a$ and $P(2L) = p_a + \rho g\, 2L \sin\alpha$; at the step $P(L) = p_0 + \rho g L \sin\alpha$. With $P_l = -dP/dx$:

$$P_{l1} = \frac{p_a - p_0 - \rho g L \sin\alpha}{L}, \qquad P_{l2} = \frac{p_0 + \rho g L \sin\alpha - p_a - 2\rho g L \sin\alpha}{L} = \frac{p_0 - p_a - \rho g L \sin\alpha}{L}$$

The profile of a gap H with the lower wall at velocity V and the upper wall at rest is the sum of Couette and Poiseuille flows (Eqs. 7.9 and 7.11, 7.70):

$$v_x = V\,\frac{H-y}{H} + \frac{P_l}{2\mu}\, y\,(H-y), \qquad Q = \int_0^H v_x\, dy = \frac{VH}{2} + \frac{P_l H^3}{12\mu}$$

$$H = 2h: \quad Q = Vh + \frac{2h^3}{3\mu}\, P_{l1}, \qquad H = h: \quad Q = \frac{Vh}{2} + \frac{h^3}{12\mu}\, P_{l2}$$

Pressure Distribution and Flux

The flux is the same in both stretches. With $\Delta = (p_0 - p_a)/L$ and $G = \rho g \sin\alpha$, $P_{l1} = -\Delta - G$ and $P_{l2} = \Delta - G$:

$$Vh - \frac{2h^3}{3\mu}\,(\Delta + G) = \frac{Vh}{2} + \frac{h^3}{12\mu}\,(\Delta - G)$$

Multiplying by $12\mu/h^3$: $6\mu V/h^2 - 8\Delta - 8G = \Delta - G$, so $9\Delta = 6\mu V/h^2 - 7G$:

$$p_0 - p_a = \frac{2}{3}\,\frac{\mu V L}{h^2} - \frac{7}{9}\,\rho g L \sin\alpha$$

The pressure $p = P - \rho g x \sin\alpha$ is linear in each stretch, because the weight term cancels (as in VF5):

$$0 < x < L: \quad p = p_a + (p_0 - p_a)\,\frac{x}{L}, \qquad L < x < 2L: \quad p = p_0 - (p_0 - p_a)\,\frac{x - L}{L}$$

(the second from $P(x) = P(L) + P_{l2}\cdot(-(x-L))$ and $p = P - \rho g x \sin\alpha$).

Volume flux. Using the second stretch with $P_{l2} = \Delta - G$:

$$Q = \frac{Vh}{2} + \frac{h^3}{12\mu}\left(\frac{2}{3}\frac{\mu V}{h^2} - \frac{16}{9}G\right) = \frac{Vh}{2} + \frac{Vh}{18} - \frac{4}{27}\frac{h^3 G}{\mu} = \frac{5}{9}Vh - \frac{4}{27}\,\frac{\rho g h^3 \sin\alpha}{\mu}$$

Minimum velocity. $Q > 0$ requires $\frac{5}{9}Vh > \frac{4}{27}\frac{\rho g h^3 \sin\alpha}{\mu}$:

$$V > V_{min} = \frac{4}{15}\,\frac{\rho g h^2 \sin\alpha}{\mu}$$

Power Needed to Drive the Belt

The liquid acts on the belt with the shear $\tau = \mu\,\partial v_x/\partial y$ at $y = 0$ (normal $+\bar{e}_y$, from the belt into the liquid):

$$\tau_1 = \mu\left(-\frac{V}{2h} + \frac{P_{l1}}{2\mu}\,2h\right) = -\frac{\mu V}{2h} + P_{l1}h, \qquad \tau_2 = -\frac{\mu V}{h} + \frac{P_{l2}h}{2}$$

$$\tau_1 + \tau_2 = -\frac{3}{2}\frac{\mu V}{h} + h\left(P_{l1} + \frac{P_{l2}}{2}\right) = -\frac{3}{2}\frac{\mu V}{h} - h\left(\frac{\Delta}{2} + \frac{3G}{2}\right)$$

The force of the liquid on the belt (per unit width) is $L(\tau_1+\tau_2)\bar{e}_x$, negative: the liquid resists. The key's boxed $W=VL(\tau_1+\tau_2)$ is the power of this force (negative); **the power that the drive must supply is its opposite**:

$$W_{drive}=-VL\left(\tau_1+\tau_2\right)=VL\left[\frac{3}{2}\frac{\mu V}{h}+\frac{3}{2}\rho g h\sin\alpha+\frac{h}{2}\,\frac{p_0-p_a}{L}\right]$$

and with $\Delta=\frac{2}{3}\frac{\mu V}{h^2}-\frac{7}{9}G$ ($\frac{h\Delta}{2}=\frac{1}{3}\frac{\mu V}{h}-\frac{7}{18}Gh$):

$$W_{drive}=VL\left[\frac{11}{6}\,\frac{\mu V}{h}+\frac{10}{9}\,\rho g h\sin\alpha\right]$$

Energy check (computed here, horizontal case $G=0$): the viscous dissipation of a gap H per unit length is $\int_0^H\mu\left(\partial v_x/\partial y\right)^2dy=\mu V^2/H+P_l^2H^3/(12\mu)$ (the cross term integrates to zero). With $P_{l1}=-\Delta=-\frac{2}{3}\mu V/h^2$ and $P_{l2}=\Delta$: stretch 1 gives $\frac{43}{54}\mu V^2/h$ and stretch 2 gives $\frac{28}{27}\mu V^2/h=\frac{56}{54}\mu V^2/h$; adding them and multiplying by L gives $\frac{99}{54}\mu V^2L/h=\frac{11}{6}\mu V^2L/h$, which is W_{drive} for $G=0$.

Answers and Checks

- Viscous dominance: $(\rho Vh/\mu)(h/L)\ll 1$.
- Pressure: $p=p_a+(p_0-p_a)x/L$ for $0<x<L$ and $p=p_0-(p_0-p_a)(x-L)/L$ for $L<x<2L$, with $p_0-p_a=\frac{2}{3}\mu VL/h^2-\frac{7}{9}\rho gL\sin\alpha$.
- Flux: $Q=\frac{5}{9}Vh-\frac{4}{27}\rho gh^3\sin\alpha/\mu$; $V_{min}=\frac{4}{15}\rho gh^2\sin\alpha/\mu$.
- Power supplied: $W=VL\left[\frac{3}{2}\frac{\mu V}{h}+\frac{3}{2}\rho gh\sin\alpha+\frac{h}{2}\frac{p_0-p_a}{L}\right]=VL\left[\frac{11}{6}\frac{\mu V}{h}+\frac{10}{9}\rho gh\sin\alpha\right]$.
- Dimensions: $\mu V^2L/h$ per unit width is $ML^{-1}T^{-1}\cdot L^2T^{-2}\cdot L/L=MLT^{-3}$, a power per unit width; $\rho gh^2/\mu$ is a velocity.
- Limits: for $\alpha=0$ and $L\to$ any, $V_{min}=0$ and $Q=\frac{5}{9}Vh$ (the pump works for any $V>0$); for h fixed and large α the weight of the column stops the pumping when $V<V_{min}$.

- Check against the official key: VF9.pdf gives the same profiles, fluxes, $p_0 - p_a$, Q, V_{min} and pressure distribution (its p is the pressure, in which the weight cancels), and $W = VL(\tau_1 + \tau_2)$, which has the sign of the force on the belt, as explained above.

PROBLEM VF10

Pulsatile Flow in a Pipe

Original statement · VF10.pdf

Oscillatory pressure gradient | Womersley parameter | Quasi-steady and Stokes layer

OFFICIAL PROBLEM STATEMENT

Consider the unidirectional periodic motion induced in an infinitely long circular pipe of radius a by an oscillatory pressure gradient $P_l = A\cos(\omega t)$. Write the conservation equation with boundary conditions that determine the axial velocity $v_x(r,t)$ and show how the problem can be solved exactly by separation of variables. Study separately the limits $a^2\omega/\nu \gg 1$ and $a^2\omega/\nu \ll 1$ and obtain the corresponding limiting solutions. In biofluid mechanics the square root $a(\omega/\nu)^{1/2}$ is called the Womersley parameter, which takes fairly large values for blood flow in large arteries, as you can see by using the values corresponding to the human aorta ($\mu/\rho \simeq 4\times10^{-2}$ cm^2/s, $a \simeq 1.2$ cm, and $\omega = 2\pi$ s^{-1}), but that decreases for flow is smaller arteries. Find how small the artery radius needs to be for Poiseuille flow to be approximately applicable.

The sheet prints "but that decreases for flow is smaller arteries" and has no figure.

Source: VF10.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: unidirectional flow, Eqs. 7.6-7.7; Stokes flow near a wall, Eqs. 7.26-7.31; the limits of an oscillating channel, Eqs. 7.33-7.35.

Worked solution

Hypotheses, Data and Equation

- Incompressible liquid of constant ρ and μ ($\nu = \mu/\rho$) in a pipe of constant section, so the flow is strictly unidirectional, $v_x(r,t)$ (Eq. 7.1), and the reduced pressure gradient $P_l(t) = A\cos\omega t$ is uniform along the pipe (Eq. 7.4).
- The solution sought is the periodic one, so no initial condition is needed.
- Aorta: $\omega \simeq 2\pi$ s^{-1} (heart rate 1 Hz), $\nu \simeq 4\times10^{-2}$ cm^2/s, $a \simeq 1.2$ cm.

Equation (7.7) with the forcing and its conditions (symmetry on the axis and no slip at the wall), with the scale of each term below it:

$$\underbrace{\rho\frac{\partial v_x}{\partial t}}_{\rho V_L\omega} = A\cos\omega t + \underbrace{\frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_x}{\partial r}\right)}_{\mu V_L/a^2}; \qquad r=0:\ \frac{\partial v_x}{\partial r}=0, \qquad r=a:\ v_x=0$$

The ratio of the local acceleration to the viscous term is $\rho V_L\omega/(\mu V_L/a^2) = a^2\omega/\nu$, the square of the Womersley parameter.

Exact Solution by Separation of Variables

With $A\cos\omega t = \mathrm{Re}\left(Ae^{\imath\omega t}\right)$, solve for a complex v^* with $v_x = \mathrm{Re}(v^*)$:

$$\rho\frac{\partial v^*}{\partial t} = Ae^{\imath\omega t} + \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v^*}{\partial r}\right), \qquad v^* = Ae^{\imath\omega t}f(r)$$

Substituting (the factor $e^{\imath\omega t}$ cancels, the variables separate):

$$\imath\omega\rho f = 1 + \frac{\mu}{r}\frac{d}{dr}\left(r\frac{df}{dr}\right); \qquad f'(0)=0, \qquad f(a)=0$$

This is a linear ordinary equation with a constant forcing. A particular solution is the constant $f_p = 1/(\imath\omega\rho)$ (the pressure gradient balances the acceleration). The homogeneous equation $f'' + f'/r - \kappa^2 f = 0$ with $\kappa^2 = \imath\omega/\nu$, $\kappa = (1+\imath)\sqrt{\omega/(2\nu)}$, is the modified Bessel equation of order zero (outside the notes, an **extension**), whose solution regular on the axis is $I_0(\kappa r)$. The wall condition fixes its coefficient:

$$f = \frac{1}{\imath\omega\rho}\left[1 - \frac{I_0(\kappa r)}{I_0(\kappa a)}\right], \qquad v_x = \mathrm{Re}\left\{\frac{A}{\imath\omega\rho}\left[1 - \frac{I_0(\kappa r)}{I_0(\kappa a)}\right]e^{\imath\omega t}\right\}$$

Check: since $I_0'' + I_0'/x = I_0$, $f'' + f'/r = -\frac{\kappa^2}{\imath\omega\rho}\frac{I_0(\kappa r)}{I_0(\kappa a)}$, and $\imath\omega\rho f - 1 = -\frac{I_0(\kappa r)}{I_0(\kappa a)}$ while $\frac{\mu}{r}(rf')' = \mu\left(f'' + \frac{f'}{r}\right) = -\frac{\mu\kappa^2}{\imath\omega\rho}\frac{I_0(\kappa r)}{I_0(\kappa a)} = -\frac{I_0(\kappa r)}{I_0(\kappa a)}$, so the equation holds; $f(a)=0$ and $f'(0)=0$ because $I_0'(0)=0$.

Limit $a^2\omega/\nu \ll 1$: Quasi-steady Poiseuille Flow

The viscous time a^2/ν is much shorter than ω^{-1}, so the unsteady term is negligible and the profile adjusts at every instant:

$$0 = A\cos\omega t + \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_x}{\partial r}\right) \implies v_x = \frac{A\cos\omega t}{4\mu}\left(a^2 - r^2\right)$$

(the integration of (7.13) with $P_l = A\cos\omega t$). From the exact solution: $I_0(x) \simeq 1 + x^2/4$, so $1 - I_0(\kappa r)/I_0(\kappa a) \simeq \kappa^2(a^2 - r^2)/4$, and $f \simeq \frac{1}{i\omega\rho}\frac{i\omega}{4\nu}\left(a^2 - r^2\right) = \frac{a^2 - r^2}{4\mu}$.

Limit $a^2\omega/\nu \gg 1$: Uniform Flow and Stokes Layer

Viscosity is negligible in most of the section, and the equation reduces to the balance of acceleration and pressure force, $\rho\,\partial v_x/\partial t = A\cos\omega t$. Integrating in time (the mean value is zero for the periodic solution):

$$v_x = \frac{A}{\rho\omega}\sin\omega t$$

a **uniform flow** (the same velocity at every radius) that lags the pressure gradient by a quarter of a period. It does not satisfy the no-slip condition: near the wall there is a thin layer where viscosity matters. The balance $\rho V_L/\omega^{-1} \sim \mu V_L/\delta^2$ gives $\delta \sim \sqrt{\nu/\omega} \ll a$. In it let $\eta = (a - r)/\sqrt{\nu/\omega}$ (the distance to the wall), $\tau = \omega t$ and $\tilde{v} = \frac{A}{\rho\omega}\sin\omega t - v_x$:

$$\frac{\partial\tilde{v}}{\partial\tau} = \frac{\partial^2\tilde{v}}{\partial\eta^2}; \qquad \eta = 0:\ \tilde{v} = \frac{A}{\rho\omega}\sin\omega t, \qquad \eta \to \infty:\ \tilde{v} \to 0$$

(the curvature of the pipe is negligible in a layer of thickness $\delta \ll a$). It is the Stokes problem (7.26) for the wall velocity $\frac{A}{\rho\omega}\sin\omega t$, whose solution (7.31), with $\sin$ in place of $\cos$, gives (computed here)

$$v_x = \frac{A}{\rho\omega}\left[\sin\omega t - e^{-\zeta}\sin(\omega t - \zeta)\right], \qquad \zeta = \frac{a-r}{\sqrt{2\nu/\omega}}$$

which is zero at the wall ($\zeta = 0$) and tends to the uniform flow away from it.

Blood Flow in the Aorta

$$\frac{a^2\omega}{\nu} = \frac{(1.2\ \text{cm})^2\, 2\pi\ \text{s}^{-1}}{4\times 10^{-2}\ \text{cm}^2/\text{s}} = \frac{1.44\times 6.283}{0.04} = 226, \qquad a\sqrt{\frac{\omega}{\nu}} = \sqrt{226} = 15$$

The Womersley parameter is about 15, so the flow in the aorta is in the limit $a^2\omega/\nu \gg 1$: a uniform core with a thin viscous layer of thickness $\sqrt{\nu/\omega} = \sqrt{0.04/6.283}\ \text{cm} = 0.0798\ \text{cm} \simeq 0.8\ \text{mm}$.

Radius for Poiseuille flow. Viscosity dominates the whole section, and the profile is the quasi-steady Poiseuille one, when $a^2\omega/\nu \lesssim 1$, that is

$$a \lesssim \sqrt{\frac{\nu}{\omega}} \simeq 0.8\ \text{mm}, \qquad a \ll 0.8\ \text{mm for a good approximation}$$

(about the size of the smallest arteries and arterioles).

Answers and Checks

- Problem: $\rho\,\partial v_x/\partial t = A\cos\omega t + \frac{\mu}{r}\partial_r\left(r\,\partial_r v_x\right), \partial_r v_x(0) = 0, v_x(a) = 0$; exact solution with I_0 (extension).
- $a^2\omega/\nu \ll 1$: $v_x = \frac{A\cos\omega t}{4\mu}(a^2 - r^2)$. $a^2\omega/\nu \gg 1$: $v_x = \frac{A}{\rho\omega}\sin\omega t$, with a Stokes layer of thickness $\sqrt{\nu/\omega}$.
- Aorta: $a^2\omega/\nu \simeq 226$, Womersley parameter $\simeq 15$; Poiseuille flow needs $a \lesssim 0.8$ mm.
- Dimensions: $A/(\rho\omega)$ is $ML^{-2}T^{-2}/(ML^{-3}T^{-1}) = LT^{-1}$; Aa^2/μ is also a velocity.

- Check against the official key: VF10.pdf gives the same equation, the complex-function problem for the separation of variables (left as the ordinary equation), both limits, the layer thickness $\sqrt{\nu/\omega}$, the Stokes problem in η and the aorta numbers (226, 15, 0.8 mm).

PROBLEM VF11

Windshield Wiper

Original statement · VF11.pdf

Couette + Poiseuille | Closed end | Force and moment

OFFICIAL PROBLEM STATEMENT

To investigate the performance of windshield wipers consider as a simplified model the L-shaped arm of the figure moving with velocity V with respect to a horizontal surface. The wiper is pulled from the end point A with a force F_x parallel to the wall, dragging in its motion the fluid located between the arm and the wall. To study the problem, we shall use a reference frame moving with the arm, as indicated in the figure. The outer pressure is p_a.

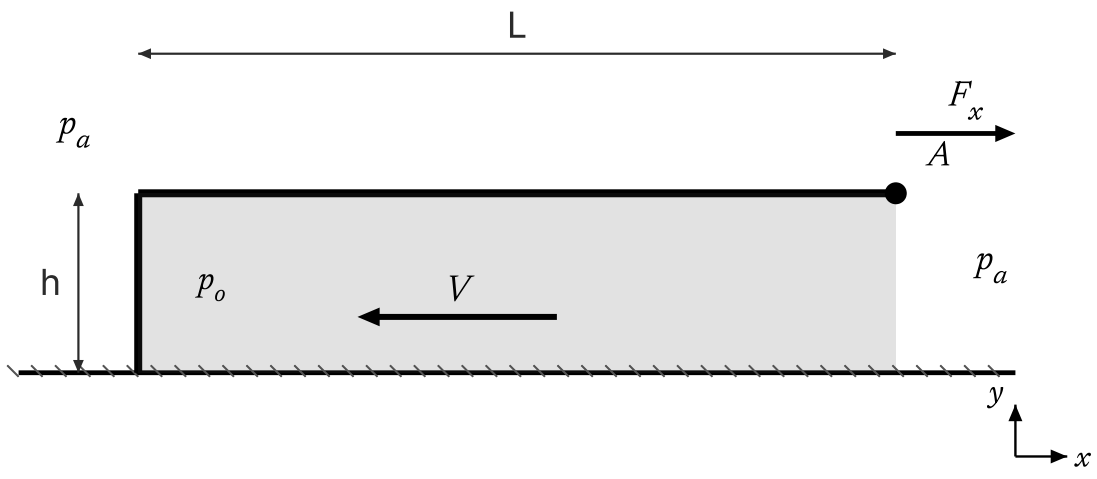
1. Determine the velocity v_x across the channel formed between the wiper and the wall.

2. Find the pressure distribution $p(x)$ including the value found at the end wall $x = 0$.

3. Compute the force F_x necessary to move the wiper.

4. Obtain the moment exerted with respect to the end point A.

5. If W is the weight of the wiper, determine the critical value of V beyond which the end $x = 0$ rises, letting the liquid flow out.



Statement figure (redrawn from VF11.pdf): an L-shaped arm whose horizontal bar of length L is at a height h above the wall and whose vertical leg closes the channel at x = 0; the wall moves with velocity V towards −x in the frame of the arm, F_x acts at A and p_o is the pressure at x = 0

Source: VF11.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.9-7.12; wedge effect, Eqs. 7.69-7.76.

Worked solution

Hypotheses, Data and Unknowns

- In the frame of the arm the wall moves with velocity $-V\bar{e}_x$ (the arm moves with $+V$ with respect to the wall) and the arm is at rest. The channel is closed at $x = 0$ by the vertical leg and open at $x = L$, where the pressure is p_a.
- The channel is slender ($h \ll L$), the flow is steady in this frame and the pressure is uniform across the gap; $P_l = -dp/dx$ is constant.
- Pressures are gauge pressures with respect to p_a on the outer faces of the wiper; gravity is not considered.

Symbol	Meaning	SI unit
L, h	length of the arm, height of the channel	m
V	velocity of the wiper with respect to the wall	m/s
p_0, p_a	pressure at $x = 0$, outer pressure	Pa
W	weight of the wiper (per unit width)	N/m

Velocity Profile

The wall at $y = 0$ moves with $-V$ and the arm at $y = h$ is at rest, so the Couette and Poiseuille flows of Eqs. (7.9) and (7.11) add (Eq. 7.12):

$$v_x = -V\,\frac{h-y}{h} + \frac{P_l}{2\mu}\,y\,(h-y)$$

The channel is closed at $x = 0$ and the flux is uniform along it (Eq. 7.72), so **the flux is zero**:

$$Q = \int_0^h v_x\,dy = -\frac{Vh}{2} + \frac{P_l h^3}{12\mu} = 0 \implies P_l = \frac{6\mu V}{h^2}$$

and the profile is $v_x = -V\frac{h-y}{h} + \frac{3V}{h^2}\,y\,(h-y)$: the liquid is dragged to the left near the wall and returns to the right near the arm.

Pressure Distribution

$P_l = -dp/dx = (p_0 - p_a)/L$ is constant with $p(L) = p_a$:

$$p_0 = p_a + \frac{6\mu V L}{h^2}, \qquad p - p_a = \frac{6\mu V}{h^2}\,(L - x)$$

an overpressure, largest at the closed end.

Force Needed to Move the Wiper

Take the wiper as the arm plus the vertical leg. The force of the liquid on it (normal pointing from the wiper into the liquid) has two parts in the x direction.

Pressure on the leg (area h per unit width, normal $+\bar{e}_x$ on its inner face, force $-(p-p_a)\bar{n}$): $-(p_0 - p_a)h = -6\mu V L/h$. The pressure on the underside of the arm is vertical.

Shear on the underside of the arm ($y = h$, $\bar{n} = -\bar{e}_y$): $\bar{\tau}\cdot\bar{n} = -\mu\,\partial v_x/\partial y$, and $\partial v_x/\partial y = V/h + \frac{P_l}{2\mu}(h - 2y)$ gives at $y = h$: $V/h - \frac{P_l h}{2\mu} = V/h - 3V/h = -2V/h$. So the force per unit area is $+2\mu V/h$ and, over the length L, $+2\mu V L/h$.

$$F_{liquid,x} = -\frac{6\mu V L}{h} + \frac{2\mu V L}{h} = -\frac{4\mu V L}{h}$$

The liquid opposes the motion. The force that must be applied at A is the opposite:

$$F_x = \frac{4\mu V L}{h}$$

(the key writes the sum of both contributions on one line, "$-\mu(\partial v_x/\partial y)_{y=h} L\,\bar{e}_x = -4\mu V L/h\,\bar{e}_x$", which is the total and not the viscous term alone; it then boxes $F_x = 4\mu V L/h$).

Moment about the End Point A

The pressure on the underside of the arm acts upward, $+(p-p_a)\,dx\,\bar{e}_y$, at the point x, whose position with respect to A is $(x-L)\bar{e}_x$:

$$\bar{M}_A = \int_0^L (x-L)\,(p-p_a)\,dx\;\bar{e}_z = -\frac{6\mu V}{h^2}\int_0^L (L-x)^2 dx\;\bar{e}_z = -\frac{6\mu V}{h^2}\,\frac{L^3}{3}\,\bar{e}_z = -\frac{2\mu V L^3}{h^2}\,\bar{e}_z$$

(since $\bar{e}_x \times \bar{e}_y = \bar{e}_z$ and $\int_0^L (L-x)^2 dx = \left[-(L-x)^3/3\right]_0^L = L^3/3$). The shear forces on the arm act at the level of A and give no moment; the pressure on the leg and the shear on it give a moment of order $\mu V L/h$, a factor $(h/L)^2$ smaller than the one above, and are neglected (the key strikes out the viscous integral).

Critical Velocity

The wiper is subjected to its weight W (at the middle, $x = L/2$), the pressure load of the film on the arm, the force at A and the reaction R of the wall on its end $x = 0$ (upward). Moments about A, with $\bar{M}_A$ the moment of the liquid:

$$0 = -\frac{2\mu V L^3}{h^2} + W\frac{L}{2} - RL \;\Longrightarrow\; R = \frac{W}{2} - \frac{2\mu V L^2}{h^2}$$

The end $x = 0$ rises when $R = 0$ (the film lifts it):

$$V_c = \frac{W h^2}{4\mu L^2}$$

For $V < V_c$ the end rests on the wall and the liquid is retained; for $V > V_c$ it rises and lets the liquid flow out.

Answers and Checks

- Velocity: $v_x = -V\frac{h-y}{h} + \frac{P_l}{2\mu}y(h-y)$ with $P_l = 6\mu V/h^2$ (zero flux).

- Pressure: $p - p_a = \frac{6\mu V}{h^2}(L - x), p_0 = p_a + \frac{6\mu V L}{h^2}$.
- Force: $F_x = 4\mu V L/h$. Moment about A: $\bar{M}_A = -\frac{2\mu V L^3}{h^2}\bar{e}_z$. Critical velocity: $V_c = Wh^2/(4\mu L^2)$.
- Dimensions: $\mu V L/h$ per unit width is $ML^{-1}T^{-1} \cdot LT^{-1} \cdot L/L = MT^{-2}$, a force per unit width; $Wh^2/(\mu L^2)$ is a velocity (W per unit width $\sim MT^{-2}$).
- Check of the force: the horizontal momentum balance of the liquid in the channel is $0 = (p_0 - p_a)h - \tau_{arm}L - \tau_{wall}L$ per unit width, with $\tau_{wall} = \mu|v'(0)| = 4\mu V/h$ ($v'(0) = V/h + P_l h/(2\mu) = V/h + 3V/h = 4V/h$ directed along $+x$) and $\tau_{arm} = 2\mu V/h$: $6\mu V L/h = 2\mu V L/h + 4\mu V L/h$.
- Check against the official key: VF11.pdf gives the same profile, pressure, $F_x = 4\mu V L/h$, $\bar{M} = -2\mu V L^3/h^2\,\bar{e}_z$ and $V = Wh^2/(4\mu L^2)$.

PROBLEM VF12

Journal Bearing

Original statement · VF12.pdf

Eccentric cylinders | Wedge effect | Force, torque, cavitation

OFFICIAL PROBLEM STATEMENT

The cylindrical bearing shown in the figure includes an external fixed casing of radius R_b and a shaft rotating with angular velocity ω of radius R such that $R_b - R \ll R$. The centers are shifted a small distance e, called exentricity, so that in between both cylinders there exists a thin circular gap of thickness $h \sim e \sim R_b - R \ll R$ full of a liquid of density ρ and viscosity μ.

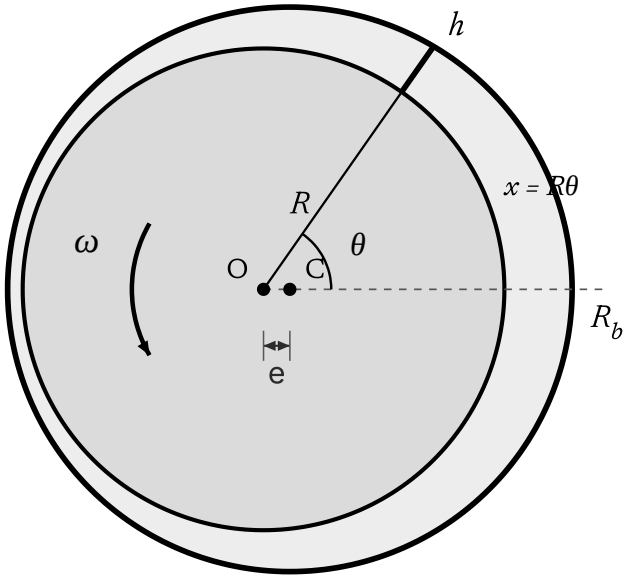
1. Show by simple trigonometric reasoning that the gap thickness can be expressed as $h(\theta) = R_b - R + e\cos\theta$.

2. Obtain the azimuthal velocity distribution v_x in terms of the distance y to the fixed casing and the unknown value of the induced pressure gradient $P_l = -R^{-1}\partial P/\partial\theta$.

3. Determine the circulating volume flux $Q = \int_0^h v_x\, dy$ as well as the pressure distribution $P(\theta) - P(0)$, where $P(0)$ is the unknown value of the pressure at $\theta = 0$, which depends on the pressurization conditions.

4. Calculate the torque and the force acting on the rotating shaft.

The sheet prints "exentricity". In the figure and in the key y is measured from the shaft surface (the key's profile is ωR at $y = 0$ and zero at $y = h$), not from the casing as the sentence of item 2 says.



Statement figure (redrawn from VF12.pdf): a shaft of radius R rotating with ω inside a fixed casing of radius R_b, their centres O and C a distance e apart; θ is measured from the line of centres, h(θ) is the gap, x = Rθ the arc along the shaft and y the distance across the gap, from the shaft to the casing

Source: VF12.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.9-7.12; the wedge effect, Eqs. 7.69-7.76.

Worked solution

Hypotheses, Data and Unknowns

- The gap is thin ($h \sim e \sim R_b - R \ll R$), so it can be unrolled: $x = R\theta$ along the shaft and y across the gap, with $y = 0$ on the shaft (moving with ωR) and $y = h$ on the fixed casing; the flow is steady and slender, with the pressure uniform across the gap and $P_l = -R^{-1}\partial P/\partial\theta$.
- Let $c = R_b - R$ and $\varepsilon = e/c$ ($0 \le \varepsilon < 1$). Forces are per unit length of the shaft.

Symbol	Meaning	SI unit
R, R_b	radii of the shaft and of the casing	m
e, $\varepsilon = e/(R_b - R)$	eccentricity and its relative value	m, 1
ω	angular velocity of the shaft	s^{-1}

Symbol	Meaning	SI unit
Q	circulating flux per unit length	$\mathrm{m^2/s}$

Gap Thickness

Place the centre of the casing at C and the centre of the shaft at O, a distance e from C, with θ the angle at O from the line OC direction; the radial line from O at the angle θ meets the casing at a point at the distance $R + h$ from O. In the triangle O, C, point of the casing, the sides are $R + h$ (from O), e (from C to O) and R_b (from C), with the angle θ at O between the sides $R + h$ and e; the law of cosines gives (the legs of the key's sketch, $(R + h)\cos\theta - e$ and $(R + h)\sin\theta$, give the same):

$$R_b^2 = (R + h)^2 - 2e(R + h)\cos\theta + e^2 \implies R + h = e\cos\theta + \sqrt{R_b^2 - e^2\sin^2\theta}$$

The square root is $R_b\left[1 - e^2\sin^2\theta/(2R_b^2) + \dots\right]$, and the second term is a factor $e/R_b \ll 1$ smaller than e, so $R + h \simeq R_b + e\cos\theta$:

$$h(\theta) = R_b - R + e\cos\theta = (R_b - R)\left[1 + \varepsilon\cos\theta\right]$$

The gap is largest at $\theta = 0$ and smallest at $\theta = \pi$.

Velocity Profile, Flux and Pressure

The shaft moves with ωR and the casing is at rest: Couette plus Poiseuille (Eq. 7.70), with the pressure gradient along $x = R\theta$:

$$v_x = \omega R\,\frac{h - y}{h} - \frac{y(h - y)}{2\mu}\,\frac{1}{R}\frac{\partial P}{\partial\theta}, \qquad Q = \int_0^h v_x\,dy = \frac{\omega R h}{2} - \frac{h^3}{12\mu R}\frac{\partial P}{\partial\theta}$$

The flux is the same at every θ (Eq. 7.72), so $\partial P/\partial\theta = \frac{12\mu R}{h^3}\left(\frac{\omega R h}{2} - Q\right)$, and integrating from 0 to θ (Eq. 7.74):

$$P(\theta)-P(0)=6\mu\omega R^2\int_0^\theta\frac{d\theta'}{h^2}-12\mu RQ\int_0^\theta\frac{d\theta'}{h^3}$$

Periodicity. $P(2\pi)=P(0)$ gives $Q=\frac{\omega R}{2}\int_0^{2\pi}\frac{d\theta}{h^2}\Big/\int_0^{2\pi}\frac{d\theta}{h^3}$. With $h=c(1+\varepsilon\cos\theta)$ and the known integrals (computed here, with $|\varepsilon|<1$)

$$\int_0^{2\pi}\frac{d\theta}{(1+\varepsilon\cos\theta)^2}=\frac{2\pi}{(1-\varepsilon^2)^{3/2}},\qquad\int_0^{2\pi}\frac{d\theta}{(1+\varepsilon\cos\theta)^3}=\frac{\pi\left(2+\varepsilon^2\right)}{(1-\varepsilon^2)^{5/2}}$$

$$Q=\frac{\omega Rc}{2}\,\frac{2\pi/(1-\varepsilon^2)^{3/2}}{\pi(2+\varepsilon^2)/(1-\varepsilon^2)^{5/2}}=\omega Rc\,\frac{1-\varepsilon^2}{2+\varepsilon^2}$$

Pressure. Substituting Q:

$$\frac{\partial P}{\partial\theta}=\frac{6\mu\omega R^2}{c^2}\left[\frac{1}{(1+\varepsilon\cos\theta)^2}-\frac{2(1-\varepsilon^2)}{(2+\varepsilon^2)(1+\varepsilon\cos\theta)^3}\right]=\frac{6\mu\omega R^2}{c^2(2+\varepsilon^2)}\,\frac{\varepsilon\left(2\cos\theta+3\varepsilon+\varepsilon^2\cos\theta\right)}{(1+\varepsilon\cos\theta)^3}$$

and this is the derivative of

$$P(\theta)-P(0)=\frac{6\mu\omega R^2}{(R_b-R)^2}\,\frac{\varepsilon\,(2+\varepsilon\cos\theta)\sin\theta}{(2+\varepsilon^2)(1+\varepsilon\cos\theta)^2}$$

(check: with $N=(2+\varepsilon\cos\theta)\sin\theta$, $N'=2\cos\theta-\varepsilon+2\varepsilon\cos^2\theta$ and $\frac{d}{d\theta}\frac{N}{(1+\varepsilon\cos\theta)^2}=\frac{N'(1+\varepsilon\cos\theta)+2\varepsilon\sin\theta\,N}{(1+\varepsilon\cos\theta)^3}=\frac{2\cos\theta+3\varepsilon+\varepsilon^2\cos\theta}{(1+\varepsilon\cos\theta)^3}$, which has the factor ε of the previous line; and the expression vanishes at $\theta=0$). The pressure is antisymmetric, $P(\theta)-P(0)=-[P(-\theta)-P(0)]$: overpressure on one side of the line of centres and underpressure on the other, which gives a force perpendicular to it.

Force and Torque on the Shaft

Torque. The shear on the shaft (normal from the shaft into the liquid, $+\bar{e}_y$) is $\tau_p = \mu\,\partial v_x/\partial y|_{y=0} = -\frac{\mu\omega R}{h} - \frac{h}{2R}\frac{\partial P}{\partial\theta}$; with $\partial P/\partial\theta = 6\mu\omega R^2/h^2 - 12\mu R Q/h^3$, $\tau_p = -\frac{4\mu\omega R}{h} + \frac{6\mu Q}{h^2}$. The torque per unit length is $M = \int_0^{2\pi} \tau_p R\cdot R\,d\theta$:

$$M = -4\mu\omega R^3\int_0^{2\pi}\frac{d\theta}{h} + 6\mu Q R^2\int_0^{2\pi}\frac{d\theta}{h^2}, \qquad \int_0^{2\pi}\frac{d\theta}{1+\varepsilon\cos\theta} = \frac{2\pi}{\sqrt{1-\varepsilon^2}}$$

$$M = -\frac{8\pi\mu\omega R^3}{c\sqrt{1-\varepsilon^2}} + \frac{12\pi\mu\omega R^3}{c(2+\varepsilon^2)\sqrt{1-\varepsilon^2}} = -\frac{4\pi\mu\omega R^3}{R_b - R}\,\frac{1+2\varepsilon^2}{\sqrt{1-\varepsilon^2}\,(2+\varepsilon^2)}$$

(the torque of the liquid on the shaft, opposing the rotation; the torque the motor applies is its opposite).

Force. The pressure force on the shaft is $\bar{F} = -\int (P-P(0))\,\bar{n}\,R\,d\theta$, with $\bar{n}$ the outward normal of the shaft, $n_z = \sin\theta$ and $n_{\parallel} = \cos\theta$ (the uniform $P(0)$ gives no net force). The component along the line of centres is zero because $P - P(0)$ is odd in θ and $\cos\theta$ is even. The perpendicular one, integrating by parts ($P(0) = P(2\pi)$ so the boundary term vanishes): $\int_0^{2\pi}(P-P(0))\sin\theta\,d\theta = \int_0^{2\pi}\cos\theta\,\frac{\partial P}{\partial\theta}\,d\theta = \frac{6\mu\omega R^2\varepsilon}{c^2(2+\varepsilon^2)}\int_0^{2\pi}\frac{\cos\theta\left(2\cos\theta+3\varepsilon+\varepsilon^2\cos\theta\right)}{(1+\varepsilon\cos\theta)^3}d\theta$, and the last integral equals $2\pi/\sqrt{1-\varepsilon^2}$ (verified numerically in the checks of this note). Hence

$$F_z = -R\int_0^{2\pi}(P-P(0))\sin\theta\,d\theta = -\frac{12\pi\mu\omega R^3}{(R_b-R)^2}\,\frac{\varepsilon}{\sqrt{1-\varepsilon^2}\,(2+\varepsilon^2)}$$

on the shaft (downward in the figure), and the opposite on the bearing (upward). The shear forces contribute to the force on the shaft at a relative order $c/R \ll 1$ and are neglected.

Limit of Large Eccentricity and Cavitation

When $\varepsilon \to 1$ (the shaft touches the casing, $h_{min} = c(1-\varepsilon) \to 0$) the force grows without bound, $F_z \propto (1-\varepsilon^2)^{-1/2}$: **a bearing can in principle bear an infinite load**. In practice it fails earlier: the underpressure side of $P(\theta) - P(0)$ reaches the vapour pressure p_v and the liquid cavitates when $P_{min} - P(0) = p_v - P(0)$. The minimum of P is where $\partial P/\partial\theta = 0$, that is $\cos\theta_{min} = -3\varepsilon/(2+\varepsilon^2)$ with $\sin\theta_{min} < 0$ ($\theta_{min} \simeq 3\pi/2$ for small ε, as the key notes).

Answers and Checks

- Gap: $h = R_b - R + e\cos\theta = (R_b - R)(1 + \varepsilon\cos\theta)$.
- Velocity: $v_x = \omega R\frac{h-y}{h} - \frac{y(h-y)}{2\mu R}\frac{\partial P}{\partial\theta}$; flux: $Q = \omega R(R_b - R)\frac{1-\varepsilon^2}{2+\varepsilon^2}$.
- Pressure: $P - P(0) = \frac{6\mu\omega R^2}{(R_b-R)^2}\frac{\varepsilon(2+\varepsilon\cos\theta)\sin\theta}{(2+\varepsilon^2)(1+\varepsilon\cos\theta)^2}$.
- Torque: $M = -\frac{4\pi\mu\omega R^3}{R_b-R}\frac{1+2\varepsilon^2}{\sqrt{1-\varepsilon^2}(2+\varepsilon^2)}$; force: $F_z = -\frac{12\pi\mu\omega R^3}{(R_b-R)^2}\frac{\varepsilon}{\sqrt{1-\varepsilon^2}(2+\varepsilon^2)}$ on the shaft, opposite on the bearing.
- Limit $\varepsilon \to 0$ (concentric): $Q = \omega Rc/2$, $P = P(0)$, $F_z = 0$ and $M = -2\pi\mu\omega R^3/c$, the Couette torque of the gap c: $\tau = \mu\omega R/c$ times $R \cdot R \cdot 2\pi$.
- Dimensions: $\mu\omega R^3/(R_b - R)$ is $ML^{-1}T^{-1}\cdot T^{-1}\cdot L^3/L = MLT^{-2}$, a torque per unit length (N m/m = N); $\mu\omega R^3/(R_b-R)^2$ is a force per unit length.
- Check against the official key: VF12.pdf gives the same geometry, profile, flux, pressure, torque and force. Its part 1 writes the gap as $(R_b - R)[1 + \varepsilon\cos\theta]$ and its flux integrals as $(1 - \varepsilon\cos\theta)$; the two give the same integrals by $\theta \to \theta + \pi$.

PROBLEM VF13

Rotating Plate over a Wall

Original statement · VF13.pdf

Opening wedge | Unsteady film | Torque

OFFICIAL PROBLEM STATEMENT

A plate of length L is initially sitting on a horizontal plane in the presence of a stagnant liquid atmosphere with reduced pressure P_∞. At a given instant, we begin to rotate the plate with constant angular velocity $\omega = d\alpha/dt$ by applying a given torque M at its left end, as sketched in the figure. For the analysis, use the approximation $h(x,t) = x\tan(\alpha) \simeq x\alpha$, valid for $\alpha \ll 1$.

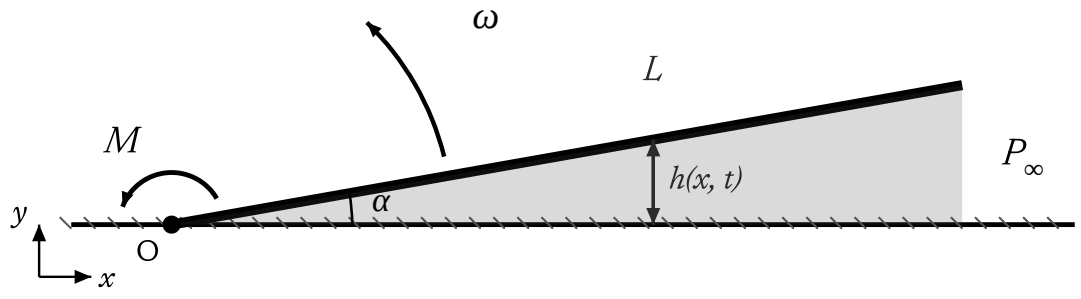
1. Demonstrate that for values of α sufficiently smaller than a critical value, to be determined, the fluid motion in the gap formed between the plate and the wall is dominated by viscosity.

2. Obtain the velocity profile v_x in the gap as a function of the unknwon value of $P_l(x,t) = -\partial P/\partial x$ as well as the associated volume flux at a given section $Q = \int_0^{\alpha x} v_x\, dy$.

3. Using continuity, write an equation linking Q and ω, and integrate it to compute the pressure distribution $P(x,t)$ (in the integration, you may anticipate that $x^3 P_l \to 0$ as $x \to 0$).

4. Determine the torque $M(t)$ needed to provide a constant angular velocity ω.

The sheet prints "unknwon" and, in item 3, a stray character before "it".



Statement figure (redrawn from VF13.pdf): a plate of length L hinged at O on a horizontal wall, rotating with ω = dα/dt under the torque M; the gap is h(x, t) = αx and the surrounding liquid is at P_∞

Source: VF13.pdf (page 1, exam of Dec 16, 2011, statement and handwritten key). Notes.pdf, Chapter 7: slender films, Eqs. 7.67-7.71; the wedge effect, Eq. 7.72; forces and moments, Eq. 7.76.

Worked solution

Hypotheses, Data and Unknowns

- The plate is hinged at the origin O and rises with the angle $\alpha(t)$ ($\alpha \ll 1$): x runs along the wall from O, y is vertical and the gap is $h = \alpha x$, increasing with time, so liquid has to flow **in** from the surroundings to fill the wedge that opens.
- The gap is slender ($h \ll x$) and the pressure is uniform across it; at the outer end $x = L$ the pressure is P_∞; at the hinge no flux ($Q = 0$ at $x = 0$).
- The plate surface moves with the velocity $v_y = \partial h/\partial t = \omega x$ (to the first order in α).

Symbol	Meaning	SI unit
$L, \alpha(t)$	length of the plate and its angle	m, rad
$\omega = d\alpha/dt$	constant angular velocity	s^{-1}
P_∞	reduced pressure of the surrounding liquid	Pa
M	torque at O (per unit width)	N

Criterion for Viscous Dominance

The scales are $h_c \sim L\alpha$ (largest gap) and $t_0 = \alpha/\omega$ (time to change the angle by its own value). The continuity equation $\partial v_x/\partial x + \partial v_y/\partial y = 0$ with $v_y \sim \omega L$ gives $V_L \sim \omega L^2/h_c = \omega L/\alpha$. Local acceleration and convective acceleration against the viscous force (Eqs. 7.67-7.68):

$$\frac{h_c^2/\nu}{t_0} \sim \frac{L^2\alpha^2\omega}{\nu\alpha} = \frac{\omega L^2\alpha}{\nu}, \qquad \frac{V_L h_c}{\nu}\frac{h_c}{L} \sim \frac{\omega L}{\alpha}\frac{L\alpha}{\nu}\,\alpha = \frac{\omega L^2\alpha}{\nu}$$

Both are the same number, and viscosity dominates if it is much smaller than one:

$$\alpha \ll \alpha_c = \frac{\nu}{\omega L^2}$$

(and of course $\alpha \ll 1$).

Velocity Profile and Flux

Equation (7.69) with the no-slip conditions at the wall ($y = 0$) and at the plate ($y = h$), both at rest in this approximation (the normal motion of the plate is the continuity term), integrated as for (7.11):

$$v_x = \frac{P_l}{2\mu}\, y\,(h - y), \qquad Q = \frac{P_l h^3}{12\mu} = -\frac{\alpha^3 x^3}{12\mu}\frac{\partial P}{\partial x}$$

Continuity and Pressure

Integrate $\partial v_x/\partial x + \partial v_y/\partial y = 0$ across the gap:

$$\int_0^h \frac{\partial v_x}{\partial x}\, dy + \int_0^h \frac{\partial v_y}{\partial y}\, dy = \frac{\partial Q}{\partial x} - \frac{dh}{dx}\, v_x(y = h) + v_y(h) - v_y(0) = 0$$

The term $\frac{dh}{dx} v_x(y = h)$ vanishes because $v_x(h) = 0$ (no slip on the plate), $v_y(0) = 0$ on the wall and $v_y(h) = \partial h/\partial t = \omega x$:

$$\frac{\partial Q}{\partial x} + \omega x = 0$$

Integrating with $Q = 0$ at $x = 0$: $Q = -\omega x^2/2$ (negative: inflow towards the hinge). Equating with the flux of the profile:

$$-\frac{\alpha^3 x^3}{12\mu}\frac{\partial P}{\partial x} = -\frac{\omega x^2}{2} \quad \Longrightarrow \quad \frac{\partial P}{\partial x} = \frac{6\mu\omega}{\alpha^3}\,\frac{1}{x}$$

(the constant of integration of the first step is zero because $x^3 P_l \to 0$ as $x \to 0$). Integrating from L to x with $P(L) = P_\infty$:

$$P - P_\infty = \frac{6\mu\omega}{\alpha^3}\,\ln\frac{x}{L}$$

negative for $x < L$: an underpressure that grows without bound near the hinge (logarithmically) and tends to hold the plate down.

Torque

The moment of the pressure on the plate about the hinge, per unit width, with the pressure acting on its underside (the force is vertical to this order, at the position x):

$$M_{liquid} = \int_0^L (P - P_\infty)\, x\, dx\, \bar{e}_z = \frac{6\mu\omega}{\alpha^3}\int_0^L x \ln\frac{x}{L}\, dx\, \bar{e}_z, \qquad \int_0^L x\ln\frac{x}{L}dx = L^2\int_0^1 u\ln u\, du = L^2\left[\frac{u^2}{2}\ln u - \frac{u^2}{4}\right]_0^1 = -\frac{L^2}{4}$$

$$M_{liquid} = -\frac{3}{2}\,\frac{\mu\omega L^2}{\alpha^3}\,\bar{e}_z$$

The liquid opposes the rotation (the rotation is counterclockwise, $+\bar{e}_z$), so the torque that must be applied is the opposite, positive:

$$\bar{M} = \frac{3}{2}\,\frac{\mu\,\omega\,L^2}{\alpha^3}\,\bar{e}_z$$

It grows as α^{-3} at the start of the motion (small α). The shear on the plate gives a moment smaller by a factor α.

Answers and Checks

- Viscous dominance: $\alpha \ll \nu/(\omega L^2)$.
- Profile and flux: $v_x = \frac{P_l}{2\mu}y(h-y)$, $Q = \frac{P_l h^3}{12\mu} = -\frac{\omega x^2}{2}$.
- Pressure: $P - P_\infty = \frac{6\mu\omega}{\alpha^3}\ln(x/L)$. Torque: $M = \frac{3}{2}\mu\omega L^2/\alpha^3$.

- Dimensions: $\mu\omega L^2/\alpha^3$ is $ML^{-1}T^{-1}\cdot T^{-1}\cdot L^2 = MT^{-2}\cdot L$, a torque per unit width (N); $\nu/(\omega L^2)$ is dimensionless.
- Consistency: the volume flowing in through $x = L$ is $|Q(L)| = \omega L^2/2$ per unit width, equal to the area of the wedge swept per unit time, $\frac{d}{dt}\left(\frac{1}{2}L\cdot \alpha L\right) = \omega L^2/2$.
- Check against the official key: VF13.pdf gives the same criterion, profile, flux, pressure (with $P(L) = P_\infty$) and torque; the key does not write the kinematic condition $v_y(h) = \omega x$, the constant of integration or the boundary condition, which are stated above.

PROBLEM VF14

Arm Approaching a Wall

Original statement · VF14.pdf

Squeeze film | Closed end | Force and torque

OFFICIAL PROBLEM STATEMENT

The L-shaped arm of length L depicted in the figure moves vertically approaching the horizontal wall, forming a thin channel of decreasing uniform thickness $h(t) \ll L$. A liquid of density ρ and viscosity μ fills the channel. Due to the arm displacement, the liquid is forced to move out, discharging to the surrounding atmosphere, where the reduced pressure is P_∞. For a known value of $h(t)$,

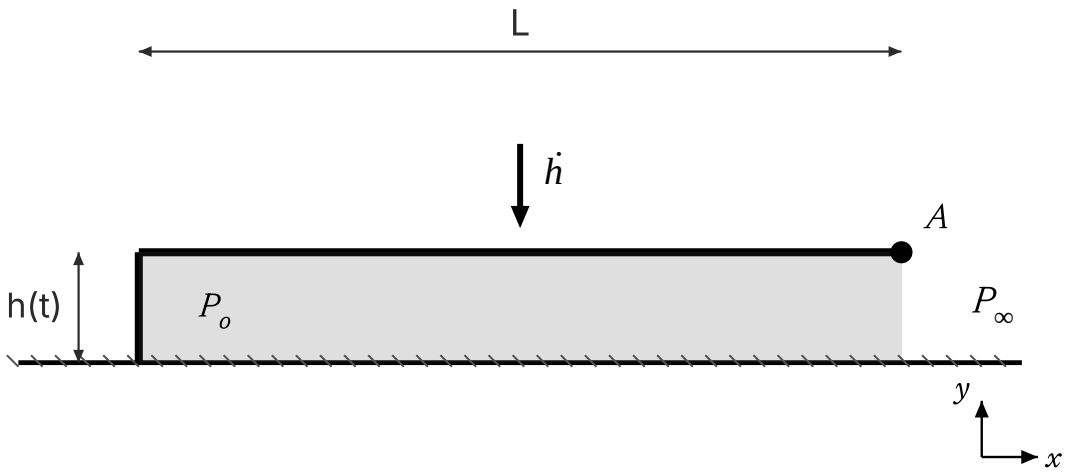
1. Give the conditions for the motion in the channel to be dominated by viscosity.

2. Determine the velocity profile across the channel $v_x(x, y, t)$, the associated volume flux $Q = \int_0^h v_x \, dy$, and the reduced pressure distribution $P(x, t)$, including the value $P_0(t)$ at $x = 0$.

3. Calculate the force acting on the arm $\bar{F} = (F_x, F_y)$.

4. Obtain the torque exerted on the arm with respect to its end point A.

The sheet writes the pressure at $x = 0$ as P_o in the figure and P_0 in the statement.



Statement figure (redrawn from VF14.pdf): an L-shaped arm whose horizontal bar of length L is at the height h(t) above a wall and whose vertical leg closes the channel at x = 0; A is the end of the bar and P_∞ the pressure outside; P_o is the pressure at x = 0

Source: VF14.pdf (page 1, exam of Jan 10, 2012, statement and handwritten key). Notes.pdf, Chapter 7: slender films, Eqs. 7.67-7.71; the wedge effect, Eq. 7.72; forces and moments, Eq. 7.76.

Worked solution

Hypotheses, Data and Unknowns

- The arm moves vertically with the velocity $\dot h = dh/dt$ (negative when it approaches the wall); the channel, $0 < x < L, 0 < y < h(t)$, is closed at $x = 0$ by the vertical leg and open at $x = L$, where the pressure is P_∞.
- Slender film ($h \ll L$), pressure uniform across it; gravity is absorbed in the reduced pressure.

Criterion for Viscous Dominance

By continuity $V_L \sim L\dot h/h$ and $t_0 \sim h/|\dot h|$. The convective and local accelerations against the viscous force (Eqs. 7.67-7.68):

$$\frac{\rho V_L h}{\mu}\frac{h}{L} \sim \frac{L\dot h}{h}\frac{h}{\nu}\frac{h}{L} = \frac{h\dot h}{\nu} \ll 1, \qquad \frac{h^2/\nu}{t_0} \sim \frac{h\dot h}{\nu} \ll 1$$

So $h\,|\dot h|/\nu \ll 1$.

Velocity, Flux and Pressure

Equation (7.69) with no slip on the wall and on the arm ($v_x = 0$ at $y = 0$ and $y = h$) gives the Poiseuille profile (7.11):

$$0 = P_l + \mu\frac{\partial^2 v_x}{\partial y^2} \implies v_x = \frac{P_l}{2\mu}\,y\,(h - y), \qquad Q = \frac{P_l h^3}{12\mu} = -\frac{\partial P}{\partial x}\frac{h^3}{12\mu}$$

Integrating the continuity equation across the channel, with $v_x(h) = 0, v_y(0) = 0$ and $v_y(h) = \dot h$ (the arm moves with $\dot h$):

$$\int_0^h \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} \right) dy = 0 \implies \frac{\partial Q}{\partial x} + \dot{h} = 0$$

so $Q = -\dot{h}x$ (zero at the closed end $x = 0$; positive for an approaching arm). Then $-\frac{\partial P}{\partial x}\frac{h^3}{12\mu} = -\dot{h}x$, that is, the first integration of the key, $\frac{\partial P}{\partial x}\frac{h^3}{12\mu} = \dot{h}x$ (the key's boxed line carries a sign that is the opposite of its next line, which is the one used):

$$\frac{\partial P}{\partial x} = \frac{12\mu\dot{h}}{h^3}\, x \implies P = \frac{6\mu\dot{h}}{h^3}\, x^2 + C_2$$

With $P(L) = P_\infty$:

$$P(x,t) = P_\infty + \frac{6\mu\dot{h}}{h^3}\left(x^2 - L^2\right), \qquad P_0(t) = P_\infty - \frac{6\mu\dot{h}L^2}{h^3}$$

For an approaching arm ($\dot{h} < 0$) there is an overpressure, largest at the closed end.

Force on the Arm

Vertical force. The liquid below the arm pushes it upward with the excess pressure (the pressure P_∞ acts on the whole surface of the arm and gives no net force):

$$F_y = \int_0^L (P - P_\infty)\, dx = \frac{6\mu\dot{h}}{h^3}\left[\frac{x^3}{3} - L^2 x\right]_0^L = \frac{6\mu\dot{h}}{h^3}\left(-\frac{2L^3}{3}\right) = -\frac{4\mu\dot{h}L^3}{h^3}$$

positive for $\dot{h} < 0$: the liquid resists the approach (it is a force of the order of $\mu|\dot{h}|L^3/h^3$, very large for a thin film).

Horizontal force. Two contributions: the pressure on the inner face of the vertical leg (area h, normal $+\bar{e}_x$ from the arm into the liquid; the force of the liquid is $-(P_0 - P_\infty)h$) and the shear on the underside of the arm ($\bar{n} = -\bar{e}_y$, force $-\mu\,\partial v_x/\partial y$ per unit area):

$$F_x = -(P_0 - P_\infty)\, h - \int_0^L \mu \left.\frac{\partial v_x}{\partial y}\right|_{y=h} dx, \qquad -(P_0 - P_\infty)h = \frac{6\mu \dot{h} L^2}{h^2}$$

With $\partial v_x/\partial y = \frac{P_l}{2\mu}(h - 2y)$, $P_l = -\partial P/\partial x = -12\mu\dot{h}x/h^3$: at $y = h$, $\mu\, \partial v_x/\partial y = -\frac{P_l h}{2} = \frac{6\mu\dot{h}x}{h^2}$, and

$$-\int_0^L \frac{6\mu\dot{h}x}{h^2}\, dx = -\frac{3\mu\dot{h}L^2}{h^2}, \qquad F_x = \frac{6\mu\dot{h}L^2}{h^2} - \frac{3\mu\dot{h}L^2}{h^2} = \frac{3\mu\dot{h}L^2}{h^2}$$

$$\bar{F} = \left(\frac{3\mu\dot{h}L^2}{h^2},\ -\frac{4\mu\dot{h}L^3}{h^3}\right), \qquad |F_x|/|F_y| \sim h/L \ll 1$$

Torque about the End Point A

The pressure on the underside of the arm gives the moment about A ($x = L$) of a vertical force at the distance $x - L$ (arm positive along $\bar{e}_x \times \bar{e}_y = \bar{e}_z$):

$$\bar{M} = -\int_0^L (P - P_\infty)(L - x)\, dx\ \bar{e}_z = -\frac{6\mu\dot{h}}{h^3}\int_0^L \left(x^2 - L^2\right)(L - x)\, dx\ \bar{e}_z$$

$$\int_0^L \left(x^2 - L^2\right)(L - x)\, dx = \int_0^L \left(Lx^2 - x^3 - L^3 + L^2 x\right) dx = L\frac{L^3}{3} - \frac{L^4}{4} - L^4 + \frac{L^4}{2} = L^4\left(\frac{1}{3} - \frac{1}{4} - 1 + \frac{1}{2}\right) = -\frac{5}{12}L^4$$

$$\bar{M} = \frac{5}{2}\,\frac{\mu\dot{h}}{h^3}\,L^4\ \bar{e}_z$$

The moment of the pressure on the leg is of order $\mu\dot{h}L^2/h$ (the pressure $\sim \mu\dot{h}L^2/h^3$ acting on the area h with the arm $\sim h$) and the moment of the shear on the arm is $\sim \mu\dot{h}L^2/h$: a factor $(h/L)^2$ smaller than the one above and neglected, as the key does.

Answers and Checks

- Viscous dominance: $h|\dot{h}|/\nu \ll 1$.
- Profile and flux: $v_x = \frac{P_l}{2\mu}y(h-y)$, $Q = -\dot{h}x$; pressure: $P = P_\infty + \frac{6\mu\dot{h}}{h^3}(x^2 - L^2)$, $P_0 = P_\infty - \frac{6\mu\dot{h}L^2}{h^3}$.
- Force: $\bar{F} = \left(\frac{3\mu\dot{h}L^2}{h^2}, -\frac{4\mu\dot{h}L^3}{h^3}\right)$; torque about A: $\bar{M} = \frac{5}{2}\frac{\mu\dot{h}L^4}{h^3}\bar{e}_z$.
- Dimensions: $\mu\dot{h}L^3/h^3$ is $ML^{-1}T^{-1}\cdot LT^{-1}\cdot L^3/L^3 = MLT^{-2}\cdot L^{-1}$, a force per unit width.
- Limits: $\dot{h}\to 0$ gives zero force, torque and overpressure; $h\to 0$ makes them grow as h^{-3} (and the film does not drain in finite time at a constant $\dot{h}$ without infinite force).
- Check against the official key: VF14.pdf gives the same criterion, profile, flux, pressure, force and torque; its boxed continuity line has the opposite sign of the next one (only the second is consistent with $\partial Q/\partial x = -\dot{h}$).

PROBLEM VF15

Belt Pump: Forces, Power and Efficiency

Original statement · VF15.pdf

Couette + Poiseuille | Forces on the belts | Efficiency

OFFICIAL PROBLEM STATEMENT

PROBLEM 4 (1 hour) The system sketched in the figure is designed to steadily pump fluid of constant density ρ and constant viscosity μ from a reservoir, where the modified pressure has a constant value P_0, to another one at pressure $P_0 + \Delta P$ ($\Delta P > 0$). To that end, a pair of parallel belts of length L, separated by a distance $h \ll L$, are driven at constant velocities V_l and V_u, dragging the liquid towards the large pressure reservoir.

The liquid flow can be assumed steady, two-dimensional (spanwise length much larger than h), and dominated by viscous forces in the gap. It proves convenient to introduce a cartersian coordinate system (x, y), where x and y are the directions parallel and perpendicular to the belts, respectively. The **only** known parameters are $(\rho, \mu, P_0, \Delta P, L, h, V_l, V_u)$.

1. (1 point). Under the assumption of dominant viscous forces, obtain the characteristic velocity, $V_{\Delta P}$, induced in the liquid by the overpressure ΔP. Show that the pump will only work if the dimensionless parameter $\Lambda = (h^2 \Delta P)/[\mu L (V_l + V_u)]$, is sufficiently small.

2. (1 point). Provide the condition(s) that must be satisfied for the flow to be dominated by viscous forces.

3. (2 points). Determine the streamwise velocity profile in the gap between both belts, $v_x(x, y)$, as well as the volume flux per unit spanwise length, q. The results must be expressed as a function of known data, and of the reduced pressure gradient, P_l.

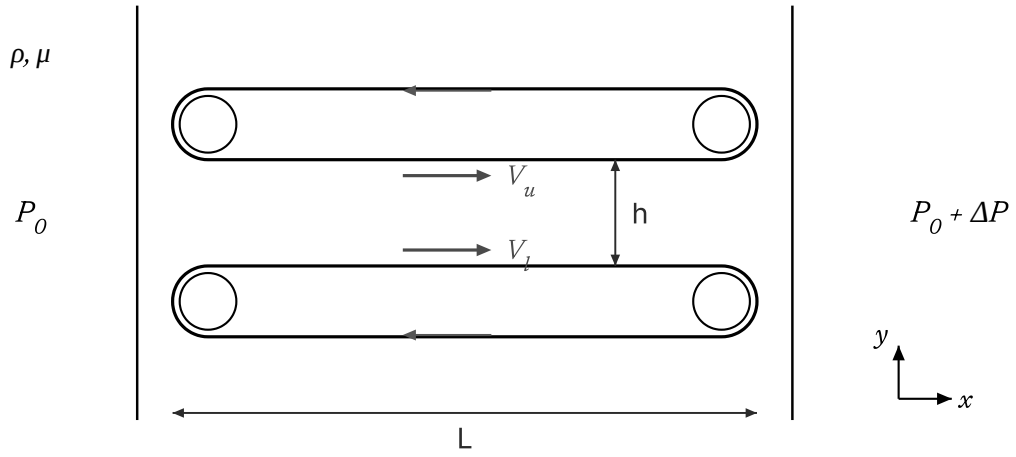
4. (2 points). Determine P_l, as well as the pressure distribution inside the channel, $P(x)$.

5. (2 points). Find the external forces per unit spanwise length in the x-direction, $F_{l,x}^{EXT}$ and $F_{u,x}^{EXT}$, that must be exerted on the lower and upper belts to keep the motion, as well as the power per unit spanwise length that needs to be supplied to drive the system, $\dot{W}$.

6. (2 points). Knowing that the net mechanical energy gained by the liquid per unit time and per unit spanwise length during the pumping process is $\dot{W}_{useful} = q \Delta P$, determine the efficiency of the pumping system, $\eta = \dot{W}_{useful}/\dot{W}$, as a function of the governing parameters. Study the particular cases a) $V_l = V_u$, and b) $V_l = 0$, demonstrating that in both cases the result can be expressed as a function of the single parameter Λ defined above. In case b), find the maximum efficiency, η_{max} as well as the corresponding value of Λ_{max}.

7. (1 additional point). Discuss the solution of Problem 3 in the light of the results obtained in Problem 4.

The sheet prints "cartersian". In the key the efficiency is item (5), the statement's item 6; the key has no text for item 7.



Statement figure (redrawn from VF15.pdf): two parallel belts of length L, a distance h apart, whose facing strands move at V_u and V_l and drag liquid of density ρ and viscosity μ from a reservoir at P_0 to one at P_0 + ΔP

Source: VF15.pdf (page 1: statement; pages 2 and 3: handwritten key; exam of Dec 18, 2012). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.8-7.12; slender flows, Eqs. 7.40-7.46.

Worked solution

Hypotheses, Data and Unknowns

- The belts are the walls $y = 0$ (lower, velocity V_l) and $y = h$ (upper, velocity V_u), both moving along $+x$ towards the high-pressure reservoir, $x = 0$ at P_0 and $x = L$ at $P_0 + \Delta P$; the pressure is reduced pressure and the flow is steady, slender and two-dimensional.
- Everything is per unit spanwise length.

Characteristic Velocity and Criterion for Pumping

The balance of the pressure and viscous forces in the gap, $\Delta P/L \sim \mu V_{\Delta P}/h^2$, gives the velocity induced by the overpressure (which pushes the liquid **backward**, from high to low pressure):

$$V_{\Delta P} \sim \frac{h^2 \Delta P}{\mu L}, \qquad \Lambda = \frac{V_{\Delta P}}{V_l + V_u} = \frac{h^2 \Delta P}{\mu L \left(V_l + V_u\right)}$$

The belts drag the liquid forward with a velocity of order $V_l + V_u$: the pump only works if this velocity beats the backflow, $V_{\Delta P}$ sufficiently smaller than $V_l + V_u$, that is, Λ small enough (Part 6 shows that the flux vanishes for a finite value of Λ).

Criterion for Viscous Dominance

The slender-flow condition (Eq. 7.41) is $(\rho V h/\mu)(h/L) \ll 1$ with the largest velocity of the problem. Depending on which one dominates:

$$\Lambda \ll 1 \; (V \sim V_l + V_u): \quad \frac{\rho \left(V_l + V_u\right) h^2}{\mu L} \ll 1; \qquad \Lambda \gg 1 \; (V \sim V_{\Delta P}): \quad \frac{\rho \, \Delta P \, h^4}{\mu^2 L^2} \ll 1$$

The pump only works for Λ small, so the first condition is the one that matters. The flow is steady, so the unsteady condition holds.

Velocity Profile and Flux

With $P_l = -\partial P/\partial x$ and Eq. (7.12) extended to two moving walls:

$$0 = P_l + \mu \frac{\partial^2 v_x}{\partial y^2}; \qquad y = 0: \; v_x = V_l, \qquad y = h: \; v_x = V_u$$

$v_x = -\frac{P_l y^2}{2\mu} + Ay + B$ with $B = V_l$ and $A = \frac{V_u - V_l}{h} + \frac{P_l h}{2\mu}$:

$$v_x = V_l + \frac{V_u - V_l}{h}\, y + \frac{P_l}{2\mu}\, y\,(h-y), \qquad q = \int_0^h v_x\, dy = \frac{(V_l + V_u)h}{2} + \frac{P_l h^3}{12\mu}$$

Pressure Gradient and Pressure

Integrating the continuity equation across the gap: $\partial q/\partial x = v_y(0) - v_y(h) = 0$ (the belts do not move normally to themselves), so q is uniform and so is P_l:

$$\frac{\partial^2 P}{\partial x^2} = 0, \quad P(0) = P_0, \quad P(L) = P_0 + \Delta P \implies P = P_0 + \Delta P\, \frac{x}{L}, \qquad P_l = -\frac{\Delta P}{L}$$

Forces on the Belts and Power

The shear stress is $\tau'_{xy} = \mu\, \partial v_x/\partial y = \mu \frac{V_u - V_l}{h} - \frac{\Delta P}{2L}(h - 2y)$ (with $P_l = -\Delta P/L$). The force of the liquid on a belt is $\bar{\tau}' \cdot \bar{n}$ with $\bar{n}$ from the belt into the liquid ($+\bar{e}_y$ on the lower belt, $-\bar{e}_y$ on the upper belt); only the x components are considered (the pressure is normal to the belts):

$$F_{lx} = \int_0^L \tau'_{xy}\big|_{y=0} dx = \frac{\mu\,(V_u - V_l)\,L}{h} - \frac{\Delta P\, h}{2}, \qquad F_{ux} = -\int_0^L \tau'_{xy}\big|_{y=h} dx = -\frac{\mu\,(V_u - V_l)\,L}{h} - \frac{\Delta P\, h}{2}$$

The external forces that keep the belts at constant velocity are the opposite:

$$F^{EXT}_{u,x} = \frac{\Delta P\, h}{2} + \frac{\mu\,(V_u - V_l)\,L}{h}, \qquad F^{EXT}_{l,x} = \frac{\Delta P\, h}{2} - \frac{\mu\,(V_u - V_l)\,L}{h}$$

and the power per unit spanwise length is the sum of force times velocity:

$$\dot{W} = F_{l,x}^{EXT} V_l + F_{u,x}^{EXT} V_u = \frac{(V_l + V_u)\, h\, \Delta P}{2} + \frac{(V_u - V_l)^2\, \mu L}{h}$$

The first term is the work against the pressure difference (the force $\Delta P\, h/2$ on each belt), the second is the viscous dissipation of the Couette part.

Efficiency

$$\eta = \frac{q\, \Delta P}{\dot{W}} = \frac{\dfrac{(V_l + V_u) h\, \Delta P}{2} - \dfrac{h^3 \Delta P^2}{12 \mu L}}{\dfrac{(V_l + V_u) h\, \Delta P}{2} + \dfrac{(V_u - V_l)^2 \mu L}{h}}$$

Dividing numerator and denominator by $(V_l + V_u) h\, \Delta P/2$, with $\frac{h^2 \Delta P}{6 \mu L (V_l + V_u)} = \frac{\Lambda}{6}$ and $\frac{2 \mu L (V_u - V_l)^2}{h^2 \Delta P (V_u + V_l)} = \frac{2}{\Lambda} \left(\frac{V_u - V_l}{V_u + V_l} \right)^2$:

$$\eta = \frac{1 - \Lambda/6}{1 + \dfrac{2}{\Lambda} \left(\dfrac{V_u - V_l}{V_u + V_l} \right)^2}$$

The flux $q = \frac{(V_l + V_u) h}{2} \left(1 - \frac{\Lambda}{6}\right)$ vanishes at $\Lambda = 6$: the pump works for $\Lambda < 6$.

a) $V_l = V_u$**.** The ratio of velocities is zero and $\eta = 1 - \Lambda/6$ (no shear dissipation: the walls move together).

b) $V_l = 0$**.** The ratio is 1, so $\eta = \dfrac{1 - \Lambda/6}{1 + 2/\Lambda} = \dfrac{\Lambda(6 - \Lambda)}{6(\Lambda + 2)}$, which is zero at $\Lambda = 0$ and at $\Lambda = 6$. The maximum: $\frac{d}{d\Lambda} \frac{6\Lambda - \Lambda^2}{6\Lambda + 12} = 0$ gives $(6 - 2\Lambda)(\Lambda + 2) - (6\Lambda - \Lambda^2) = 0$, that is $\Lambda^2 + 4\Lambda - 12 = 0$, $(\Lambda + 6)(\Lambda - 2) = 0$:

$$\Lambda_{max} = 2, \qquad \eta_{max} = \frac{2 \cdot 4}{6 \cdot 4} = \frac{1}{3}$$

Discussion of Problem 3 (Item 7)

The key has no text for this item; the following is computed here. Problem 3 of the exam ([[Problem - DA10 Two-Belt Viscous Pump]]) is this pump with $V_l = 0$, calibrated with glycerol ($h = 1$ cm, $L = 20$ cm, $\mu = 1.41$ kg/(m s), $\Delta P = 1$ kPa) at $V_u = 0.10$ to 0.35 m/s, so $\Lambda = h^2 \Delta P/(\mu L V_u) = 0.355/V_u$ (with V_u in m/s) goes from 3.55 to 1.01. The formula of case b) gives $\eta = 0.26, 0.33, 0.33, 0.32, 0.30, 0.28$ for $V_u = 10, 15, 20, 25, 30, 35$ cm/s, the values of the efficiency column of the calibration table; the flux $q = \frac{V_u h}{2}(1 - \Lambda/6)$ gives 2.04 cm^2/s at $V_u = 10$ cm/s (the table has 2), and the power $\dot{W} = \frac{V_u h \Delta P}{2} + \frac{V_u^2 \mu L}{h}$ gives $0.5 + 0.28 = 0.78$ W/m at the same speed (the table: 0.8). The calibration points are around the optimum $\Lambda = 2$ ($V_u = 17.8$ cm/s), where η is highest; and since η depends only on Λ, the oil, with $\Delta P_{oil}/\Delta P_g = 2$ and $\mu_{oil}/\mu_g = 3.55$, works at the same Λ and the same efficiency for the speed $V_{u,oil} = 0.564\, V_{u,g}$, the similarity condition of Problem 3.

Answers and Checks

- $V_{\Delta P} \sim h^2 \Delta P/(\mu L)$; the pump works for $\Lambda \ll 1$ (in fact $\Lambda < 6$). Viscous dominance: $\rho(V_l + V_u)h^2/(\mu L) \ll 1$.
- Velocity: $v_x = V_l + \frac{V_u - V_l}{h} y + \frac{P_l}{2\mu} y(h - y)$; flux: $q = \frac{(V_l + V_u)h}{2} + \frac{P_l h^3}{12\mu}$; $P_l = -\Delta P/L$, $P = P_0 + \Delta P x/L$.
- Forces: $F_{u,x}^{EXT} = \frac{\Delta P h}{2} + \frac{\mu(V_u - V_l)L}{h}$, $F_{l,x}^{EXT} = \frac{\Delta P h}{2} - \frac{\mu(V_u - V_l)L}{h}$; power: $\dot{W} = \frac{(V_l + V_u)h\Delta P}{2} + \frac{(V_u - V_l)^2 \mu L}{h}$.
- Efficiency: $\eta = \frac{1 - \Lambda/6}{1 + (2/\Lambda)[(V_u - V_l)/(V_u + V_l)]^2}$; a) $\eta = 1 - \Lambda/6$; b) $\Lambda_{max} = 2$, $\eta_{max} = 1/3$.
- Dimensions: $\Delta P\, h$ is $ML^{-1}T^{-2} \cdot L = MT^{-2}$, a force per unit width; Λ is $L^2 \cdot ML^{-1}T^{-2}/(ML^{-1}T^{-1} \cdot L \cdot LT^{-1}) = 1$.
- Energy check: $\dot{W} - q\Delta P$ must be the viscous dissipation per unit span, $L \int_0^h \mu\, (\partial v_x/\partial y)^2\, dy$. With $\partial v_x/\partial y = \frac{V_u - V_l}{h} + \frac{P_l}{2\mu}(h - 2y)$ the cross term integrates to zero and the integral is $\mu \frac{(V_u - V_l)^2}{h} L + \frac{P_l^2 h^3}{12\mu} L$; and $\dot{W} - q\Delta P = \frac{(V_u - V_l)^2 \mu L}{h} + \frac{h^3 \Delta P^2}{12 \mu L}$, the same (with $P_l^2 L = \Delta P^2/L$).
- Check against the official key: VF15.pdf gives the same criterion, profile, flux, pressure, forces, power, efficiency, both cases and $\Lambda_{max} = 2$, $\eta_{max} = 1/3$.

PROBLEM VF16

Piston Falling over a Liquid Layer

Original statement · VF16.pdf

Radial squeeze film | Piston dynamics | Quasi-steady piston

OFFICIAL PROBLEM STATEMENT

PROBLEM 4 (1 hour) A cylindrical piston of weight W and radius R, whose upper side is exposed to the atmosphere, slides without friction inside a coaxial container of the same radius. The initial distance between the lower side of the piston and the bottom of the tank is $h_0 \ll R$. Below the piston, the container is filled with liquid of density ρ and viscosity μ. The piston falls due to its own weight, and forces the liquid to leave the tank, discharging to the atmosphere through a circular hole of radius $\alpha R \ll R$.

The known data are $(\rho, \mu, h_0, R, \alpha, W, g, p_a)$. The objective is to determine the vertical position of the piston as a function of time, $h(t)$, under the assumption of dominant viscous forces in the liquid. Gravitational forces in the liquid can be neglected ($P \approx p$). The following steps are suggested:

1. (2 points). Using the radial momentum equation, find the liquid velocity profile, $v_r(z,t)$, in the slender region $\alpha R \leq r \leq R$, $0 \leq z \leq h(t)$, as a function of $h(t)$, and of the pressure gradient $P_l = -\partial p/\partial r$.

2. (2 points). Obtain $p(r,t)$ as a function of $h(t)$, $\dot{h}(t) = dh/dt$, and known data. To that end, integrate the continuity equation across the gap (in the z direction), finding the second order differential equation satisfied by $p(r,t)$. To solve the equation, note that the pressure drop across the non-slender turning region $0 \leq r \leq \alpha R$, $0 \leq z \leq h(t)$, can be neglected, so that $p(r = \alpha R) \approx p_a$. Note also that $v_r = 0$ at the container wall $r = R$.

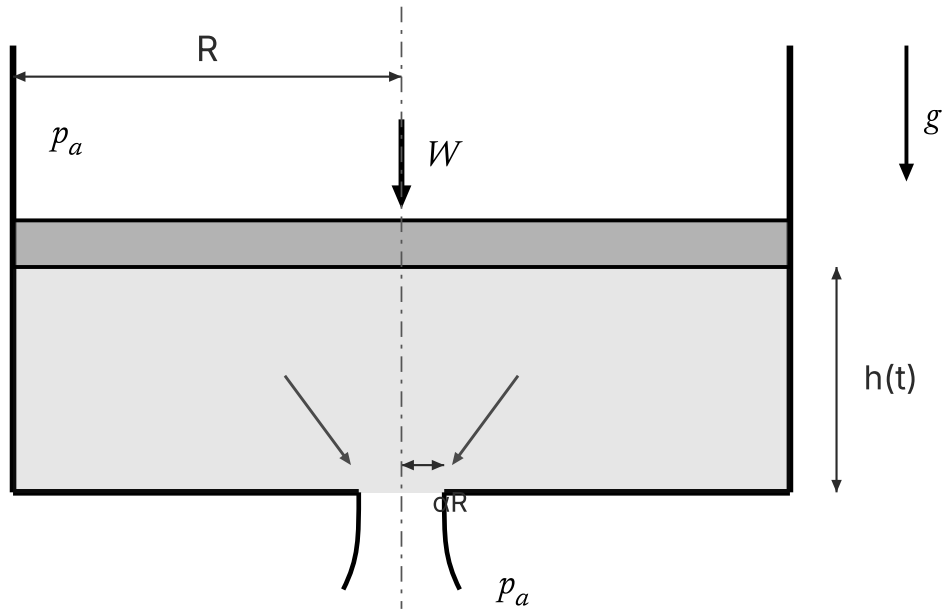
3. (2 points). Find the vertical force acting on the piston due to the liquid and the external atmosphere, F, as a function of $h(t)$, $\dot{h}(t)$, and known data. Note that it proves convenient to introduce a function $f(\alpha)$ to express the dependence of F on α.

4. (2 points). Using Newton's second law applied to the piston in the vertical direction, find the second-order ODE with initial conditions satisfied by $h(t)$.

5. (2 points). Obtain $h(t)$, as well as the volume flux through the hole, $Q(t)$, under the assumption of negligible inertia of the piston (acceleration of the piston much smaller than g). Note that, in this approximation, the pressure forces balance the weight of the piston, the ODE becomes of first order, and only the initial condition $h(0) = h_0$ can be satisfied.

6. (2 additional points). Provide the condition that must be accomplished for viscous forces to be dominant in the liquid, as well as the criterion of negligible inertia of the piston, in terms of the governing parameters $(\rho, \mu, h_0, R, \alpha, W, g, p_a)$.

The sheet gives the hint $\int x \ln x \, dx = \frac{x^2}{2}\left(\ln x - \frac{1}{2}\right)$.



Statement figure (redrawn from VF16.pdf): a piston of weight W and radius R at the height h(t) above the bottom of a container; the liquid below it leaves through a central hole of radius αR at the pressure p_a (axisymmetric section, r radial and z vertical)

Source: VF16.pdf (page 1: statement; page 2: handwritten key; exam of Jan 14, 2013). Notes.pdf, Chapter 7: slender films, Eqs. 7.67-7.71; the wedge effect, Eq. 7.72; Newton's second law.

Worked solution

Hypotheses, Data and Unknowns

- The film between the piston and the bottom is slender ($h \ll R$), axisymmetric, with z the height above the bottom and r the radius. The velocity at the piston is $v_z = \dot{h}$ (< 0 while it falls), at the bottom $v_z = 0$.
- The liquid has no weight effects, so $P \approx p$; the pressure at the hole edge is p_a (the non-slender turning region has a negligible pressure drop).
- The piston has its weight W downward and the atmospheric pressure on its upper face; the liquid pressure acts on its lower face.

Velocity Profile

Radial momentum (Eq. 7.69 in cylindrical coordinates) with no slip at the bottom and the piston:

$$0 = -\frac{\partial p}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2} = P_l + \mu \frac{\partial^2 v_r}{\partial z^2}; \qquad z = 0: \; v_r = 0, \qquad z = h: \; v_r = 0 \implies v_r = \frac{P_l(r,t)}{2\mu} z\,(h - z)$$

Pressure

Integrate the continuity equation $\frac{\partial (r v_r)}{\partial r} + \frac{\partial (r v_z)}{\partial z} = 0$ across the gap:

$$\frac{\partial}{\partial r}\left[r \int_0^h v_r \, dz \right] + r\,[v_z(h) - v_z(0)] = 0 \implies \frac{\partial}{\partial r}\left[r\,\frac{P_l h^3}{12\mu} \right] + r\dot{h} = 0$$

and with $P_l = -\partial p/\partial r$: $\frac{\partial}{\partial r}\left(r \frac{\partial p}{\partial r} \right) = \frac{12\mu\dot{h}}{h^3}\, r$. First integration: $r\frac{\partial p}{\partial r} = \frac{6\mu\dot{h}}{h^3} r^2 + A$. At the container wall $v_r = 0$ for every z, and since $v_r \propto P_l$ this means $\partial p/\partial r = 0$ at $r = R$:

$$A = -\frac{6\mu\dot{h}R^2}{h^3}, \qquad \frac{\partial p}{\partial r} = \frac{6\mu\dot{h}}{h^3}\left(r - \frac{R^2}{r} \right)$$

Second integration, with $p(\alpha R) = p_a$:

$$p = \frac{6\mu\dot{h}}{h^3}\left(\frac{r^2}{2} - R^2 \ln r \right) + B \implies p - p_a = -\frac{6\mu\dot{h}}{h^3}\left[R^2 \ln \frac{r}{\alpha R} - \frac{r^2 - (\alpha R)^2}{2} \right]$$

For $\dot{h} < 0$ it is an overpressure.

Force on the Piston

The excess force is $F = 2\pi \int_{\alpha R}^{R} (p - p_a)\, r\, dr$ (the part $r < \alpha R$, in front of the hole, is at about p_a). With $r = R\eta$, $r\,dr = R^2 \eta\, d\eta$:

$$F = -\frac{12\pi\mu \dot{h}}{h^3}\, R^4 f(\alpha), \qquad f(\alpha) = \int_{\alpha}^{1} \eta \left[\ln\eta - \ln\alpha - \frac{\eta^2 - \alpha^2}{2} \right] d\eta$$

Evaluating the three pieces with Barrow's rule (using the hint $\int \eta \ln\eta\, d\eta = \frac{\eta^2}{2}\left(\ln\eta - \frac{1}{2}\right)$):

$$\int_{\alpha}^{1} \eta\ln\eta\, d\eta = -\frac{1}{4} - \frac{\alpha^2}{2}\ln\alpha + \frac{\alpha^2}{4}, \qquad -\ln\alpha \int_{\alpha}^{1} \eta\, d\eta = -\frac{1-\alpha^2}{2}\ln\alpha, \qquad -\frac{1}{2}\int_{\alpha}^{1} (\eta^3 - \alpha^2\eta)\, d\eta = -\frac{1-\alpha^4}{8} + \frac{\alpha^2(1-\alpha^2)}{4}$$

$$f(\alpha) = -\frac{1}{2}\ln\alpha - \frac{3}{8} + \frac{\alpha^2}{2} - \frac{\alpha^4}{8}$$

Discrepancy with the official solution

The key gives "$f(\alpha) = -\frac{1}{2}\ln\alpha - \frac{5}{12} + \frac{3}{4}\alpha^2 - \frac{1}{3}\alpha^3$" after "computing the integral". That expression is not the value of the integral above (it is not even a function of α^2, as the integral is): at $\alpha = 0.5$ the integral is 0.0888 and the key's polynomial gives 0.0757. Both vanish at $\alpha = 1$ and both behave as $-\frac{1}{2}\ln\alpha$ for $\alpha \to 0$. The value above is used from here on; its derivative is $f'(\alpha) = -(1-\alpha^2)^2/(2\alpha)$, which is negative: f decreases from infinity to zero as the hole grows.

The force is positive (upward) for $\dot{h} < 0$: the liquid resists the fall.

Equation of the Piston and Solution

Newton's second law for the piston in the vertical direction (the weight W downward, the force F of the liquid and of the atmosphere upward; h upward):

$$\frac{W}{g}\,\ddot{h} = F - W, \qquad h(0) = h_0, \quad \dot{h}(0) = 0$$

(with $F = -12\pi\mu\dot{h}R^4 f(\alpha)/h^3$, a second-order ODE).

Negligible inertia, $|\ddot{h}| \ll g$. Then $F = W$ and the equation is of first order, with only $h(0) = h_0$:

$$\frac{dh}{dt} = -\frac{W}{12\pi\mu R^4 f(\alpha)}\,h^3 \equiv -k\,h^3$$

Separating variables, $\frac{dh}{h^3} = -k\,dt$, and integrating (Barrow's rule): $\left[-\frac{1}{2h^2}\right]_{h_0}^{h} = -kt$, so $\frac{1}{h^2} = \frac{1}{h_0^2} + 2kt$:

$$\bar{h} = \frac{h}{h_0} = (1 + 2\tau)^{-1/2}, \qquad \tau = \frac{W h_0^2}{12\pi\mu R^4 f(\alpha)}\,t$$

Volume flux through the hole. The liquid expelled equals the volume swept by the piston, $Q = -\pi R^2 \dot{h} = \pi R^2 k h^3$ (computed here):

$$Q(t) = \frac{W\,h^3}{12\mu R^2 f(\alpha)} = \frac{W h_0^3}{12\mu R^2 f(\alpha)}\,(1 + 2\tau)^{-3/2}$$

The piston never reaches the bottom in a finite time: $h \to 0$ as $t^{-1/2}$ (it needs to squeeze an ever thinner film).

Criteria

Viscous dominance. The scale of the velocity of the piston follows from the time scale $t_c \sim \mu R^4/(W h_0^2)$ (the factor of τ): $\dot{h}_c \sim h_0/t_c = W h_0^3/(\mu R^4)$. The slender-film conditions (Eqs. 7.67-7.68) reduce, as in VF14, to

$$\frac{h_0 \dot{h}_c}{\nu} \ll 1 \implies \frac{\rho W h_0^4}{\mu^2 R^4} \ll 1$$

(besides $h_0 \ll R$).

Negligible inertia of the piston. The acceleration scale is $\ddot{h}_c \sim h_0/t_c^2 = h_0 \left(\frac{W h_0^2}{\mu R^4}\right)^2 = \frac{W^2 h_0^5}{\mu^2 R^8}$, so the condition $\ddot{h}_c \ll g$ is

$$\frac{W^2 h_0^5}{\mu^2 R^8 g} \ll 1$$

Answers and Checks

- Velocity: $v_r = \frac{P_l}{2\mu} z(h-z)$; pressure: $p - p_a = -\frac{6\mu \dot{h}}{h^3}\left[R^2 \ln \frac{r}{\alpha R} - \frac{r^2 - (\alpha R)^2}{2}\right]$.
- Force: $F = -\frac{12\pi\mu\dot{h}}{h^3} R^4 f(\alpha)$, $f(\alpha) = -\frac{1}{2}\ln\alpha - \frac{3}{8} + \frac{\alpha^2}{2} - \frac{\alpha^4}{8}$.
- Equation: $\frac{W}{g}\ddot{h} = F - W$, $h(0) = h_0$, $\dot{h}(0) = 0$.
- Quasi-steady piston: $\bar{h} = (1+2\tau)^{-1/2}$, $\tau = \frac{W h_0^2 t}{12\pi\mu R^4 f(\alpha)}$; $Q(t) = \frac{W h^3}{12\mu R^2 f(\alpha)}$.
- Criteria: $\rho W h_0^4/(\mu^2 R^4) \ll 1$ and $W^2 h_0^5/(\mu^2 R^8 g) \ll 1$.
- Dimensions: $W h_0^2/(\mu R^4)$ is $MLT^{-2}\cdot L^2/(ML^{-1}T^{-1}L^4) = T^{-1}$, so τ is dimensionless; $W h^3/(\mu R^2)$ is $L^3 T^{-1}$.
- Limit $\alpha \to 1$ (a hole as wide as the piston): $f \to 0$ and $F \to 0$, so the liquid offers no resistance, as it can escape freely; $\alpha \to 0$ (a tiny hole): $f \simeq -\frac{1}{2}\ln\alpha \to \infty$ and the fall is slowed down.

- Check against the official key: VF16.pdf gives the same profile, pressure, the form of F with $f(\alpha)$ as the integral above, Newton's law, $\bar{h} = (1 + 2\tau)^{-1/2}$ and both criteria; the closed form of $f(\alpha)$ differs, as noted; the key does not write $Q(t)$.

PROBLEM VF17

Thin Plate in a Couette Channel

Original statement · VF17.pdf

Two channels in parallel | Forces and power | Energy balance

OFFICIAL PROBLEM STATEMENT

PROBLEM 3 (1 hour) A perfect liquid of constant density ρ and constant viscosity μ fills the space between two infinite parallel walls separated by a distance $2h$, that move parallel to themselves in the same direction with a constant velocity U. As shown in the figure, a very thin solid plate of length $L \gg h$ is placed at the center of the channel. In the regions far upstream and far downstream from the plate the liquid moves with uniform velocity U due to the motion of the confining walls, and the modified pressure takes uniform values $P(x \to -\infty) \to P_{-\infty}$ and $P(x \to \infty) \to P_{\infty}$, respectively.

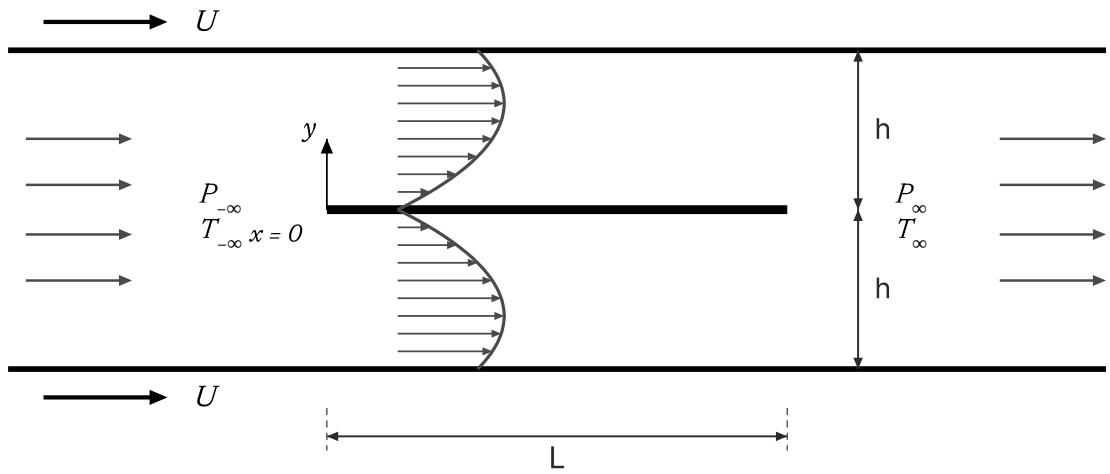
1. Provide the criterion that must be accomplished for the flow to be dominated by viscous forces. (**1 point**)

2. Obtain the velocity field in the region occupied by the plate, in terms of the unknown pressure gradient $P_l = -\partial P/\partial x$. (**3 points**)

3. Obtain the pressure drop $P_{-\infty} - P_{\infty}$, as well as the pressure gradient P_l. (**3 points**)

4. Determine the forces (per unit spanwise length) that the liquid exerts on the confining walls, F_w, and on the inner plate, F_p, as well as the power that must be applied to drive the walls. (**3 points**)

5. If all the inner walls are thermally isolated, determine the temperature increase in the liquid between two points situated far downstream and far upstream, $T_{\infty} - T_{-\infty}$. To that end, apply the energy equation in integral form to an appropriately defined control volume (**1 extra point**).



Statement figure (redrawn from VF17.pdf): a channel between two walls at y = ±h moving with velocity U, with a thin plate of length L on the centre line from x = 0 to x = L; the liquid far upstream and downstream moves with U at the pressures and temperatures P_{−∞}, T_{−∞} and P_∞, T_∞; the sketched profiles are those found in the solution

Source: VF17.pdf (page 1, exam of June 27, 2013; statement only, no key). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.8-7.12; slender flows, Eqs. 7.40-7.46.

Worked solution

Hypotheses, Data and Unknowns

- The walls are at $y = \pm h$ and move with U; the plate lies on $y = 0$ for $0 < x < L$ and is at rest. Over the plate the channel is divided into two identical half-channels of height h, symmetric with respect to $y = 0$.
- Steady flow, slender ($L \gg h$) and two-dimensional; the entrance and exit regions, of length of order h, are neglected: the pressure losses in them are small against the loss along the plate (Notes.pdf, Eq. 7.63).
- The liquid has constant c (specific heat), and the inner walls are insulated, so there is no heat flux across the boundaries. Per unit spanwise length.

Symbol	Meaning	SI unit
h, L	half-height of the channel, length of the plate	m
U	velocity of the walls and of the far flow	m/s
$P_{-\infty}$, P_{∞}	reduced pressures far upstream and downstream	Pa

Symbol	Meaning	SI unit
$T_{-\infty}, T_{\infty}, c$	temperatures far away, specific heat	K, J/(kg K)

Criterion for Viscous Dominance

In the half-channel of height h and length L with the velocity scale U (Eq. 7.41):

$$\frac{\rho U h}{\mu}\,\frac{h}{L} \ll 1$$

The flow is steady. (If the pressure-driven velocity, computed below, is of the order of U the same criterion holds.)

Velocity Field in the Plate Region

In the upper half-channel, $0 < y < h$, with the plate at rest ($y = 0$) and the wall moving with U ($y = h$), the profile is Couette plus Poiseuille (Eq. 7.12):

$$v_x = U\,\frac{y}{h} + \frac{P_l}{2\mu}\,y\,(h - y), \qquad q_{half} = \int_0^h v_x\,dy = \frac{Uh}{2} + \frac{P_l h^3}{12\mu}$$

The lower half is the mirror image, $v_x(-y) = v_x(y)$.

Pressure Drop and Pressure Gradient

The flux is conserved along the channel. Far from the plate it is $2h\,U$ (uniform velocity). Over the plate it is the sum of the two halves:

$$2hU = 2\left[\frac{Uh}{2} + \frac{P_l h^3}{12\mu}\right] = Uh + \frac{P_l h^3}{6\mu} \implies P_l = \frac{6\mu U}{h^2}$$

The couette flux alone (Uh) is only half of what must pass, so the pressure must push: $P_l > 0$, the pressure falls in the direction of the flow. The gradient is uniform (the section is constant), so

$$P_{-\infty} - P_{\infty} = P_l L = \frac{6\mu U L}{h^2}$$

(the losses at the ends of the plate are of order $\mu U/h$, a factor h/L smaller). With this value the velocity is

$$v_x = U\frac{y}{h} + \frac{3U}{h^2}\, y\,(h-y) = \frac{U\,y}{h^2}\,(4h - 3y), \qquad 0 \le y \le h$$

which is U at the wall, zero at the plate and has a maximum $\frac{4}{3}U$ at $y = \frac{2}{3}h$ (the jet seen in the sketch of the sheet).

Forces and Power

The shear stress is $\mu\,\partial v_x/\partial y = \mu\left[\frac{U}{h} + \frac{P_l}{2\mu}(h-2y)\right] = \frac{\mu U}{h^2}\,(4h-6y)$. The force of the liquid on a surface is $\bar{\tau}' \cdot \bar{n}$ with $\bar{n}$ pointing from the surface into the liquid.

Walls. On the upper wall ($y = h$, $\bar{n} = -\bar{e}_y$): $\tau = -\mu \frac{\partial v_x}{\partial y}\Big|_h = -\frac{\mu U}{h^2}(4h - 6h) = \frac{2\mu U}{h}$ in $+x$: the liquid in the gap moves faster than the wall and drags it forward. Over the length L, for each wall (the walls outside the plate region are in uniform flow, with no shear):

$$F_w = \frac{2\mu U L}{h} \text{ on each wall,} \quad \text{in the direction of its motion;} \qquad \text{both walls: } \frac{4\mu U L}{h}$$

Plate. On each face ($y = 0^{\pm}$, $\bar{n} = \pm\bar{e}_y$): $\tau = \mu \frac{\partial v_x}{\partial y}\Big|_0 = \frac{4\mu U}{h}$ along $+x$. Two faces and the length L:

$$F_p = 2 \cdot \frac{4\mu U}{h}\, L = \frac{8\mu U L}{h}$$

(the drag on the plate, in the direction of the flow).

Check by momentum. For the liquid in the plate region (no change of momentum flux, uniform U in and out), the forces on it add to zero: the pressure difference on the section $2h$, $2h(P_{-\infty} - P_{\infty}) = 12\mu UL/h$, minus the reactions on the two walls, $4\mu UL/h$, minus the reaction on the plate, $8\mu UL/h$: $12 - 4 - 8 = 0$.

Power. The liquid pushes the walls forward with the total force $4\mu UL/h$, so to keep them at the constant velocity U an external force $-F_w$ must hold each one back. The power that the external agent supplies to the walls is

$$\dot{W}_{walls} = -2F_w U = -\frac{4\mu U^2 L}{h}$$

negative: the walls do not have to be driven, they **receive** the power $4\mu U^2 L/h$ from the liquid (which is being pushed through the plate region by the imposed pressure drop); a brake has to absorb it. Its magnitude is $2F_w U$.

Temperature Increase

Control volume: the liquid between two sections far upstream and far downstream, bounded by the walls and the plate (all insulated: no heat). Steady energy balance in integral form: net outflow of $\rho(e + v^2/2)$ plus the work of the pressure at the two sections equals the work of the viscous forces on the moving boundaries (the plate is at rest, the walls move with U). The kinetic energy flux is the same in and out (uniform U); with $e = cT$:

$$\rho c\,(2hU)\,(T_{\infty} - T_{-\infty}) = \underbrace{2hU\,(P_{-\infty} - P_{\infty})}_{\text{work of the pressure}} - \underbrace{2F_w U}_{\text{work done on the walls}} = \frac{12\mu U^2 L}{h} - \frac{4\mu U^2 L}{h} = \frac{8\mu U^2 L}{h}$$

(the work of the force that the walls exert on the liquid is $-F_w U$ per wall; the reduced pressure is used, since the flow is horizontal and the heights are the same). Hence

$$T_{\infty} - T_{-\infty} = \frac{8\mu U^2 L/h}{2\rho c\, hU} = \frac{4\mu UL}{\rho c\, h^2}$$

Check with the dissipation: $\Phi = \mu(\partial v_x/\partial y)^2 = \mu U^2(4h-6y)^2/h^4$, and $\int_0^h \Phi\, dy = \frac{\mu U^2}{h^4}\left[-\frac{(4h-6y)^3}{18}\right]_0^h = \frac{\mu U^2}{h^4}\,\frac{64h^3+8h^3}{18} = \frac{4\mu U^2}{h}$; two halves and the length L: $\frac{8\mu U^2 L}{h}$, the same, as it must be (all the mechanical power that is not stored becomes heat).

Answers and Checks

- Viscous dominance: $(\rho U h/\mu)(h/L) \ll 1$.
- Velocity over the plate: $v_x = \frac{U\,y}{h^2}(4h-3y)$ for $0 \le y \le h$ (symmetric in y); $P_l = 6\mu U/h^2$; $P_{-\infty} - P_{\infty} = 6\mu UL/h^2$.
- Forces: $F_w = 2\mu UL/h$ on each wall (forward), $F_p = 8\mu UL/h$ on the plate (forward). Power at the walls: $2F_wU = 4\mu U^2 L/h$, received from the liquid.
- Temperature: $T_{\infty} - T_{-\infty} = 4\mu UL/(\rho c h^2)$.
- Dimensions: $\mu UL/h$ per unit width is MT^{-2}, a force per unit width; $\mu UL/(\rho c h^2)$ is $ML^{-1}T^{-1}\cdot LT^{-1}\cdot L/(ML^{-3}\cdot L^2T^{-2}\Theta^{-1}\cdot L^2) = \Theta$.
- Limit $L \to 0$: all the quantities vanish, as they should without a plate.
- Check against the official key: none; the sheet has no solution, so all the results of this note are computed here and checked by the momentum, flux and energy balances above.

PROBLEM VF18

Liquid Dragged up a Gap by a Moving Wall

Original statement · VF18.pdf

Couette + Poiseuille | Rising column | Critical velocity

OFFICIAL PROBLEM STATEMENT

A liquid of density ρ and viscosity μ is contained in an open vessel limited laterally by a vertical wall. A thin flat plate is placed parallel to the wall with its lower end at a depth H, forming a gap of thickness d, as shown in the figure. At the initial instant of time, the vertical wall begins to move upwards with a constant velocity V, dragging the liquid into the gap, where a liquid column of height $h(t)$ rises. To solve the problem, assume that the flow in the gap is dominated by viscosity. Address the following questions in order:

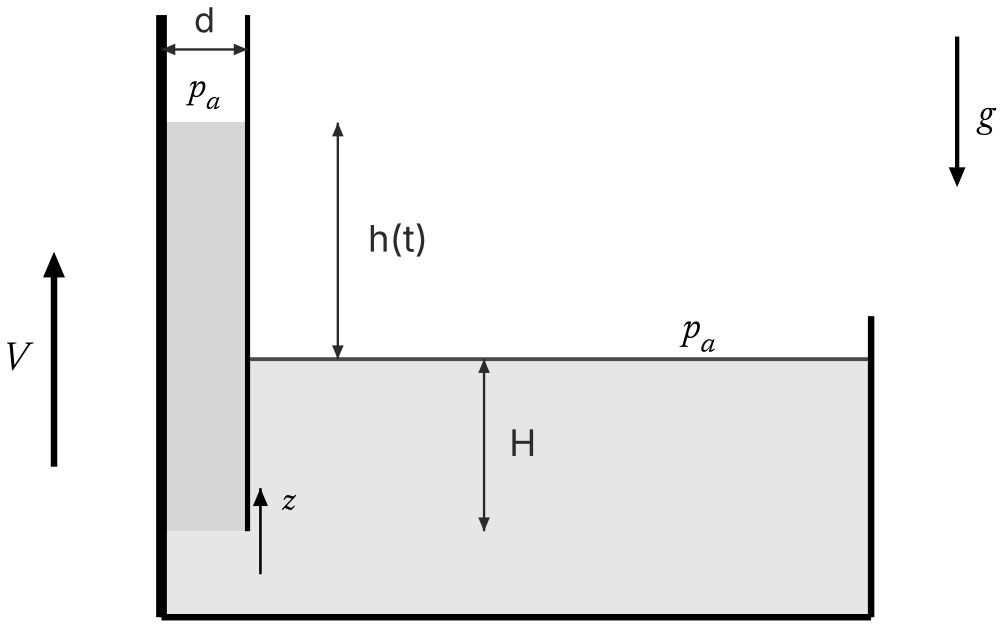
1. Obtain the velocity profile across the gap $v_z(y)$.

2. Calculate the volume flow rate that enters the gap $Q(t) = \int_0^d v_z \, dy$ as well as the distribution of reduced pressure $P = p + \rho g z$ along the liquid column and its associated gradient $P_l = -\partial P/\partial z$. In particular, write expressions for the instantaneous values of $Q(t)$ and $P_l(t)$ in terms of $h(t)$.

3. Write the differential equation with initial condition that determines the variation with time of $h(t)$ and compute $h(t)$ by integration.

4. Investigate the solution that appears for large times. In particular, show that for values of the wall velocity V below a critical value V_c the liquid column settles at a constant height h_∞, whereas for $V > V_c$ the column keeps rising, with a rate dh/dt that approaches a constant value $dh/dt = \dot{h}_\infty$ for large times. Give the values of V_c, h_∞, and $\dot{h}_\infty$.

5. For $V < V_c$, compute the power needed to keep the wall in motion when the steady solution is reached.



Statement figure (redrawn from VF18.pdf, panel t > 0): a vertical wall that moves up with velocity V and a fixed plate at the distance d, with its lower end at the depth H below the free surface; the gap is filled with liquid up to the height h(t) above the free surface; p_a acts on the free surface and on the top of the column

Source: VF18.pdf (page 1, prefinal exam of Dec 16, 2013; statement and handwritten key). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.9-7.12; wedge effect, Eqs. 7.69-7.75.

Worked solution

Hypotheses, Data and Unknowns

- The wall ($y = 0$) moves up with V; the plate ($y = d$) is at rest. z is measured upwards from the free surface of the vessel; the gap extends from the lower end of the plate, $z = -H$, up to the top of the column, $z = h(t)$, with $h(0) = 0$.
- The free surface of the vessel and the top of the column are at p_a. The hydrostatic pressure at the lower end of the plate: $p = p_a + \rho g H$.
- The flow is slender and quasi-steady (viscosity dominates), so the profile is Couette plus Poiseuille at each instant, with the reduced pressure uniform across the gap and a uniform gradient $P_l(t)$ along it.

Symbol	Meaning	SI unit
V	velocity of the wall	m/s
d, H	width of the gap, depth of the lower end of the plate	m
$h(t)$	height of the column above the free surface	m
$Q(t)$	volume flux entering the gap (per unit width)	m^2/s

Velocity Profile

Equation (7.69) with $P_l = -\partial P/\partial z$, the wall at $y = 0$ moving with V and the plate at $y = d$ at rest (Eq. 7.12):

$$v_z = V\left(1 - \frac{y}{d}\right) + \frac{P_l}{2\mu}\, d^2\, \frac{y}{d}\left(1 - \frac{y}{d}\right)$$

Flux and Pressure

$$Q = \int_0^d v_z\, dy = \frac{Vd}{2} + \frac{P_l d^3}{12\mu}, \qquad \frac{\partial P}{\partial z} = \frac{12\mu}{d^3}\left(\frac{Vd}{2} - Q\right)$$

The reduced pressure at the lower end of the plate ($z = -H$, $p = p_a + \rho g H$) is $p_a + \rho g H - \rho g H = p_a$, and at the top of the column ($z = h$, $p = p_a$) it is $p_a + \rho g h$. Since $\partial P/\partial z$ is uniform in the gap, integrating:

$$P - p_a = \frac{12\mu}{d^3}\left(\frac{Vd}{2} - Q\right)(z + H), \qquad P_l = -\frac{p_a + \rho g h - p_a}{h + H} = -\frac{\rho g h}{h + H}$$

Equating the two expressions of P_l (P_l from the integral and from the flux):

$$Q = \frac{Vd}{2} - \frac{\rho g h}{h+H}\,\frac{d^3}{12\mu}$$

(the key writes the three results without the intermediate step: with $z = h$, $P - p_a = \rho g h$ gives $\frac{12\mu}{d^3}\left(\frac{Vd}{2} - Q\right) = \frac{\rho g h}{h+H}$, and $P_l = -\partial P/\partial z$).

Evolution of the Column

The column is a liquid of cross-section d (per unit width), so its top rises with the mean velocity Q/d: $d\,\dot{h} = Q$, that is

$$\frac{dh}{dt} = \frac{V}{2} - \frac{\rho g d^2}{12\mu}\,\frac{h}{h+H} = \frac{1}{2}\left[V - V_c\,\frac{h}{h+H}\right], \qquad V_c = \frac{\rho g d^2}{6\mu}, \quad h(0) = 0$$

Writing $A = V - V_c$, $\frac{dh}{dt} = \frac{1}{2}\,\frac{Ah+VH}{h+H}$ and $dt = \frac{2(h+H)}{Ah+VH}\,dh$. Since $h + H = \frac{1}{A}(Ah + VH) - \frac{HV_c}{A}$ (as $H - \frac{VH}{A} = -\frac{HV_c}{A}$), the integral from 0 to h (Barrow's rule) is

$$\frac{t}{2} = \frac{h}{A} - \frac{HV_c}{A^2}\ln\frac{Ah+VH}{VH} \implies h - \frac{V_c H}{V - V_c}\ln\frac{(V-V_c)h + VH}{VH} = (V - V_c)\frac{t}{2}$$

(for $V \neq V_c$; for $V = V_c$ the equation is $dh/dt = \frac{V}{2}\,\frac{H}{h+H}$ and $h + \frac{h^2}{2H} = \frac{Vt}{2}$).

Large Times

- $V > V_c$. $A > 0$ and the logarithm grows slower than h: $h \to \infty$ as $t \to \infty$ and $dh/dt \to \frac{1}{2}\,(V - V_c)$, since $h/(h+H) \to 1$:

$$\dot{h}_\infty = \frac{V - V_c}{2}$$

- $V < V_c$. $A < 0$: dh/dt goes to zero when $Ah + VH = 0$, so $Q \to 0$ and the column settles at

$$h_\infty = \frac{HV}{V_c - V}$$

(the argument of the logarithm tends to zero, the column approaches h_∞ asymptotically). - **Critical velocity:** $V_c = \dfrac{\rho g d^2}{6\mu}$: the wall velocity for which the Couette flux $Vd/2$ equals the maximum backflow that the weight of an infinitely long column can drive, $\rho g d^3/(12\mu)$.

Power in the Steady State, $V < V_c$

At the end $Q = 0$, so $P_l = -6\mu V/d^2$ (from $Q = 0$: $\frac{Vd}{2} + \frac{P_l d^3}{12\mu} = 0$), and the profile is $v_z = V\left(1 - \frac{y}{d}\right) - \frac{3V}{d^2}\, y\,(d - y)$. The shear on the moving wall ($\bar{n} = +\bar{e}_y$ from the wall into the liquid):

$$\tau_w = \mu \frac{\partial v_z}{\partial y}\Big|_{y=0} = -\frac{\mu V}{d} + \frac{P_l d}{2} = -\frac{\mu V}{d} - \frac{3\mu V}{d} = -\frac{4\mu V}{d}$$

(the key writes the same, $\left(-\frac{V}{d} - \frac{\rho g h}{h+H}\frac{d}{2\mu}\right)\mu = -\frac{4\mu V}{d}$ at $h = h_\infty$, where $\frac{\rho g h_\infty}{h_\infty + H} = \frac{6\mu V}{d^2}$). This is in $-z$, resisting the motion. The wall is in contact with the liquid of the gap over the whole filled length from $z = -H$ to $z = h_\infty$, that is $h_\infty + H = \frac{HV_c}{V_c - V}$. The power that must be applied is the force times the velocity:

$$\dot{W} = \frac{4\mu V}{d}\,(h_\infty + H)\,V = \frac{4\mu V^2 H}{d}\,\frac{V_c}{V_c - V}$$

(per unit width). **Energy check:** with $Q = 0$ the liquid does not gain potential energy nor receive any net pressure work, so the whole power is dissipated: $\mu \int_0^d (\partial v_z/\partial y)^2 dy = \frac{\mu V^2}{d}\int_0^1 (4 - 6s)^2 ds = \frac{4\mu V^2}{d}$ per unit length, times the length $h_\infty + H$, the same.

Discrepancy with the official solution

The key boxes $\dot{W} = \frac{4\mu V^3}{d}\frac{H}{V_c - V}$, which is $\frac{4\mu V}{d}\, V\, h_\infty$: the wetted length it uses is h_∞, the height of the column above the free surface. The gap, and so the wall in contact with the liquid of the film, extends from the lower end of the plate, $z = -H$, to the top of the column, $z = h_\infty$, a length $h_\infty + H$, and the profile (with the same shear) is the same in the part below the free surface. The correct length is $h_\infty + H = HV_c/(V_c - V)$, and $\dot{W} = 4\mu V^2 H\, V_c/\left[d\,(V_c - V)\right]$, larger than the key's by the factor V_c/V. The dissipation check above confirms it. (If the part of the wall below the free surface is not counted, because the liquid there is in the open vessel, the key's value would follow; the statement does not say so, and the gap is filled from the lower end of the plate.)

Answers and Checks

- Velocity: $v_z = V\left(1 - \frac{y}{d}\right) + \frac{P_l}{2\mu} d^2 \frac{y}{d}\left(1 - \frac{y}{d}\right)$; $Q = \frac{Vd}{2} - \frac{\rho g h}{h+H}\frac{d^3}{12\mu}$; $P_l = -\frac{\rho g h}{h+H}$.
- Equation: $\frac{dh}{dt} = \frac{V}{2} - \frac{\rho g d^2}{12\mu}\frac{h}{h+H}$, $h(0) = 0$; solution $h - \frac{V_c H}{V - V_c}\ln\frac{(V-V_c)h + VH}{VH} = (V - V_c)\frac{t}{2}$.
- $V_c = \rho g d^2/(6\mu)$; for $V > V_c$, $\dot{h}_\infty = (V - V_c)/2$; for $V < V_c$, $h_\infty = HV/(V_c - V)$.
- Power: $\dot{W} = \frac{4\mu V^2}{d}\left(h_\infty + H\right) = \frac{4\mu V^2 H V_c}{d(V_c - V)}$.
- Dimensions: $\rho g d^2/\mu$ is $ML^{-3}\cdot LT^{-2}\cdot L^2/(ML^{-1}T^{-1}) = LT^{-1}$, a velocity; $\mu V^2 H/d$ is $ML^{-1}T^{-1}\cdot L^2T^{-2} = MLT^{-3}$ per unit width, a power per unit width.
- Limits: for $H \to 0$, $h_\infty \to 0$ and $\dot{W} \to 0$; for $V \to V_c^-$, $h_\infty \to \infty$ and $\dot{W} \to \infty$.
- Check against the official key: VF18.pdf gives the same profile, Q, P_l, differential equation, implicit solution, V_c, $\dot{h}_\infty$, h_∞ and the wall shear; its power differs by the length used, as noted.

PROBLEM VF19

Viscous Pump with a Moving Rod

Original statement · VF19.pdf

Annulus | Moving rod | Flux and minimum velocity

OFFICIAL PROBLEM STATEMENT

Problem 4: To fill an open container of depth H located at a height L above the surface of a reservoir we use the viscous pumping system sketched in the figure, including a pipe or radius R_2 connecting the surface of the reservoir with the bottom of the container and an infinitely long, inner, coaxial rod of radius R_1 that moves vertically with velocity U, dragging the water into the inter-cylinder gap.

1. Give the condition for the flow in the gap to be dominated by viscosity.

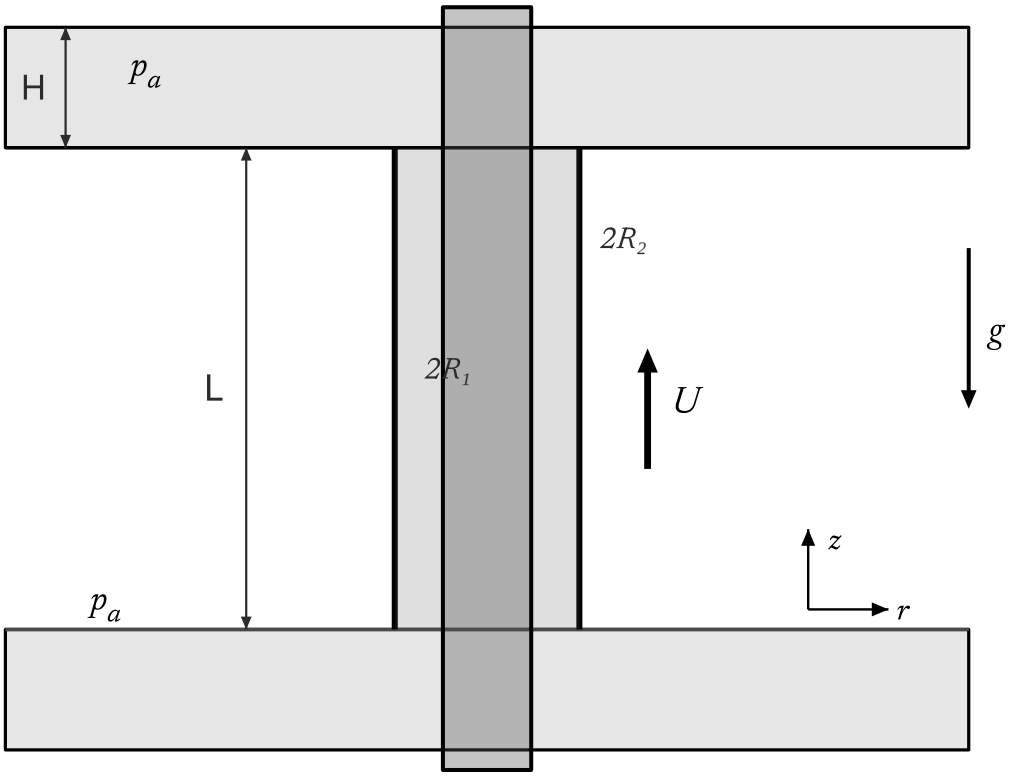
2. Obtain the velocity profile $v_z(r)$ across the gap.

3. Determine the loss of reduced pressure per unit length $P_l = -\partial P/\partial z$.

4. Compute the force acting on the moving shaft and the power needed to move it.

5. Determine the volume flow rate Q of the pumping system, including the minimum velocity U_{min} required for $Q > 0$.

The sheet prints "a pipe or radius R_2" ("or" for "of").



Statement figure (redrawn from VF19.pdf): a reservoir at p_a, a vertical pipe of radius R_2 and length L up to the bottom of an open container of depth H, and a coaxial rod of radius R_1 moving up with velocity U inside the pipe; z is vertical and r radial

Source: VF19.pdf (page 1, final exam of Jan 21, 2014; statement and handwritten key). Notes.pdf, Chapter 7: steady flow in a circular pipe, Eqs. 7.13-7.16; slender flows, Eqs. 7.40-7.46.

Worked solution

Hypotheses, Data and Unknowns

- The gap between the rod ($r = R_1$, moving with $+U\bar{e}_z$) and the pipe ($r = R_2$, at rest) has the length L and is slender, $R_2 - R_1 \ll L$; the flow is steady and axisymmetric, $v_z(r)$.
- z is the height above the reservoir surface. At the entrance of the pipe ($z = 0$) $p = p_a$; at its exit ($z = L$, the bottom of the container, under a depth H of water at p_a on top) $p = p_a + \rho g H$. The reduced pressure is $P = p + \rho g z$.

Symbol	Meaning	SI unit
R_1, R_2	radii of the rod and of the pipe	m

Symbol	Meaning	SI unit
L, H	length of the pipe, depth of the container	m
U	velocity of the rod (upward)	m/s
$\Delta = R_2^2 - R_1^2$, $\Lambda = \ln(R_2/R_1)$	abbreviations	m^2, 1

Criterion for Viscous Dominance

The slender-flow condition (Eq. 7.41), with the width $R_2 - R_1$ of the gap, the velocity U and the length L:

$$\frac{\rho U\,(R_2 - R_1)}{\mu}\,\frac{R_2 - R_1}{L} = \frac{\rho U\,(R_2 - R_1)^2}{\mu L} \ll 1$$

(the same with the velocity driven by the pressure, of order $P_l (R_2 - R_1)^2/\mu$, if it is larger than U).

Velocity Profile

Equation (7.7) in steady state with no slip on both walls:

$$0 = P_l + \frac{\mu}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right); \qquad r = R_1:\ v_z = U, \qquad r = R_2:\ v_z = 0$$

As in VF3, the solution is $v_z = \frac{P_l}{4\mu}\left(R_2^2 - r^2\right) + C\ln\frac{R_2}{r}$, which is zero at $r = R_2$; the rod condition gives $U = \frac{P_l}{4\mu}\Delta + C\Lambda$:

$$v_z = \frac{P_l}{4\mu}\left(R_2^2 - r^2\right) + \left[U - \frac{P_l}{4\mu}\left(R_2^2 - R_1^2\right)\right]\frac{\ln(R_2/r)}{\ln(R_2/R_1)}$$

Pressure Loss per Unit Length

The gap has constant section, so P_l is uniform. The reduced pressure at the entrance is p_a and at the exit $p_a + \rho g H + \rho g L$:

$$P_l = -\frac{\partial P}{\partial z} = \frac{p_a - [p_a + \rho g (H + L)]}{L} = -\frac{\rho g (H + L)}{L}$$

negative: the pressure (reduced) increases along the flow, opposing it; the rod has to push the water against the weight of the whole column from the reservoir to the top of the container.

Force on the Rod and Power

The force of the liquid on the rod (normal from the rod into the liquid, $\bar{n} = +\bar{e}_r$) per unit area is $\tau_w = \mu \, \partial v_z / \partial r |_{r=R_1}$. With $\frac{dv_z}{dr} = -\frac{P_l}{2\mu} r - \frac{C}{r}$:

$$\tau_w = -\frac{P_l R_1}{2} - \frac{\mu C}{R_1} = -\frac{\mu U}{R_1 \Lambda} - \frac{P_l R_1}{2} \left[1 - \frac{1}{2} \, \frac{R_2^2 / R_1^2 - 1}{\ln (R_2 / R_1)} \right]$$

(using $C = \left(U - P_l \Delta / (4\mu) \right) / \Lambda$). With $P_l = -\rho g (H + L) / L$ both terms are negative: the liquid resists the motion of the rod, because $\frac{\Delta}{2 R_1^2 \Lambda} = \frac{x^2 - 1}{2 \ln x} > 1$ for $x = R_2 / R_1 > 1$. The force on the rod over the length L is $F_z = 2\pi R_1 L \, \tau_w \, \bar{e}_z$, downward, and the power that must be supplied to move it is minus the power of that force (the key writes $W = U \, 2\pi R_1 L \, \tau_w$, the power of the force of the liquid, negative):

$$W = -2\pi R_1 L U \tau_w = 2\pi R_1 L U \left[\frac{\mu U}{R_1 \ln (R_2 / R_1)} + \frac{\rho g (H + L) R_1}{2L} \left(\frac{R_2^2 - R_1^2}{2 R_1^2 \ln (R_2 / R_1)} - 1 \right) \right]$$

The key's lines end at the expression of τ_w without writing the final value.

Volume Flux and Minimum Velocity

$Q = \int_{R_1}^{R_2} 2\pi r\, v_z\, dr$. Barrow's rule for the two terms:

$$\int_{R_1}^{R_2} r\left(R_2^2 - r^2\right) dr = \left[\frac{R_2^2 r^2}{2} - \frac{r^4}{4}\right]_{R_1}^{R_2} = \frac{\left(R_2^2 - R_1^2\right)^2}{4}, \qquad \int_{R_1}^{R_2} r \ln\frac{R_2}{r}\, dr = \left[\frac{r^2}{2}\ln\frac{R_2}{r} + \frac{r^2}{4}\right]_{R_1}^{R_2} = \frac{R_2^2 - R_1^2}{4} - \frac{R_1^2}{2}\ln\frac{R_2}{R_1}$$

$$Q = \frac{\pi P_l \Delta^2}{8\mu} + \frac{\pi}{2}\left(U - \frac{P_l \Delta}{4\mu}\right)\frac{\Delta - 2R_1^2\Lambda}{\Lambda}$$

Discrepancy with the official solution

The key gives $Q = -2\pi\left\{\frac{P_l}{16\mu}\left(R_2^4 - R_1^4\right) + \frac{U - P_l\Delta/(4\mu)}{4\ln(R_2/R_1)}\left[R_2^2\left(2\ln R_2 - 1\right) - R_1^2\left(2\ln R_1 - 1\right)\right]\right\}$, which is not the integral of its own velocity profile. The first term should have $\left(R_2^2 - R_1^2\right)^2$ (the integral of $r(R_2^2 - r^2)$, as above), not $R_2^4 - R_1^4$; and the second keeps only the antiderivative of $r\ln r$ and drops the constant term that comes from $\ln R_2$ (a logarithm of a dimensional length appears, which is dimensionally meaningless); the overall sign is also the opposite. As a consequence the key's minimum velocity, solved from its own $Q = 0$, is not the one of the integral. The results above are used.

Checks of Q: for $R_1 \to 0, \Delta \to R_2^2, R_1^2\Lambda \to 0, \Lambda \to \infty$ and the second term vanishes: $Q \to \frac{\pi P_l R_2^4}{8\mu}$, the Poiseuille flux (7.15). For a thin gap $g = R_2 - R_1 \ll R_1$, $(\Delta - 2R_1^2\Lambda)/\Lambda \simeq 2R_1 g \cdot \frac{g}{R_1} \cdot R_1$ (to first order $2R_1^2\epsilon$ with $\epsilon = g/R_1$), so the second term is $\pi R_1^2 \epsilon U = \pi R_1 g U$, the Couette flux $\frac{Ug}{2} \cdot 2\pi R_1$.

Minimum velocity. Setting $Q = 0$: $\left(U - \frac{P_l\Delta}{4\mu}\right) = -\frac{P_l\Delta^2}{4\mu}\frac{\Lambda}{\Delta - 2R_1^2\Lambda}$, so

$$U_{min} = \frac{P_l\Delta}{4\mu}\left[1 - \frac{\Delta\,\Lambda}{\Delta - 2R_1^2\Lambda}\right] = \frac{\rho g(H+L)\left(R_2^2 - R_1^2\right)}{4\mu L}\left[\frac{\left(R_2^2 - R_1^2\right)\ln(R_2/R_1)}{R_2^2 - R_1^2 - 2R_1^2\ln(R_2/R_1)} - 1\right]$$

positive, since $\frac{\Delta\Lambda}{\Delta - 2R_1^2\Lambda} > 1$. For $U > U_{min}$ the rod pumps water to the container ($Q > 0$).

Answers and Checks

- Viscous dominance: $\rho U(R_2-R_1)^2/(\mu L) \ll 1$.
- Velocity: $v_z = \frac{P_l}{4\mu}(R_2^2-r^2) + \left[U - \frac{P_l}{4\mu}(R_2^2-R_1^2)\right]\frac{\ln(R_2/r)}{\ln(R_2/R_1)}$; $P_l = -\rho g(H+L)/L$.
- Force on the rod: $F_z = 2\pi R_1 L \tau_w$ with τ_w above; power supplied: $W = -2\pi R_1 L U \tau_w > 0$.
- Flux: $Q = \frac{\pi P_l \Delta^2}{8\mu} + \frac{\pi}{2}\left(U - \frac{P_l\Delta}{4\mu}\right)\frac{\Delta - 2R_1^2\Lambda}{\Lambda}$; U_{min} as above.
- Dimensions: $\rho g(H+L)\Delta/(\mu L)$ is $ML^{-3}\cdot LT^{-2}\cdot L^2/(ML^{-1}T^{-1}) = LT^{-1}$, a velocity; $\Delta^2 P_l/\mu$ is $L^4\cdot ML^{-2}T^{-2}/(ML^{-1}T^{-1}) = L^3T^{-1}\cdot L^{-1}$, a flux per unit length times L (as in (7.15)).
- Check against the official key: VF19.pdf gives the same criterion, profile, P_l and shear; its Q and U_{min} differ, as noted, and it does not write the final value of the power.

PROBLEM VF20

Cylinder Levitated by a Jet

Original statement · VF20.pdf

Radial Poiseuille flow | Thin film | Force balance

OFFICIAL PROBLEM STATEMENT

FLUID MECHANICS I. PROBLEM 2 (10 POINTS). 16-01-2019. Time: 75 minutes. The cylinder of radius R and height $2R$ shown in the figure is submerged in an infinite tank of liquid of density ρ and viscosity μ. The cylinder, of density $\rho_s > \rho$, is sustained at a constant height $h \ll R$ by the effect of a narrow jet of volumetric flow rate Q and radius of the order of h, injected at the center of the base of the cylinder (see the sketch). Assuming the fluid movement is negligible in the bulk of the tank (P_0 = const.) and the flow below the cylinder is dominated by viscosity, we want to determine the dependence of h as a function of Q. To this end, the following sequence is suggested:

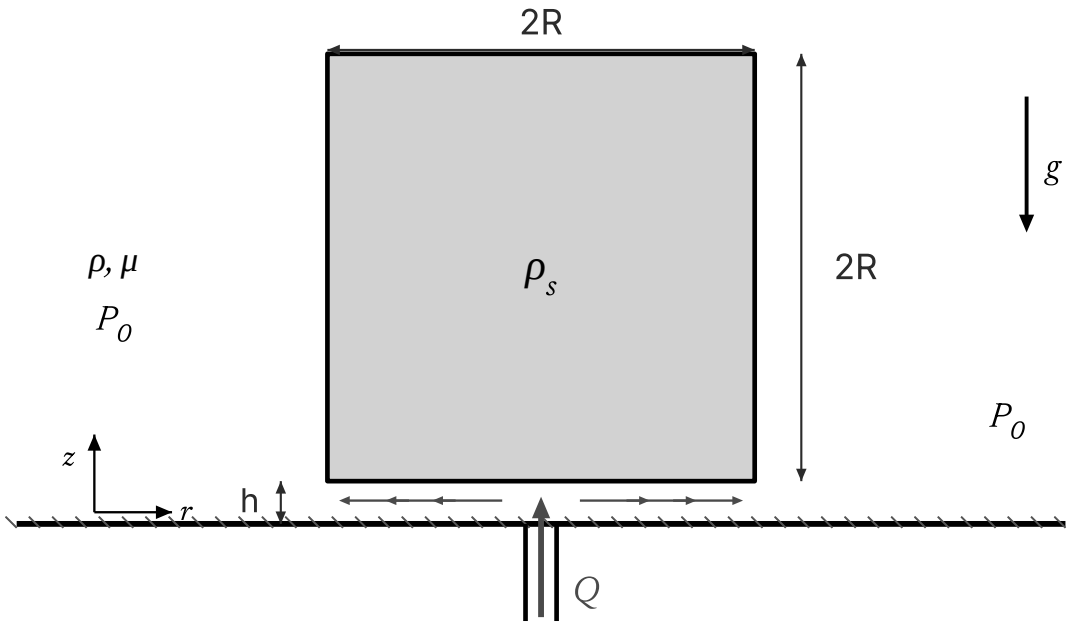
1. Give the criterion, as a function of the known parameters, for the flow in the gap of width h between the cylinder and the bottom of the tank to be dominated by viscosity.

2. Determine the velocity profile as a function of the local reduced pressure gradient $\partial P/\partial r$.

3. Obtain the local flow rate $q(r)$ at a generic radial position r

4. Use the continuity equation to derive an equation for the reduced pressure P, with all needed boundary conditions. Obtain P.

5. Determine h



Statement figure (redrawn from VF20.pdf): a cylinder of radius R, height 2R and density ρ_s hovering at the height h above the bottom of a tank of liquid (ρ, μ, pressure P_0 far away) on a jet of flow rate Q injected at the centre of its base; z is vertical and r radial

Source: VF20.pdf (page 1: statement; pages 2 and 3: handwritten key; exam of 16-01-2019). Notes.pdf, Chapter 7: slender films, Eqs. 7.67-7.71; the wedge effect, Eq. 7.72.

Worked solution

Hypotheses, Data and Unknowns

- The gap between the base of the cylinder ($z = h$) and the bottom of the tank ($z = 0$) is slender, axisymmetric and steady; the liquid enters it through the jet at the centre ($r \lesssim h$) and leaves at $r = R$ to the tank, where the reduced pressure is P_0 (the liquid is at rest there).
- Reduced pressure $P = p + \rho g z$. The cylinder is in equilibrium: its weight $W = 2\pi R^3 \rho_s g$ (volume $2\pi R^3$) balances the force of the liquid.

Symbol	Meaning	SI unit
R, $2R$	radius and height of the cylinder	m
ρ_s, ρ, μ	density of the cylinder, of the liquid, viscosity	kg/m^3, kg/m^3, kg/(m s)

Symbol	Meaning	SI unit
h, Q	height of the gap, flux of the jet	m, m^3/s

Criterion for Viscous Dominance

The convective acceleration is negligible against the viscous force (Eq. 7.41) when $\frac{\rho V_z c h}{\mu}\frac{h}{R} \ll 1$, with V_{zc} the velocity scale of the jet, of radius $\sim h$, so $V_{zc} \sim Q/h^2$. Continuity in the gap, $V_{rc}/R \sim V_{zc}/h$, gives $V_{rc} \sim V_{zc}h/R \ll V_{zc}$, and the radial velocity in the gap is $v_r \sim Q/(2\pi r h)$, equal to $\sim Q/h^2$ at $r \sim h$. Then

$$\frac{\rho V_{zc} h}{\mu}\,\frac{h}{R} \sim \frac{\rho Q}{\mu h}\,\frac{h}{R} = \frac{\rho Q}{\mu R} \ll 1$$

(the key states $V_r \sim Q/h^2$ for $r \sim h$ and boxes the result; at larger r the velocity is smaller, so the condition is the most demanding at the entrance of the gap).

Velocity Profile and Flux

Radial momentum in the gap (Eq. 7.69 in cylindrical coordinates), with no slip at the bottom of the tank ($z = 0$) and at the base of the cylinder ($z = h$):

$$0 = -\frac{\partial P}{\partial r} + \mu\frac{\partial^2 v_r}{\partial z^2}; \qquad z = 0:\ v_r = 0, \qquad z = h:\ v_r = 0$$

Integrating twice, $v_r = \frac{1}{2\mu}\frac{\partial P}{\partial r}z^2 + Az$ ($B = 0$ by the condition at $z = 0$), and $0 = \frac{1}{2\mu}\frac{\partial P}{\partial r}h^2 + Ah$ gives A:

$$v_r = -\frac{1}{2\mu}\,\frac{\partial P}{\partial r}\left(hz - z^2\right)$$

The local flux through the cylinder of radius r and height h is

$$q(r) = \int_0^h 2\pi r\, v_r\, dz = -\frac{2\pi r}{2\mu}\frac{\partial P}{\partial r}\left[\frac{hz^2}{2} - \frac{z^3}{3}\right]_0^h = -\frac{\pi h^3 r}{6\mu}\frac{\partial P}{\partial r}$$

Pressure

Continuity in the gap: no liquid enters or leaves through the bottom or the base (both are walls), so $q(r) = Q$ for every r:

$$Q = -\frac{\pi h^3 r}{6\mu}\frac{\partial P}{\partial r} \implies \frac{\partial P}{\partial r} = -\frac{6\mu Q}{\pi h^3 r}$$

Integrating in r with the boundary condition $P = P_0$ at $r = R$:

$$P = P_0 - \frac{6\mu Q}{\pi h^3}\,\ln\frac{r}{R}$$

an overpressure for $r < R$, which grows (logarithmically) towards the centre, where the slender approximation stops being valid (at $r \sim h$).

Height of the Cylinder

The force of the liquid on the cylinder is the difference of the pressures on its base and on its top, over the area πR^2 (the lateral pressure is horizontal). At the top, $z = h + 2R$, the liquid is at rest with the reduced pressure P_0, so $p_t = P_0 - \rho g(h + 2R)$; at the base, $p_b = P - \rho g h$:

$$F = \int_0^R 2\pi r\,(p_b - p_t)\,dr = \int_0^R 2\pi r\,[2\rho g R + (P - P_0)]\,dr = 2\pi\rho g R^3 - \frac{12\mu Q}{h^3}\int_0^R r\ln\frac{r}{R}\,dr$$

With $r = Rs$, $\int_0^R r\ln\frac{r}{R}dr = R^2\int_0^1 s\ln s\,ds = R^2\left[\frac{s^2}{2}\ln s - \frac{s^2}{4}\right]_0^1 = -\frac{R^2}{4}$:

$$F = 2\pi\rho g R^3 + \frac{3\mu Q R^2}{h^3}$$

The first term is the buoyancy (the weight of the displaced liquid, $\rho g \cdot 2\pi R^3$) and the second is the lift of the jet. The cylinder floats at the height h when $F = W = 2\pi R^3 \rho_s g$:

$$\frac{3\mu Q R^2}{h^3} = 2\pi R^3 g\left(\rho_s - \rho\right) \implies h = \left[\frac{3\mu Q R^2}{2\pi R^3 g\left(\rho_s - \rho\right)}\right]^{1/3} = \left[\frac{3\mu Q}{2\pi R g\left(\rho_s - \rho\right)}\right]^{1/3}$$

Answers and Checks

- Viscous dominance: $\rho Q/(\mu R) \ll 1$ (and $h \ll R$).
- Profile: $v_r = -\frac{1}{2\mu}\frac{\partial P}{\partial r}(hz - z^2)$; flux: $q = -\frac{\pi h^3 r}{6\mu}\frac{\partial P}{\partial r} = Q$; pressure: $P = P_0 - \frac{6\mu Q}{\pi h^3}\ln\frac{r}{R}$.
- Height: $h = \left[\frac{3\mu Q}{2\pi R g(\rho_s - \rho)}\right]^{1/3}$.
- Dimensions: $\mu Q/(Rg\Delta\rho)$ is $ML^{-1}T^{-1}\cdot L^3T^{-1}/(L\cdot LT^{-2}\cdot ML^{-3}) = L^3$, so h is a length; $\rho Q/(\mu R)$ is $ML^{-3}L^3T^{-1}/(ML^{-1}T^{-1}L) = 1$.
- Limits: $Q \to 0$ gives $h \to 0$ (the cylinder rests on the bottom) and $h \propto Q^{1/3}$; $\rho_s \to \rho$ makes the cylinder neutrally buoyant and $h \to \infty$ (it floats with any Q).
- Check against the official key: VF20.pdf gives the same criterion (stated without the step $V_{zc} \sim Q/h^2$), profile, q, P, force ($2\pi\rho g R^3 + 3\mu Q R^2/h^3$) and h (left as $\left[3\mu Q R^2/(2\pi R^3 g(\rho_s - \rho))\right]^{1/3}$).